popgenVCF quickstart example: chromosome 22 subset (160 samples, 8 populations)

popgenVCF

2026-08-11

Contents

1 Executive summary

This PDF file is a portable, human-readable analysis report. Figures are embedded directly in the file. The HTML version can be opened in a standard web browser, while the PDF version is optimized for compact exchange, printing, and email. The accompanying `analysis_results.rds` file is a reproducibility archive for R users; it is not required to read this report.

Generated with **popgenVCF v0.10.0**.

The analysis retained **160 samples** from **8 populations**, with **1969 QC-passing SNPs** and **357 LD-pruned SNPs**.

The global Weir-Cockerham F_{ST} estimate was **0.0915**.

2 Quality control

step	variants	removed_at_step	retained_percent
Input biallelic	98922	0	100.00
After MAF	63585	35337	64.28
After missingness	63585	0	64.28
After LD pruning	357	63228	0.36

3 Genetic diversity

population	n_samples	n_loci	polymorphic_loci	polymorphic_fraction	observed_heterozygosity	expected_heterozygosity	inbreeding_coefficient	mean_minor_allele_frequency	mean_locus_call_rate	hwe_tested_loci	hwe_significant_loci	hwe_significant_loci_fdr	private_allele_loci	mean_allelic_richness	mean_effective_alleles
CHB	20	1969	1653	0.8395	0.2683	0.2855	0.0603	0.2133	1	1653	149	55	0	1.8395	1.5054
GBR	20	1969	1736	0.8817	0.2764	0.2827	0.0226	0.2004	1	1736	99	31	0	1.8817	1.4800
ITU	20	1969	1829	0.9289	0.2736	0.2919	0.0626	0.2059	1	1829	100	43	0	1.9289	1.4893
LWK	20	1969	1622	0.8238	0.2638	0.2727	0.0326	0.1963	1	1622	66	14	0	1.8238	1.4688
PEL	20	1969	1800	0.9142	0.2330	0.2579	0.0968	0.1851	1	1800	147	45	0	1.9142	1.4362
PUR	20	1969	1844	0.9365	0.2748	0.2730	-0.0067	0.1871	1	1844	81	21	0	1.9365	1.4487
STU	20	1969	1788	0.9081	0.2561	0.2863	0.1056	0.2013	1	1788	131	33	0	1.9081	1.4807
YRI	20	1969	1636	0.8309	0.2731	0.2705	-0.0096	0.1944	1	1636	92	18	0	1.8309	1.4637

4 Principal component analysis

sample	vcf_sample	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	population
NA18536	NA18536	-0.0575	0.0756	-0.1958	-0.0176	0.0470	0.0646	-0.0320	-0.0509	-0.0829	0.0169	CHB
NA18538	NA18538	-0.0665	0.1310	-0.0958	-0.0006	0.0158	-0.0224	0.1012	0.0074	-0.2047	-0.0051	CHB
NA18541	NA18541	-0.0654	0.0991	-0.0443	-0.0362	0.0684	0.0917	0.0875	-0.0995	-0.0353	0.0002	CHB
NA18552	NA18552	-0.0714	0.1650	-0.1811	-0.1448	0.0223	0.0384	0.0671	0.0537	0.0328	-0.0593	CHB
NA18553	NA18553	-0.0616	-0.0660	-0.1590	0.0237	-0.2193	-0.0004	-0.1910	-0.0064	-0.1380	0.0197	CHB
NA18555	NA18555	-0.0312	0.0930	-0.1309	-0.1100	0.0308	0.0717	-0.0562	0.0124	0.0901	-0.1218	CHB
NA18557	NA18557	-0.0893	0.2057	-0.1264	-0.0737	0.0509	0.0121	0.0892	0.0377	-0.0913	0.0984	CHB
NA18562	NA18562	-0.0381	0.0209	-0.0813	0.0459	0.0625	-0.0312	0.1107	-0.1086	0.0245	-0.1299	CHB
NA18564	NA18564	-0.0305	0.0841	-0.1174	-0.1036	-0.0883	-0.0264	-0.1778	0.0857	0.2782	-0.1910	CHB
NA18573	NA18573	-0.0685	0.0633	0.0378	0.0394	-0.0765	0.0588	-0.0318	-0.1708	0.0201	-0.0102	CHB
NA18574	NA18574	-0.0873	0.1260	-0.1222	-0.0647	0.0656	0.0098	0.0426	-0.0156	-0.0750	-0.0113	CHB
NA18606	NA18606	-0.0584	0.0752	-0.1100	0.0288	0.0297	-0.0415	0.1097	0.1042	-0.0327	-0.0739	CHB
NA18611	NA18611	-0.0717	0.1273	-0.0419	-0.0468	-0.0176	0.1518	-0.0208	0.0271	-0.1508	-0.0482	CHB
NA18612	NA18612	-0.0504	-0.0424	-0.2118	0.1398	-0.1597	-0.0342	-0.1189	-0.0957	-0.0276	0.1465	CHB
NA18613	NA18613	-0.0583	0.1108	-0.0705	-0.0469	0.0092	0.0932	0.0875	0.0691	-0.1390	0.1136	CHB
NA18616	NA18616	-0.0173	0.0510	-0.1403	-0.1915	0.0842	0.1383	-0.0259	-0.1606	0.0728	-0.0742	CHB
NA18617	NA18617	-0.0307	0.1223	-0.1382	-0.1085	-0.0277	0.0449	-0.0945	0.0113	0.0748	-0.1828	CHB
NA18638	NA18638	-0.0722	0.1224	-0.0679	-0.0460	-0.0106	0.1791	-0.0069	-0.1120	-0.1554	0.0068	CHB
NA18643	NA18643	-0.0437	0.0467	-0.1922	0.0077	0.0101	-0.0009	0.0329	-0.0462	0.0704	-0.0114	CHB
NA18647	NA18647	-0.0558	0.1011	-0.1989	-0.0603	0.1633	0.1660	0.0094	-0.0507	0.0072	0.0560	CHB

5 PCA SNP loadings (top 20 by |loading|; see 31_PCA_loadings.tsv for the complete top-N list)

axis	snp_id	chromosome	position	contribution	magnitude
PC7	10937	22	20437004	0.2848	0.2848
PC7	10502	22	20371297	0.2541	0.2541
PC7	11374	22	20502771	0.2524	0.2524
PC8	13593	22	20753041	0.2191	0.2191
PC6	10937	22	20437004	-0.2191	0.2191
PC8	13990	22	20765538	0.2178	0.2178
PC8	14246	22	20773711	0.2034	0.2034
PC5	4933	22	20159049	-0.2010	0.2010
PC3	15906	22	20832523	0.1980	0.1980
PC8	13078	22	20733667	0.1952	0.1952
PC7	10415	22	20360856	0.1932	0.1932
PC8	14080	22	20768038	-0.1925	0.1925
PC9	3724	22	20121080	-0.1900	0.1900
PC6	10502	22	20371297	-0.1887	0.1887
PC10	11570	22	20630754	0.1883	0.1883
PC9	1516	22	20047587	-0.1869	0.1869
PC3	14848	22	20794119	0.1865	0.1865
PC9	2006	22	20065314	-0.1859	0.1859
PC8	14377	22	20777533	-0.1806	0.1806
PC10	11593	22	20639324	0.1763	0.1763

6 Population differentiation

population_1	population_2	n_1	n_2	fst	nm	jost_d	jost_d_n_snps
CHB	GBR	20	20	0.1210	1.8153	0.0547	1969
CHB	ITU	20	20	0.0631	3.7098	0.0274	1969
CHB	LWK	20	20	0.1631	1.2832	0.0754	1969
CHB	PEL	20	20	0.0794	2.9000	0.0322	1969
CHB	PUR	20	20	0.1101	2.0199	0.0480	1969
CHB	STU	20	20	0.0865	2.6390	0.0379	1969
CHB	YRI	20	20	0.1631	1.2824	0.0751	1969
GBR	ITU	20	20	0.0232	10.5031	0.0096	1969
GBR	LWK	20	20	0.1198	1.8364	0.0524	1969
GBR	PEL	20	20	0.0874	2.6113	0.0355	1969
GBR	PUR	20	20	0.0267	9.1026	0.0106	1969
GBR	STU	20	20	0.0350	6.8881	0.0144	1969
GBR	YRI	20	20	0.1344	1.6100	0.0594	1969
ITU	LWK	20	20	0.1057	2.1159	0.0465	1969
ITU	PEL	20	20	0.0400	5.9960	0.0158	1969
ITU	PUR	20	20	0.0269	9.0319	0.0109	1969
ITU	STU	20	20	0.0052	48.2538	0.0021	1969
ITU	YRI	20	20	0.1189	1.8530	0.0528	1969
LWK	PEL	20	20	0.1737	1.1891	0.0759	1969
LWK	PUR	20	20	0.1176	1.8764	0.0500	1969
LWK	STU	20	20	0.1112	1.9987	0.0485	1969
LWK	YRI	20	20	0.0052	47.5634	0.0020	1969
PEL	PUR	20	20	0.0621	3.7736	0.0239	1969
PEL	STU	20	20	0.0609	3.8555	0.0242	1969
PEL	YRI	20	20	0.1799	1.1393	0.0788	1969
PUR	STU	20	20	0.0339	7.1152	0.0136	1969
PUR	YRI	20	20	0.1226	1.7887	0.0522	1969
STU	YRI	20	20	0.1187	1.8565	0.0520	1969

7 Nei's standard genetic distance between populations (see 46_population_genetic_distance.tsv; a neighbour-joining tree is written alongside the individual-level tree when at least three populations are present)

population	CHB	GBR	ITU	LWK	PEL	PUR	STU	YRI
CHB	0.0000	0.0661	0.0378	0.0880	0.0418	0.0588	0.0486	0.0876
GBR	0.0661	0.0000	0.0197	0.0633	0.0452	0.0202	0.0244	0.0707
ITU	0.0378	0.0197	0.0000	0.0573	0.0251	0.0207	0.0122	0.0639
LWK	0.0880	0.0633	0.0573	0.0000	0.0879	0.0606	0.0594	0.0112
PEL	0.0418	0.0452	0.0251	0.0879	0.0000	0.0332	0.0337	0.0910
PUR	0.0588	0.0202	0.0207	0.0606	0.0332	0.0000	0.0234	0.0629
STU	0.0486	0.0244	0.0122	0.0594	0.0337	0.0234	0.0000	0.0629
YRI	0.0876	0.0707	0.0639	0.0112	0.0910	0.0629	0.0629	0.0000

8 Genome scan FST outliers (top 20 windows by FST; see 39_genome_scan_fst.tsv for the complete list)

chromosome	window_start	window_end	n_snps	global_fst
22	20800428	20850427	104	0.1718
22	20750428	20800427	213	0.1319
22	20850428	20900427	113	0.1118
22	20250428	20300427	102	0.1097
22	20900428	20950427	138	0.1076
22	20150428	20200427	196	0.1007
22	20200428	20250427	111	0.0952
22	20000428	20050427	160	0.0875
22	20700428	20750427	135	0.0845
22	20400428	20450427	24	0.0838
22	20100428	20150427	115	0.0694
22	20950428	21000427	193	0.0693
22	20350428	20400427	64	0.0666
22	20450428	20500427	48	0.0639
22	20050428	20100427	131	0.0512
22	20600428	20650427	38	0.0510
22	20500428	20550427	9	0.0403
22	20300428	20350427	44	0.0351
22	20650428	20700427	31	0.0194

9 Linkage disequilibrium decay (mean r-squared by physical distance bin; see 43_LD_decay.tsv for the complete list)

distance_bin_start	distance_bin_end	n_pairs	mean_r2
0	4999	29649	0.2378
5000	9999	26955	0.1428
10000	14999	26344	0.1186
15000	19999	24701	0.1005
20000	24999	19601	0.0945
25000	29999	14745	0.0945
30000	34999	12472	0.0908
35000	39999	8631	0.1008
40000	44999	5739	0.1013
45000	49999	3807	0.0782
50000	54999	2603	0.0340
55000	59999	1467	0.0194
60000	64999	1189	0.0399
65000	69999	1594	0.0293
70000	74999	1744	0.0346
75000	79999	983	0.0459
80000	84999	1010	0.0379
85000	89999	915	0.0645
90000	94999	779	0.0601
95000	99999	748	0.0492
100000	104999	402	0.0710
105000	109999	349	0.0425
110000	114999	266	0.0550
115000	119999	107	0.0121
120000	124999	111	0.0139
125000	129999	72	0.0122
130000	134999	158	0.0092
135000	139999	192	0.0176
140000	144999	139	0.0142
145000	149999	263	0.0208
150000	154999	372	0.0142
155000	159999	241	0.0104
160000	164999	317	0.0113
165000	169999	363	0.0157
170000	174999	455	0.0130
175000	179999	245	0.0136
180000	184999	308	0.0087
185000	189999	190	0.0078
190000	194999	318	0.0091
195000	199999	65	0.0104
200000	204999	214	0.0084
205000	209999	216	0.0067
210000	214999	116	0.0069
215000	219999	243	0.0055
220000	224999	150	0.0072
225000	229999	173	0.0058
230000	234999	63	0.0073
235000	239999	657	0.0133
240000	244999	1	0.0077

10 LD-based effective population size (per population; see 45_Ne_LD.tsv for full diagnostics)

population	n_samples	n_snps	n_chromosomes	n_pairs	harmonic_mean_n	mean_r2	mean_r2_drift	ne	ne_status
CHB	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes
GBR	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes
ITU	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes
LWK	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes
PEL	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes
PUR	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes
STU	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes
YRI	20	0	0	0	NA	NA	NA	NA	fewer_than_two_chromosomes

11 Close relatives (up to 20 shown; see 36_close_relatives.tsv for the complete list)

sample_1	sample_2	population_1	population_2	IBS0	kinship	relationship_degree
HG03873	HG03998	ITU	STU	0.0000	0.4525	duplicate/MZ twin
NA19331	NA19334	LWK	LWK	0.0028	0.4459	duplicate/MZ twin
HG01938	HG02278	PEL	PEL	0.0056	0.2781	1st-degree
HG00742	HG01054	PUR	PUR	0.0140	0.2468	1st-degree
NA19172	NA19247	YRI	YRI	0.0168	0.2448	1st-degree
NA19360	NA18516	LWK	YRI	0.0112	0.2313	1st-degree
HG01067	HG01182	PUR	PUR	0.0140	0.2225	1st-degree
HG01067	HG01086	PUR	PUR	0.0196	0.2125	1st-degree
HG00099	HG00141	GBR	GBR	0.0112	0.2110	1st-degree
HG01067	HG01305	PUR	PUR	0.0168	0.2100	1st-degree
HG00122	HG01308	GBR	PUR	0.0140	0.1971	1st-degree
NA18516	NA19222	YRI	YRI	0.0140	0.1959	1st-degree
HG01396	HG03740	PUR	STU	0.0084	0.1944	1st-degree
NA18516	NA19210	YRI	YRI	0.0196	0.1910	1st-degree
HG00253	HG01067	GBR	PUR	0.0168	0.1900	1st-degree
HG01938	HG02345	PEL	PEL	0.0084	0.1875	1st-degree
NA18516	NA19247	YRI	YRI	0.0196	0.1856	1st-degree
HG00250	HG00253	GBR	GBR	0.0168	0.1852	1st-degree
HG00141	HG00240	GBR	GBR	0.0168	0.1825	1st-degree
HG00253	HG01396	GBR	PUR	0.0168	0.1818	1st-degree

12 Sex check (reported vs inferred sex from X-chromosome heterozygosity and Y-chromosome call rate; up to 20 shown, discordant and ambiguous calls first; see 42_sex_check.tsv for the complete list)

sample	vcf_sample	n_x_snps	x_heterozygosity_F	x_inferred_sex	n_y_snps	y_call_rate	y_inferred_sex	inferred_sex	reported_sex	status
NA18907	NA18907	1148	0.9205	male	60468	0	female	discordant	female	discordant
HG04059	HG04059	1148	0.9277	male	60468	0	female	discordant	female	discordant
HG00250	HG00250	1148	1.0637	male	60468	0	female	discordant	female	discordant
HG01061	HG01061	1148	1.0872	male	60468	0	female	discordant	female	discordant
HG01303	HG01303	1148	1.1831	male	60468	0	female	discordant	female	discordant
HG02425	HG02425	1148	-0.2899	female	60468	0	female	female	female	match
HG03873	HG03873	1148	-0.2706	female	60468	0	female	female	female	match
HG03760	HG03760	1148	-0.2472	female	60468	0	female	female	female	match
HG01942	HG01942	1148	-0.2301	female	60468	0	female	female	female	match
HG01566	HG01566	1148	-0.2275	female	60468	0	female	female	female	match
HG03888	HG03888	1148	-0.2272	female	60468	0	female	female	female	match
HG00125	HG00125	1148	-0.1966	female	60468	0	female	female	female	match
HG01098	HG01098	1148	-0.1649	female	60468	0	female	female	female	match
HG00099	HG00099	1148	-0.1573	female	60468	0	female	female	female	match
HG01396	HG01396	1148	-0.1504	female	60468	0	female	female	female	match
HG03858	HG03858	1148	-0.1370	female	60468	0	female	female	female	match
HG00258	HG00258	1148	-0.1324	female	60468	0	female	female	female	match
HG03857	HG03857	1148	-0.1303	female	60468	0	female	female	female	match
HG04070	HG04070	1148	-0.1297	female	60468	0	female	female	female	match
HG02215	HG02215	1148	-0.1258	female	60468	0	female	female	female	match

13 Runs of homozygosity (highest FROH samples, up to 20 shown; see 38_ROH_sample_summary.tsv for the complete list)

sample	population	n_runs	total_length_bp	mean_length_bp	longest_run_bp	froh
HG02146	PEL	4	788356	197089.00	574060	0.7884
HG01942	PEL	4	595694	148923.50	235561	0.5957
HG01933	PEL	4	537370	134342.50	311897	0.5374
HG01578	PEL	6	528877	88146.17	194872	0.5289
HG03989	STU	2	469894	234947.00	265232	0.4699
HG00151	GBR	4	438830	109707.50	207650	0.4388
HG01950	PEL	4	438096	109524.00	217282	0.4381
NA18611	CHB	3	437054	145684.67	249970	0.4371
HG00637	PUR	4	407361	101840.25	155344	0.4074
HG00123	GBR	3	406869	135623.00	221114	0.4069
NA18612	CHB	3	400265	133421.67	226775	0.4003
NA18616	CHB	5	391340	78268.00	268046	0.3914
HG01054	PUR	3	379034	126344.67	228087	0.3790
HG01086	PUR	3	364023	121341.00	162866	0.3640
HG03836	STU	2	359185	179592.50	284841	0.3592
HG03990	STU	3	349620	116540.00	175138	0.3496
NA18647	CHB	3	339679	113226.33	182839	0.3397
HG01917	PEL	3	337023	112341.00	261589	0.3370
HG01927	PEL	2	330751	165375.50	191744	0.3308
NA18564	CHB	3	323282	107760.67	181407	0.3233

14 Discriminant analysis of principal components

K	n_pca	n_da	BIC	assignment_accuracy	mean_success	calinski_harabasz	davies_bouldin	replicate_max_rmse
2	40	1	NA	0.2500	0.9958	15.6223	2.7833	0.0971
3	10	2	NA	0.2938	0.9966	13.3767	3.0860	0.0000
4	10	3	NA	0.3562	0.9648	11.6552	2.8675	0.3841
5	30	4	NA	0.4062	0.9361	10.2898	2.6773	0.3512
6	10	5	NA	0.3938	0.9306	9.5564	2.5154	0.2944
7	10	6	NA	0.4812	0.9407	9.0366	2.4878	0.0727
8	10	7	NA	0.4625	0.9058	8.3942	2.4244	0.1614
9	10	8	NA	0.4688	0.9239	7.8689	2.6212	0.1830
10	10	9	NA	0.3812	0.8975	7.3769	2.6614	0.2373

15 DAPC SNP loadings (top 20 by contribution across all K; see 22f_DAPC_loadings_K.tsv for the complete per-K top-N lists)

K	axis	snp_id	chromosome	position	contribution
3	LD2	10525	22	20376408	0.0202
3	LD2	10527	22	20376436	0.0199
3	LD2	10765	22	20396584	0.0184
4	LD2	10525	22	20376408	0.0184
3	LD2	10675	22	20388698	0.0184
3	LD2	10757	22	20394994	0.0182
3	LD2	10807	22	20401090	0.0182
4	LD2	10527	22	20376436	0.0181
4	LD2	10807	22	20401090	0.0181
3	LD2	10806	22	20401066	0.0180
3	LD2	10758	22	20395128	0.0180
4	LD2	10765	22	20396584	0.0179
4	LD2	10758	22	20395128	0.0179
4	LD2	10823	22	20402404	0.0179
3	LD2	10937	22	20437004	0.0178
4	LD2	10806	22	20401066	0.0177
3	LD2	10823	22	20402404	0.0177
3	LD2	10561	22	20378905	0.0175
4	LD2	10757	22	20394994	0.0174
3	LD2	10812	22	20401719	0.0174

16 Analysis of molecular variance

component	Sigma	%
Variations Between population	27.1959	9.0406
Variations Between samples Within population	12.8319	4.2657
Variations Within samples	260.7906	86.6937
Total variations	300.8184	100.0000

statistic	Phi
Phi-samples-total	0.1331
Phi-samples-population	0.0469
Phi-population-total	0.0904

17 Mantel test and isolation by distance

mantel_r	mantel_p	slope	r_squared	partial_mantel_r	partial_mantel_p
0.3129	0.001	0.0028	0.1366	0.1627	0.001

18 Admixture-model cross-validation

K	cv_error
2	0.47688
3	0.46091
4	0.45539
5	0.45427
6	0.45225
7	0.45211
8	0.45249
9	0.45470

19 Chromosome-specific results

chromosome	qc_snps	ld_snps	global_fst	pc1_percent
22	1969	357	0.0915	50.9697

20 Figure gallery

The gallery contains one version of every analysis figure. The HTML report prefers browser-friendly SVG or PNG files. The PDF report prefers the compact vector PDF source for each figure. Original files remain available in the sibling `figures/` directory.

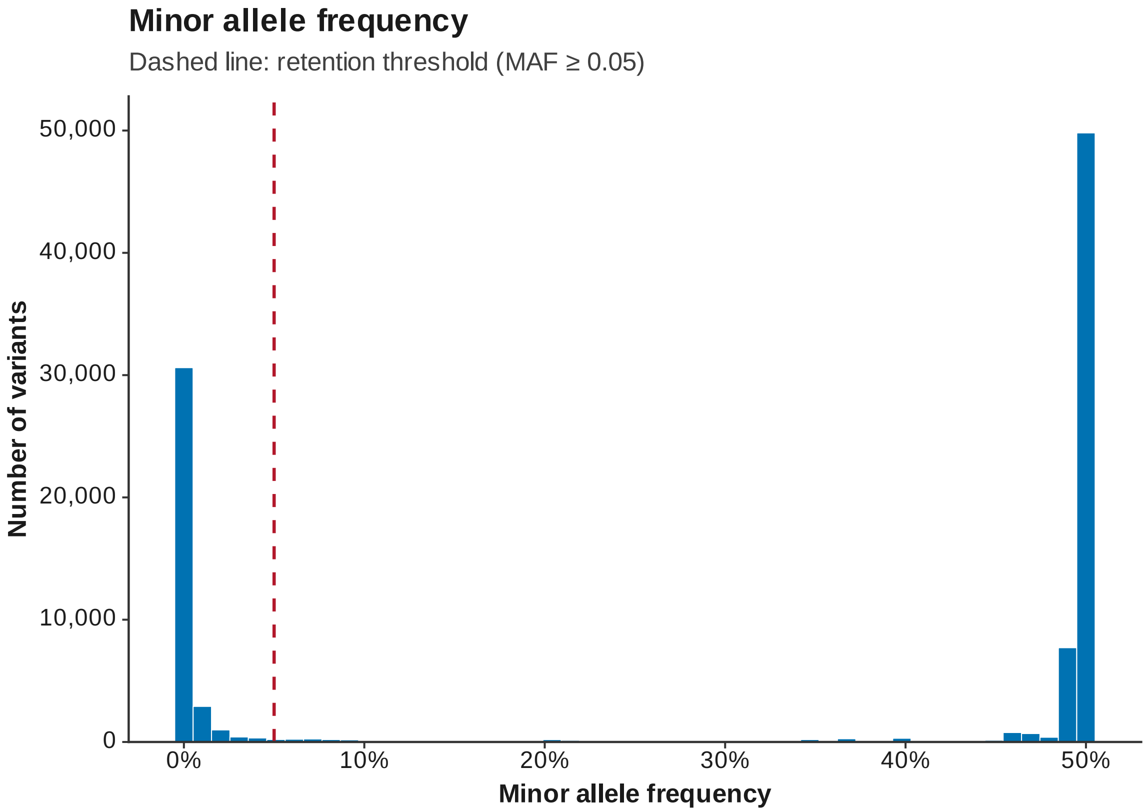
Minor allele frequency
Dashed line: retention threshold (MAF ≥ 0.05)
50,000
40,000
30,000
20,000
10,000
0
Number of variants
0%
10%
20%
30%
40%
50%
Minor allele frequency

Figure 1: MAF

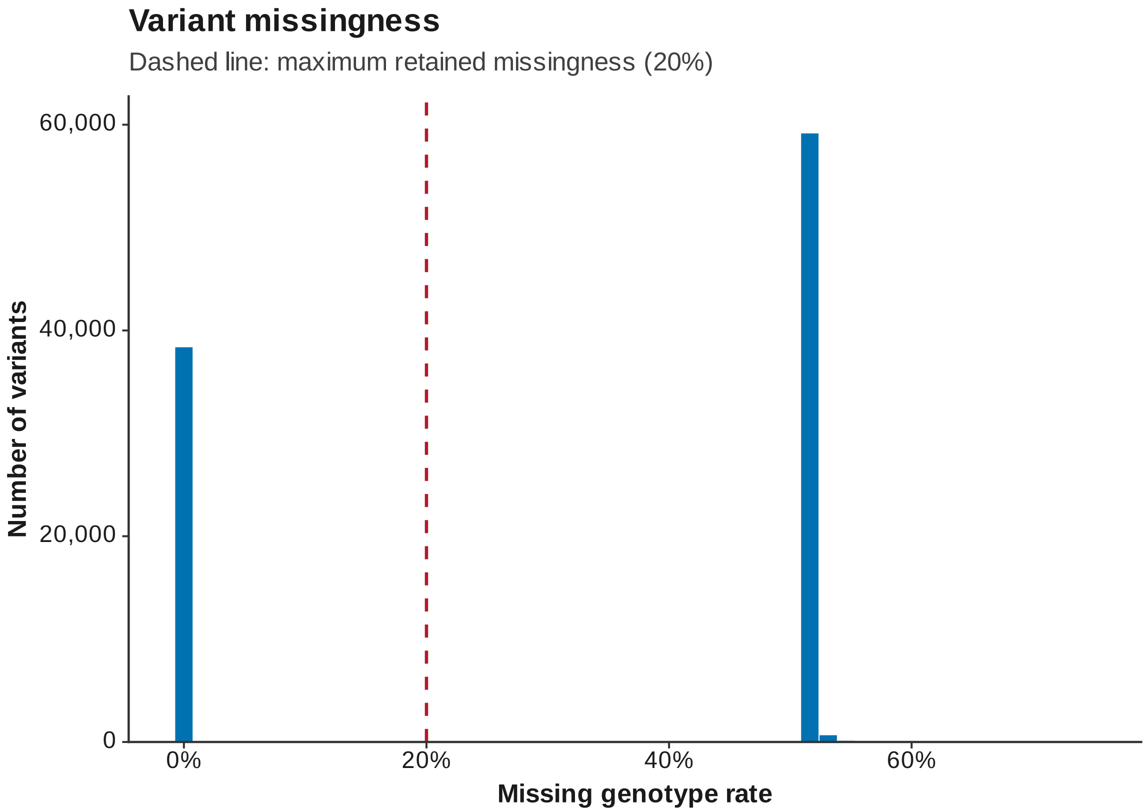

Figure 2: Variant missingness

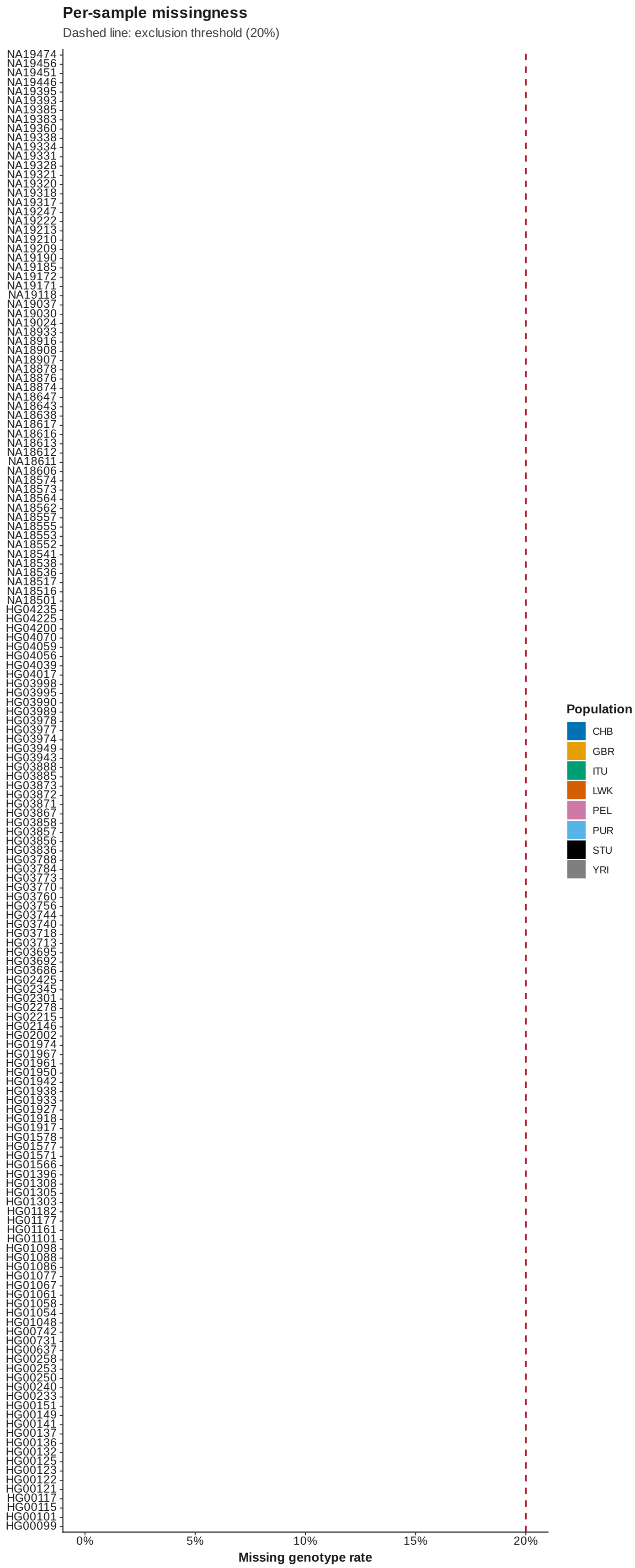

Figure 3: Sample missingness

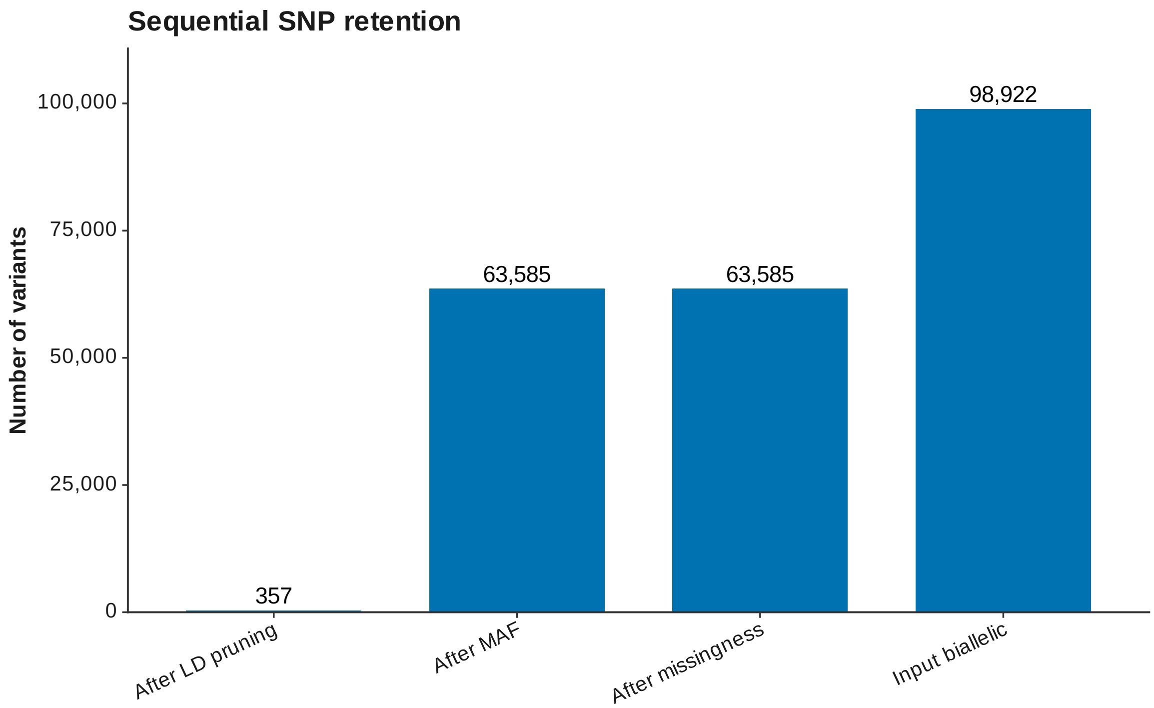

Figure 4: SNP retention

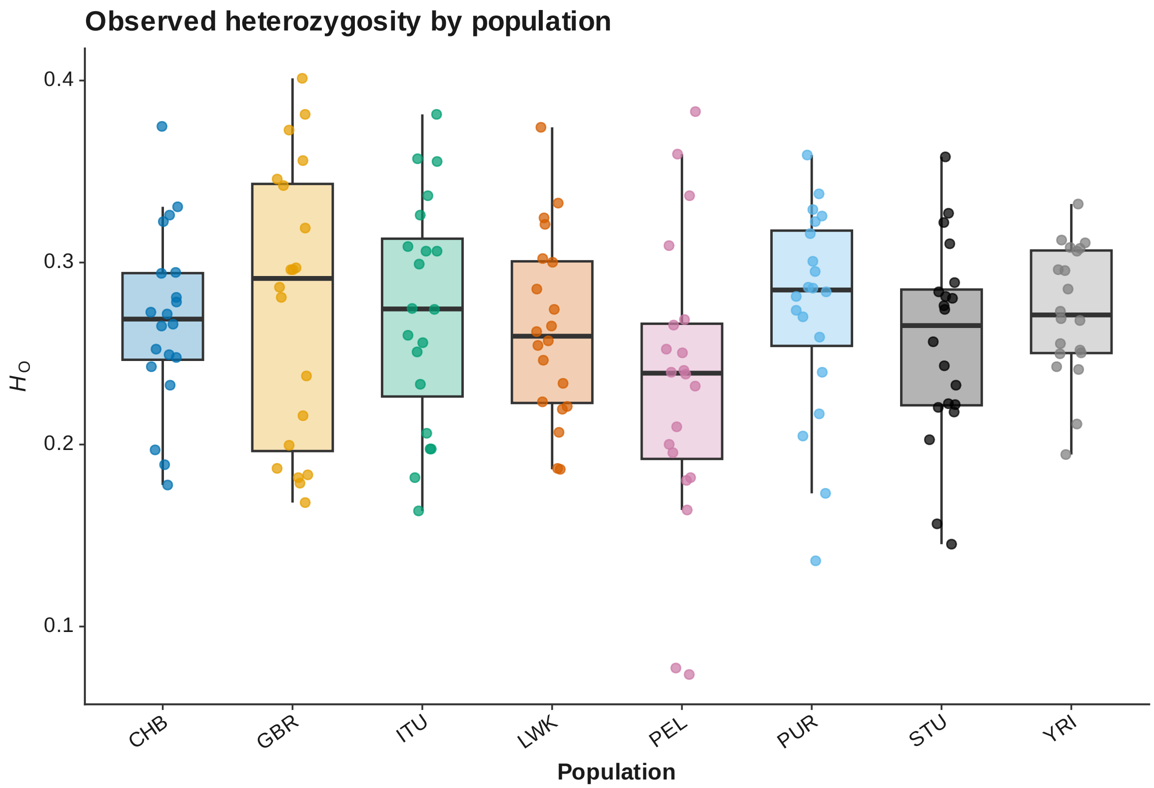

Figure 5: Sample heterozygosity

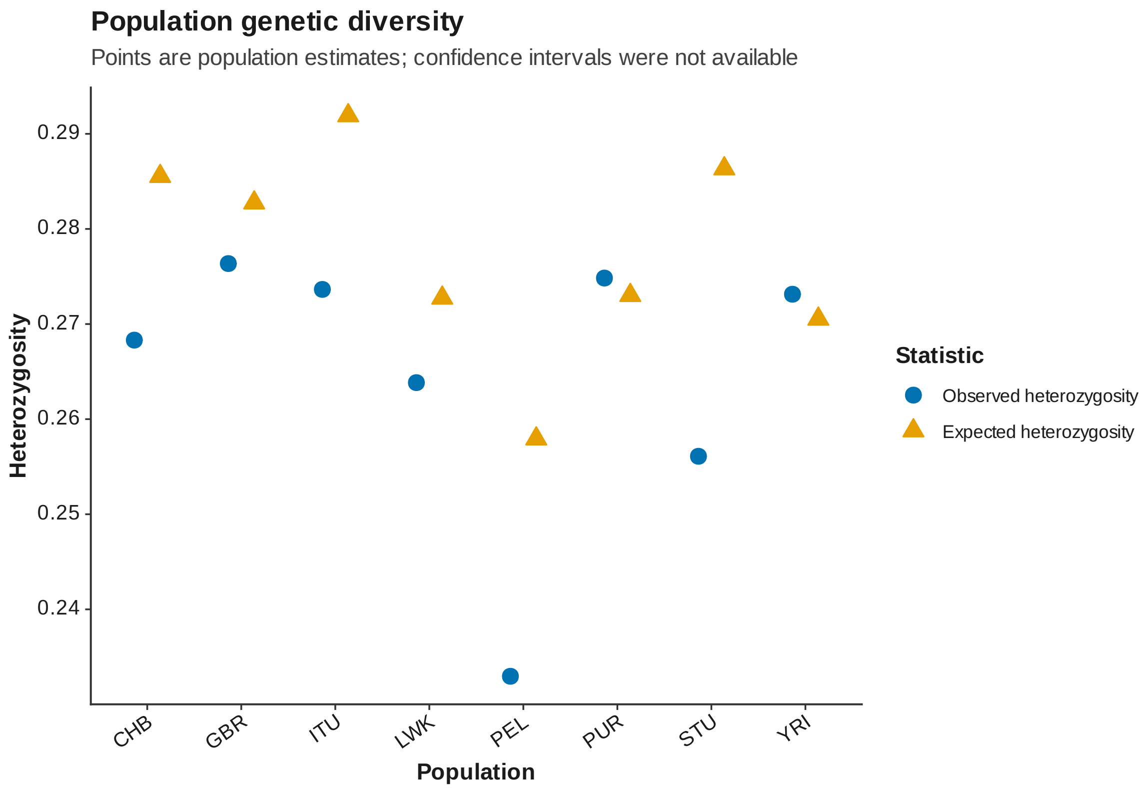

Figure 6: Population diversity

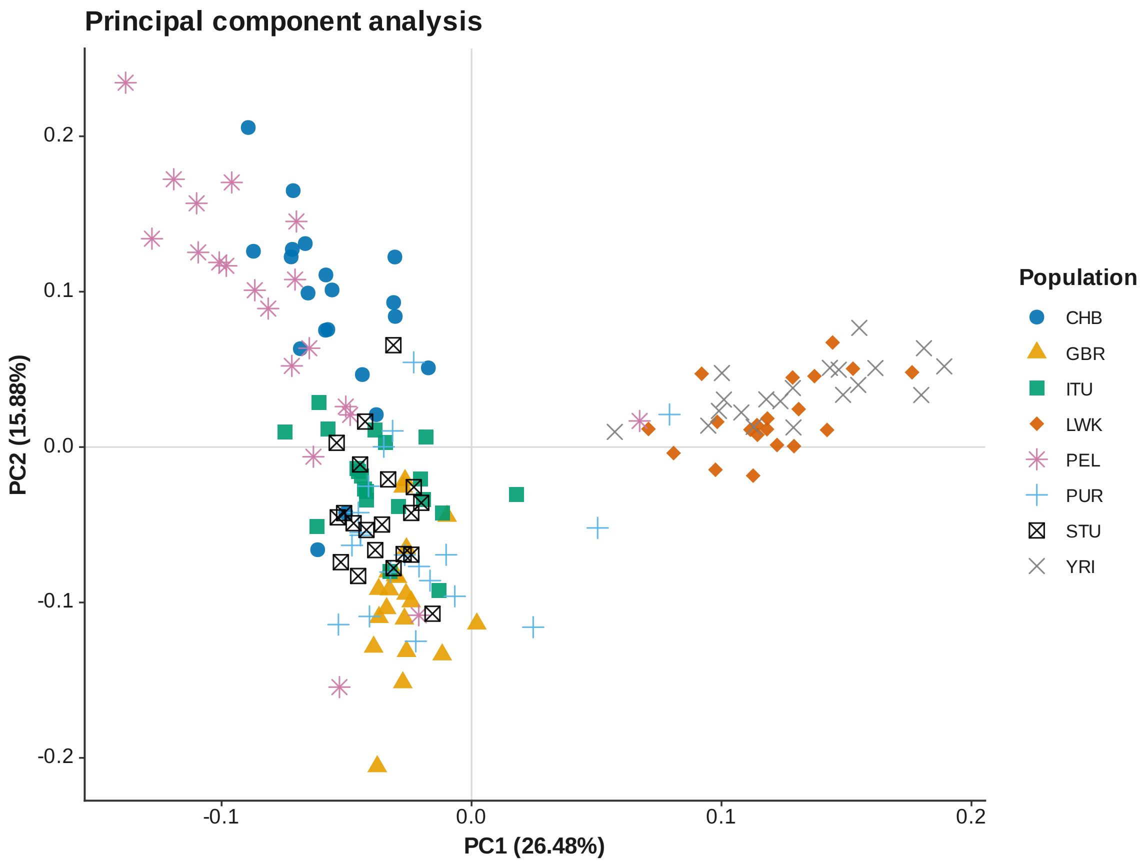

Figure 7: PCA PC 1 PC 2

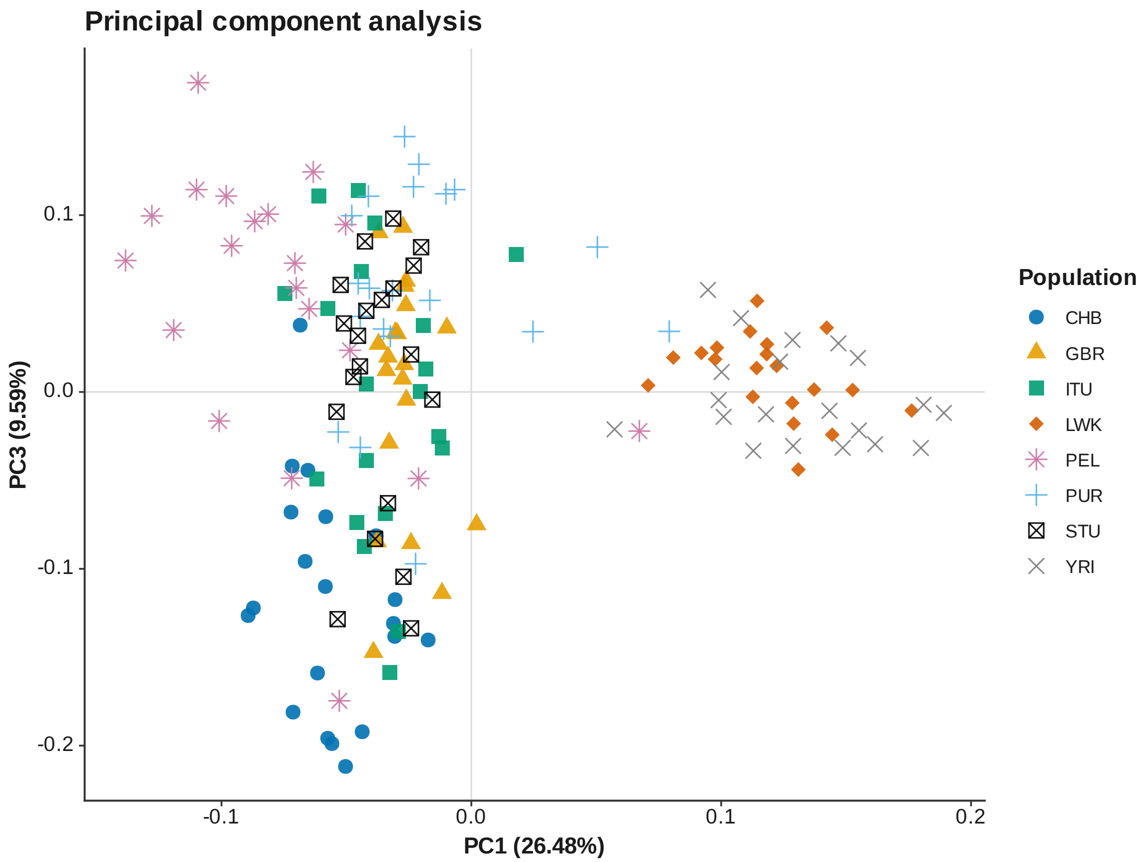

Figure 8: PCA PC 1 PC 3

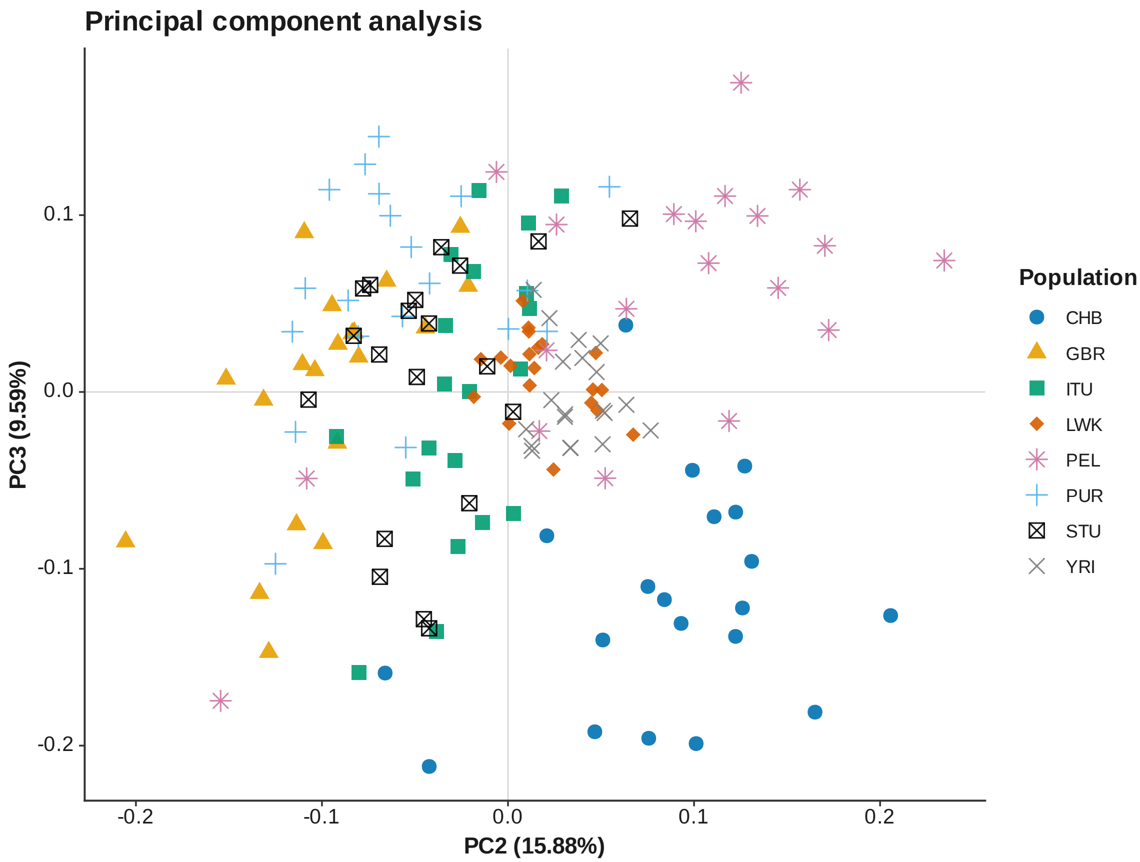

Figure 9: PCA PC 2 PC 3

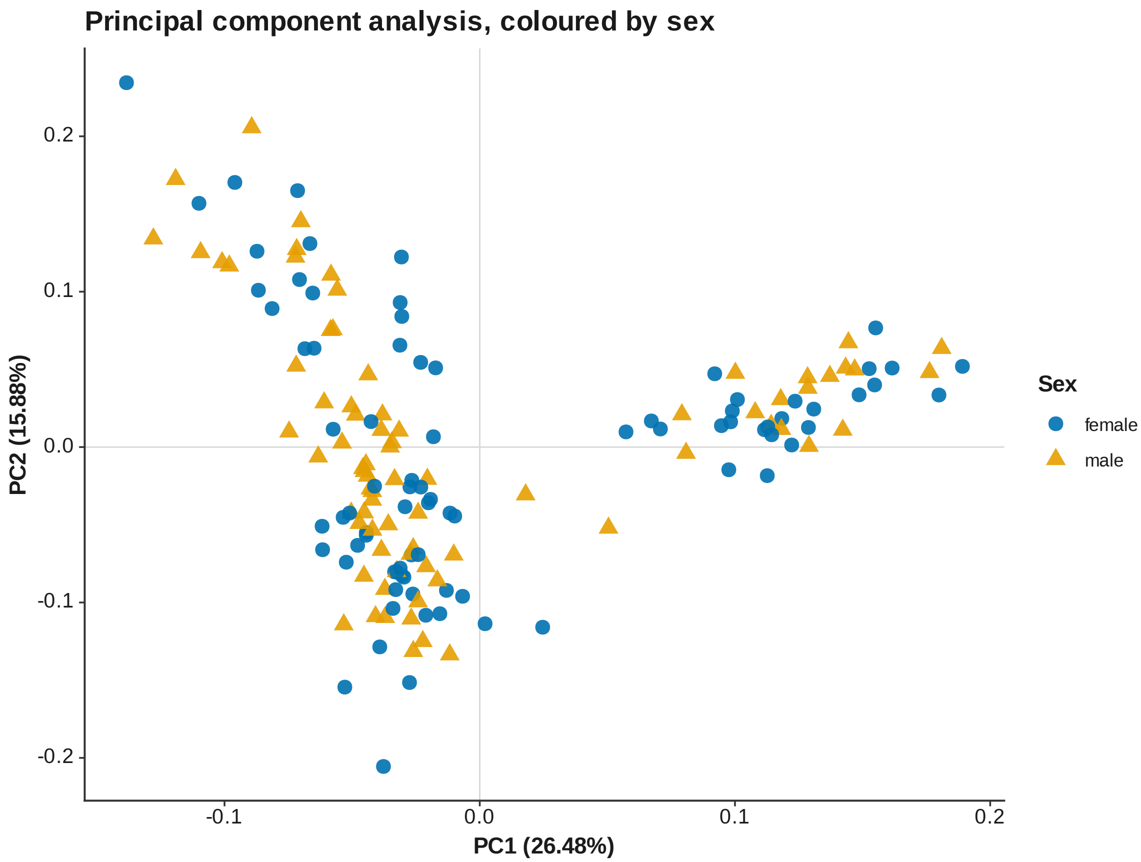

Figure 10: PCA PC 1 PC 2 by sex

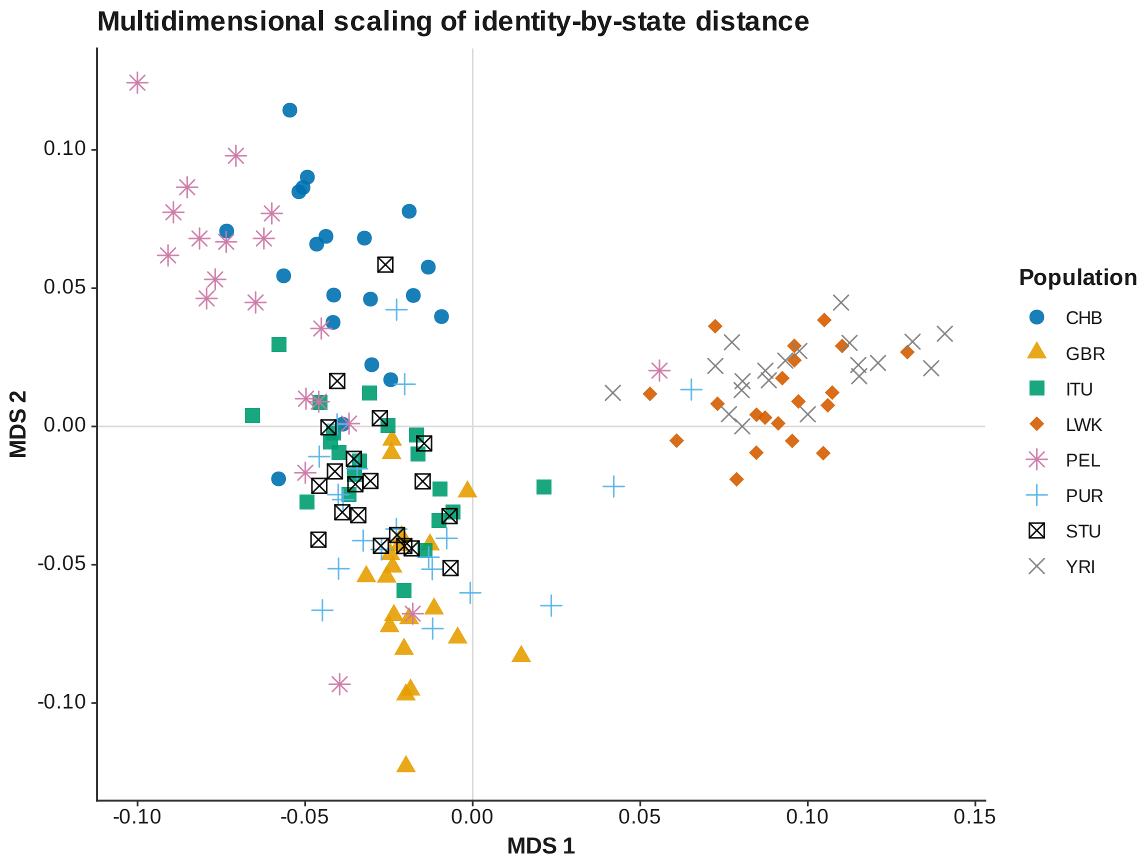

Figure 11: IBS MDS

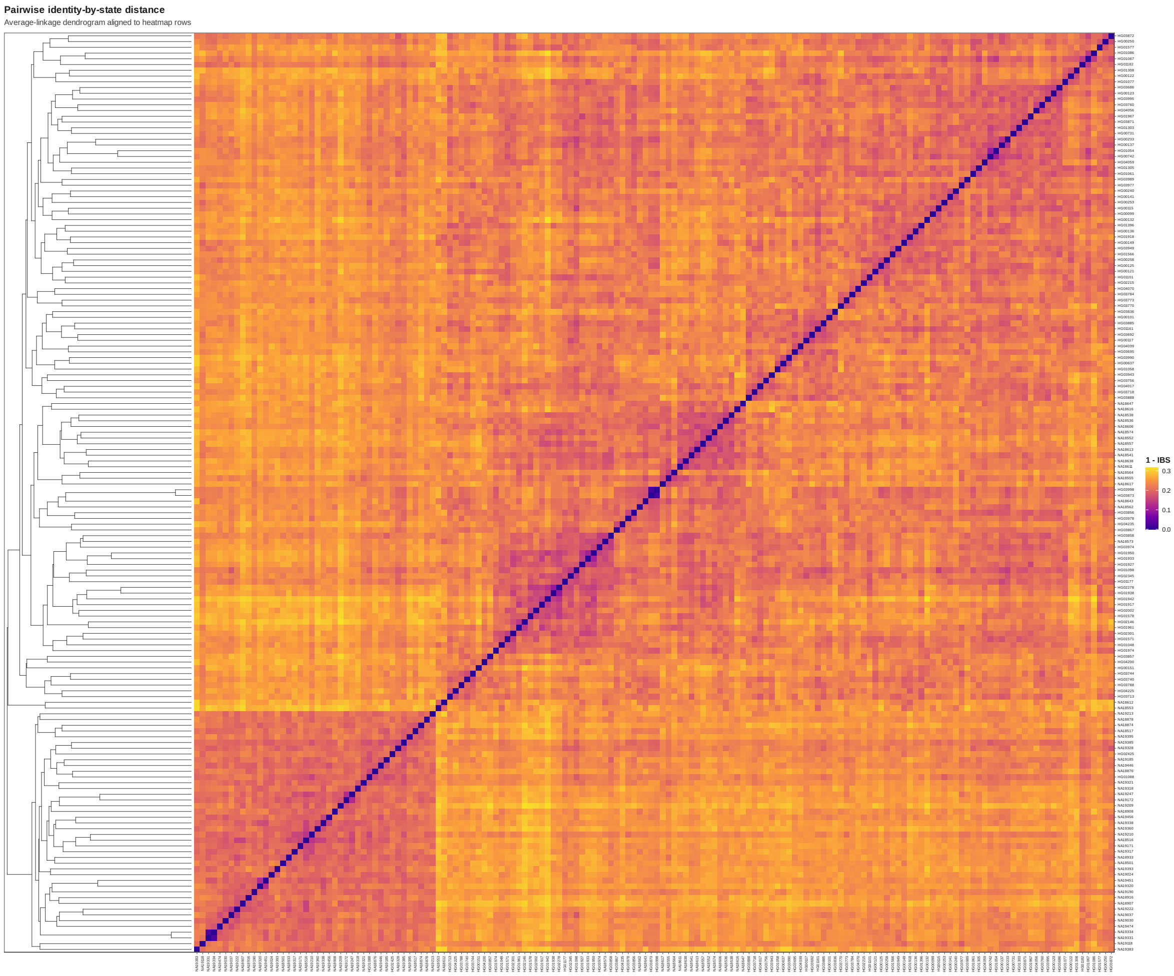

Figure 12: IBS heatmap

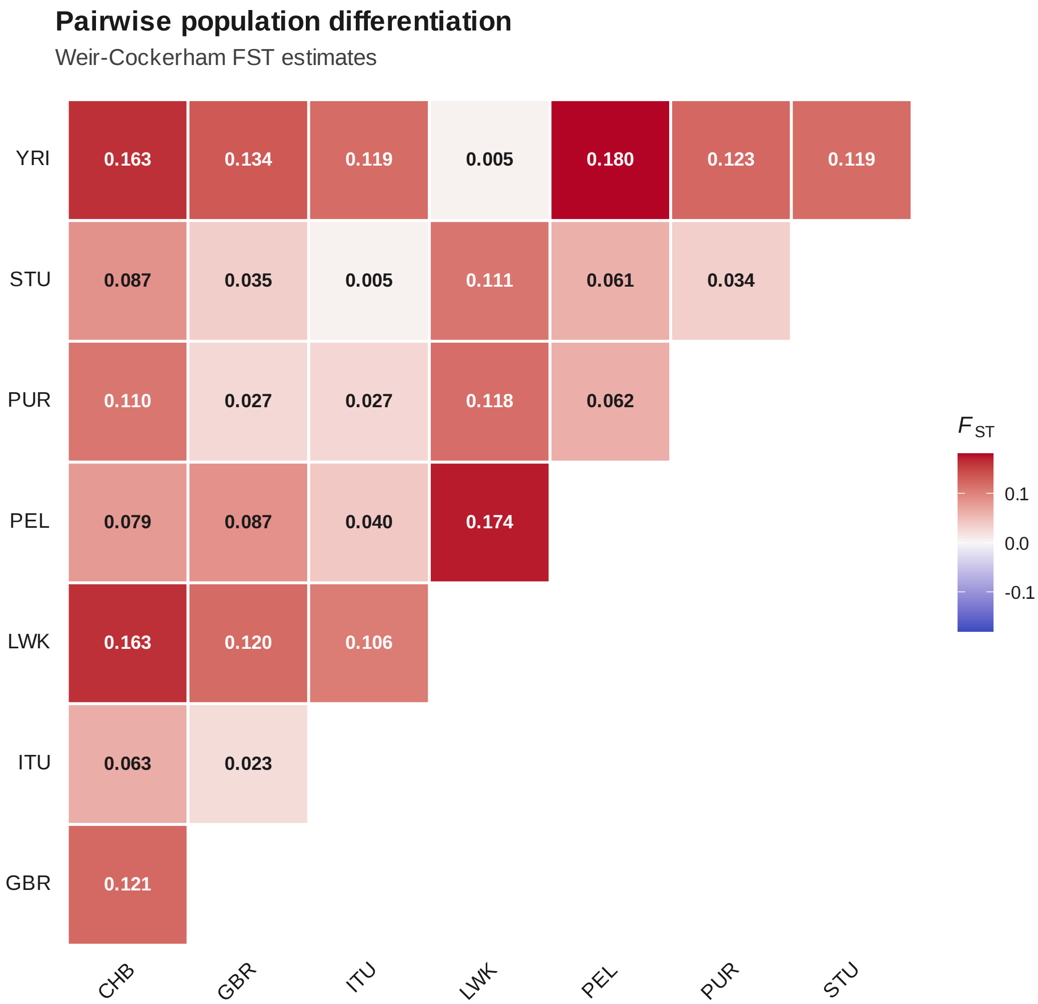

Figure 13: Pairwise FST

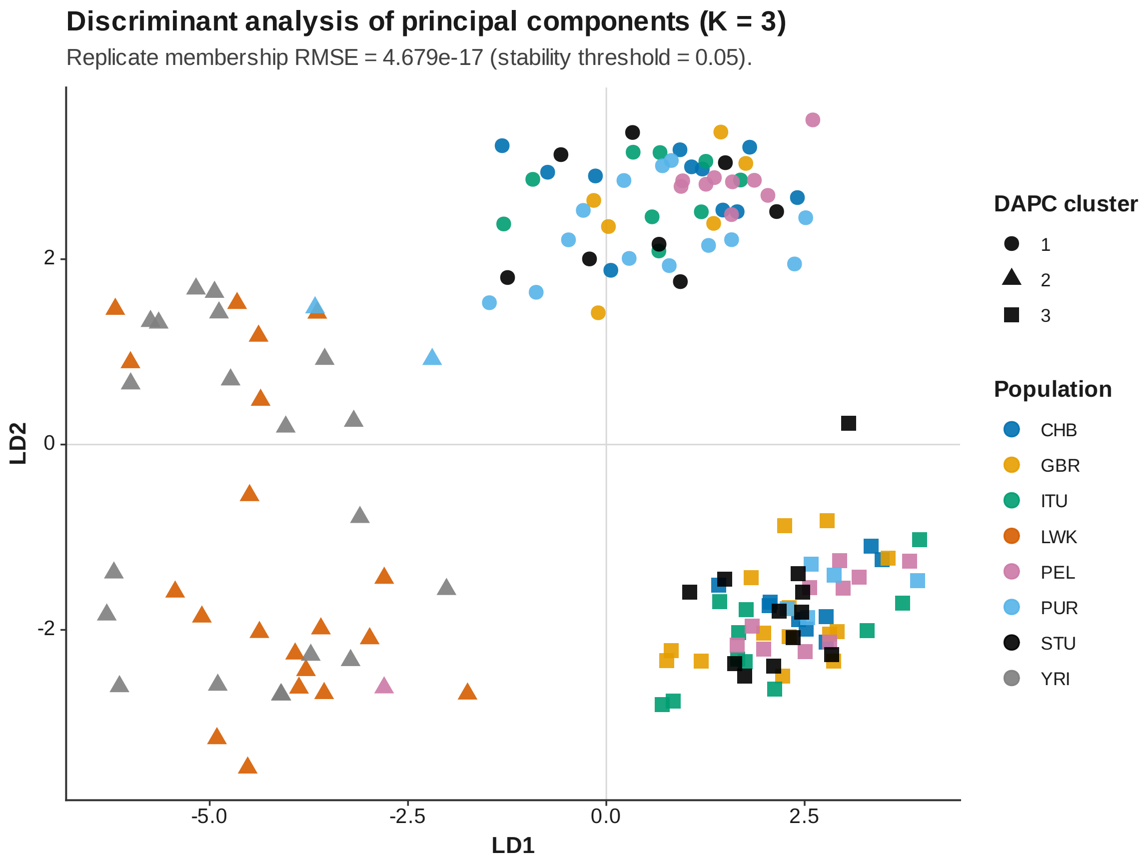

Figure 14: DAPC K 3

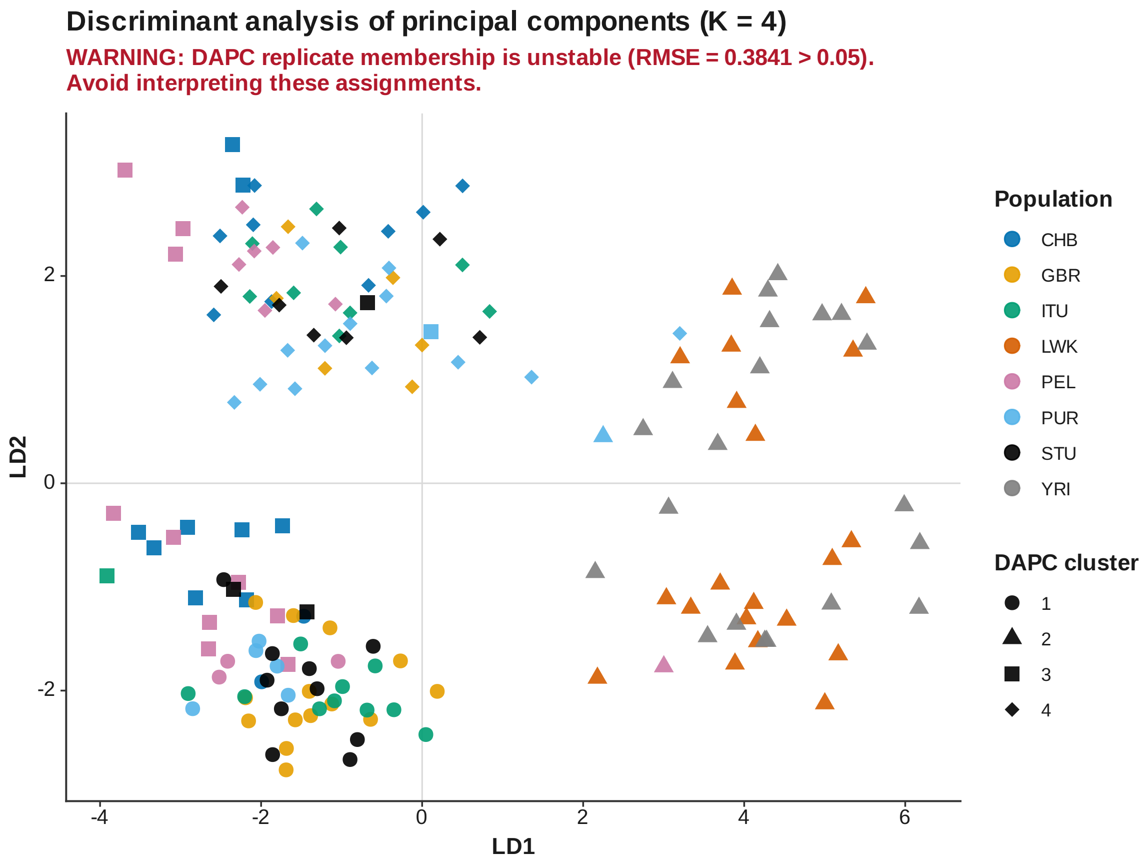

Figure 15: DAPC K 4

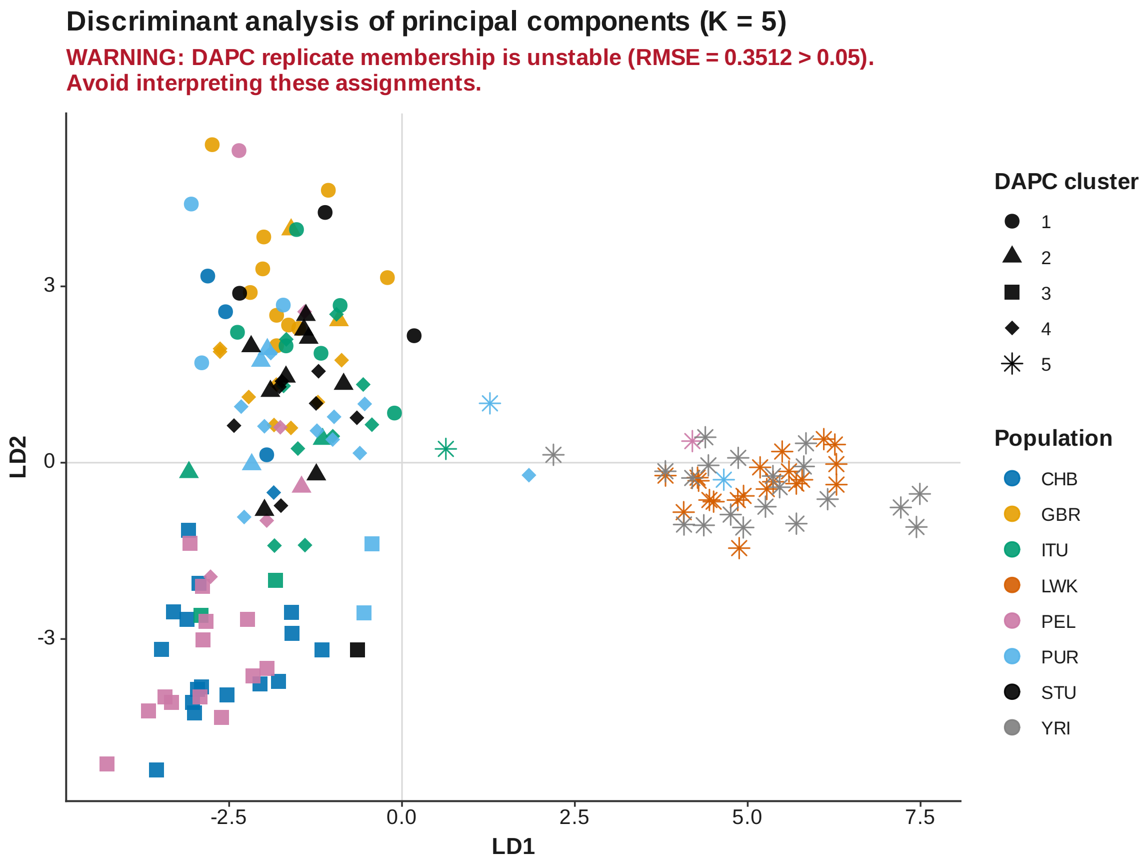

Figure 16: DAPC K 5

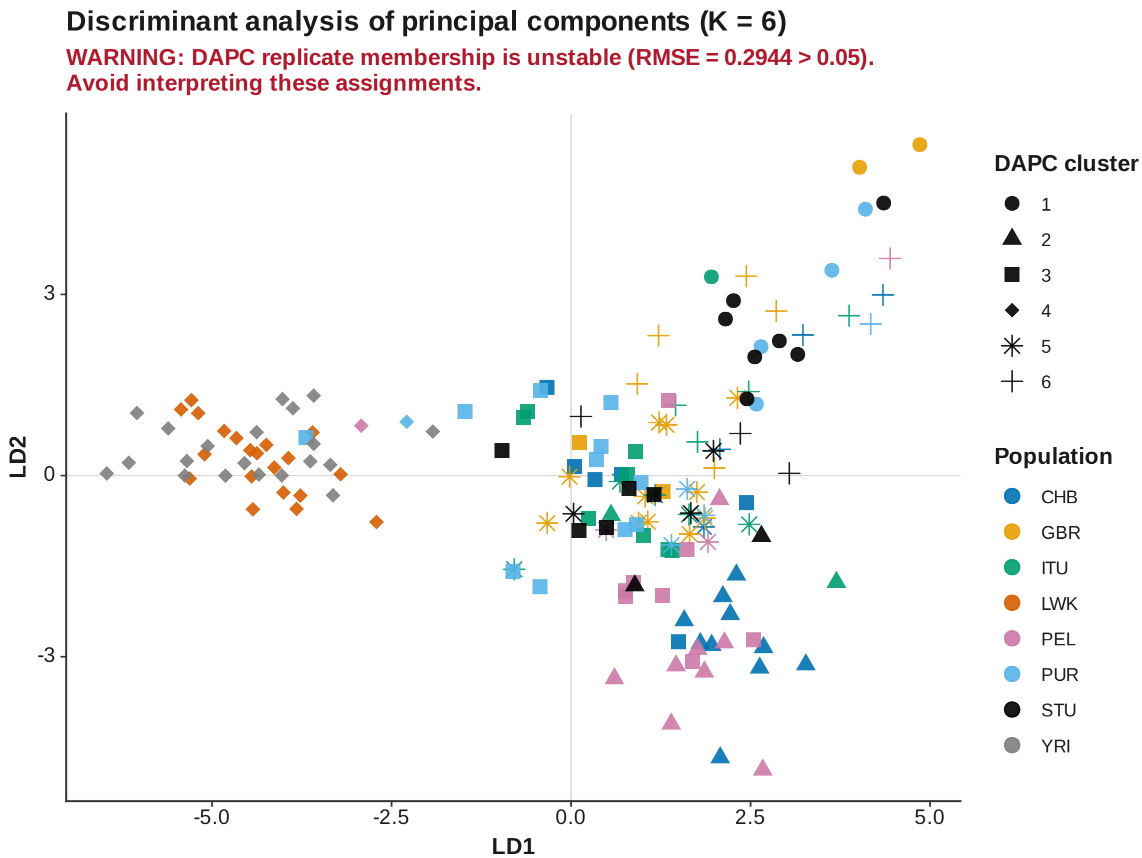

Figure 17: DAPC K 6

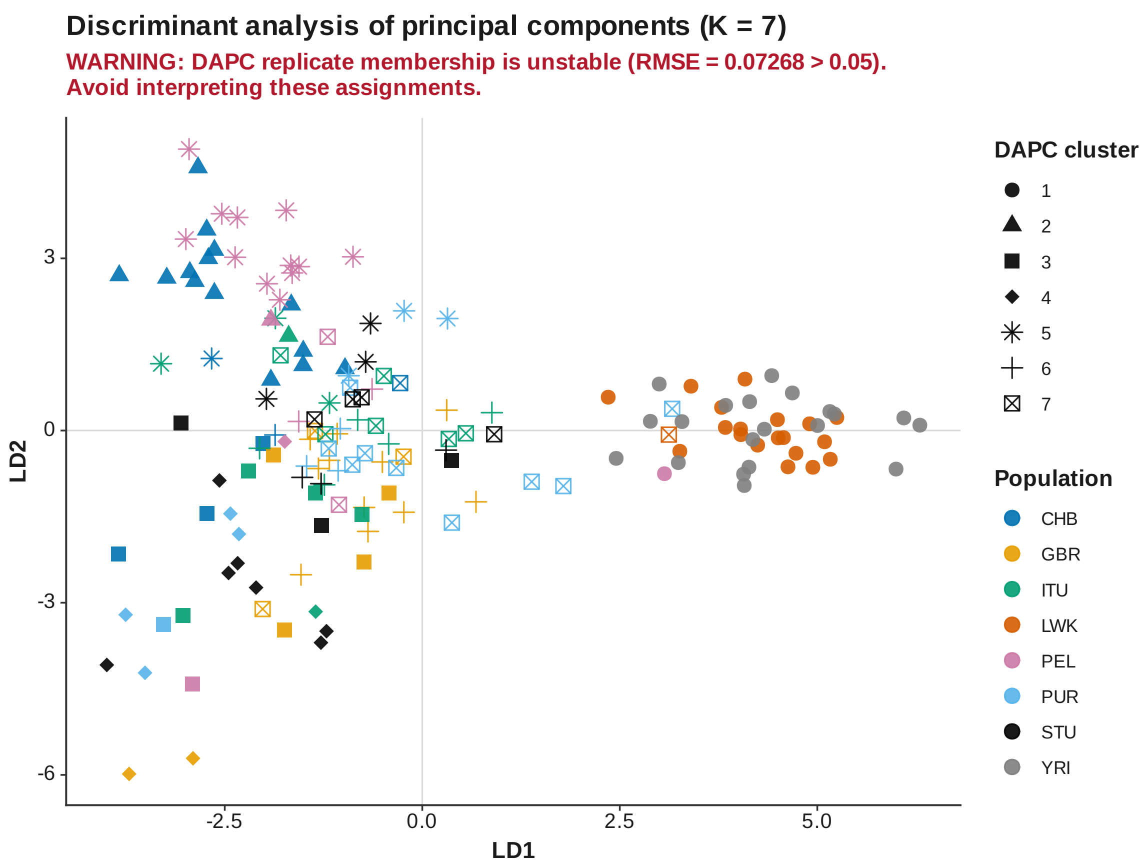

Figure 18: DAPC K 7

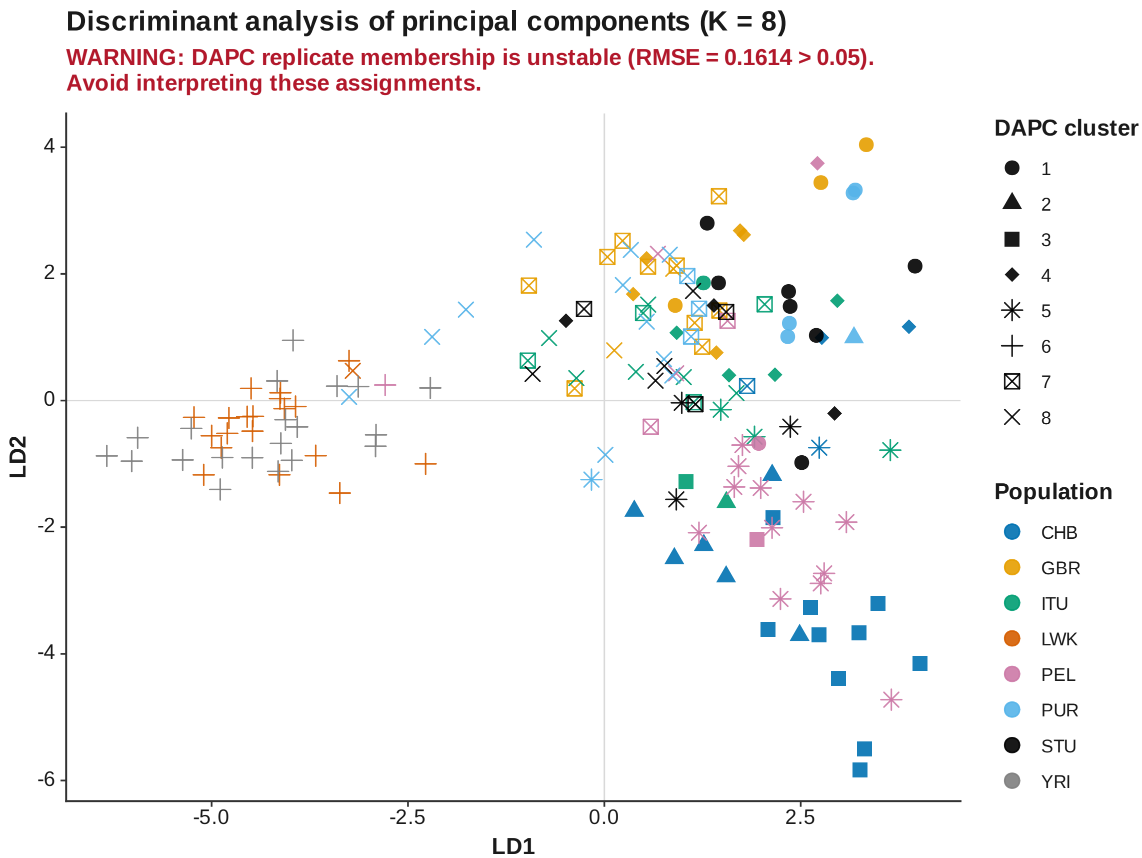

Figure 19: DAPC K 8

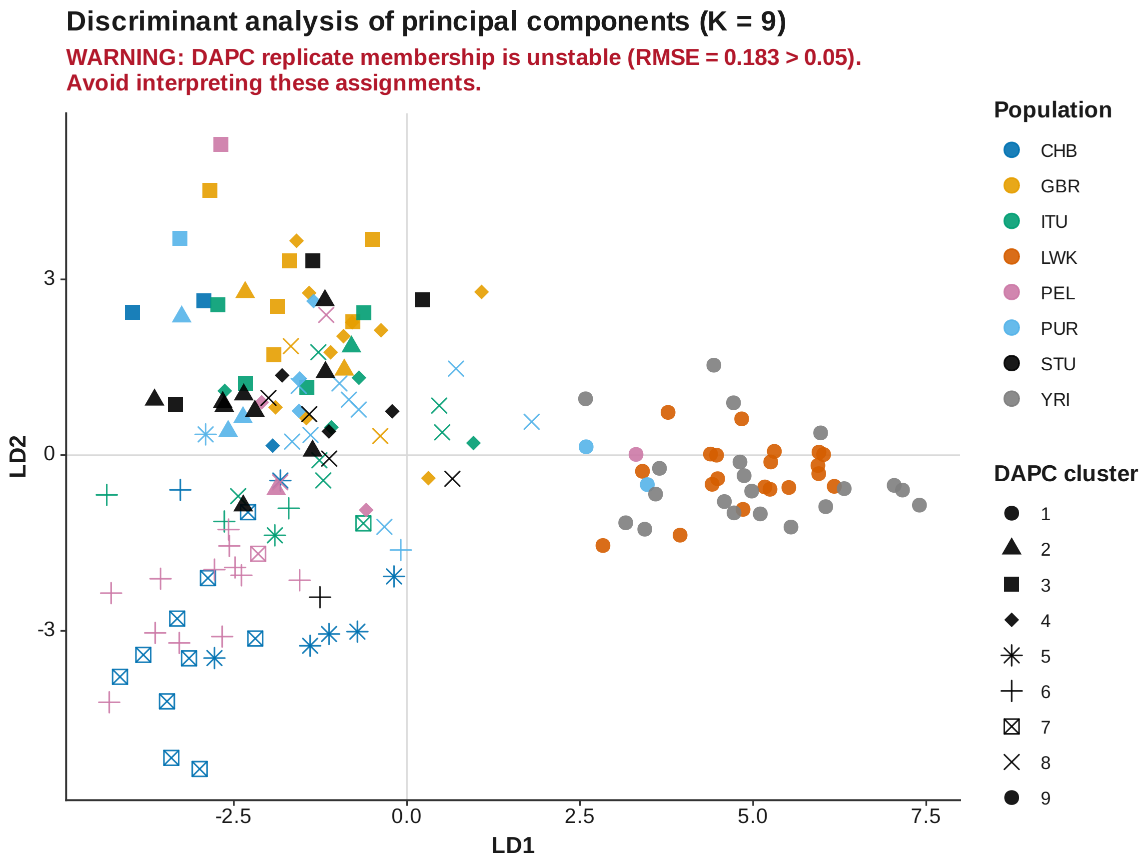

Figure 20: DAPC K 9

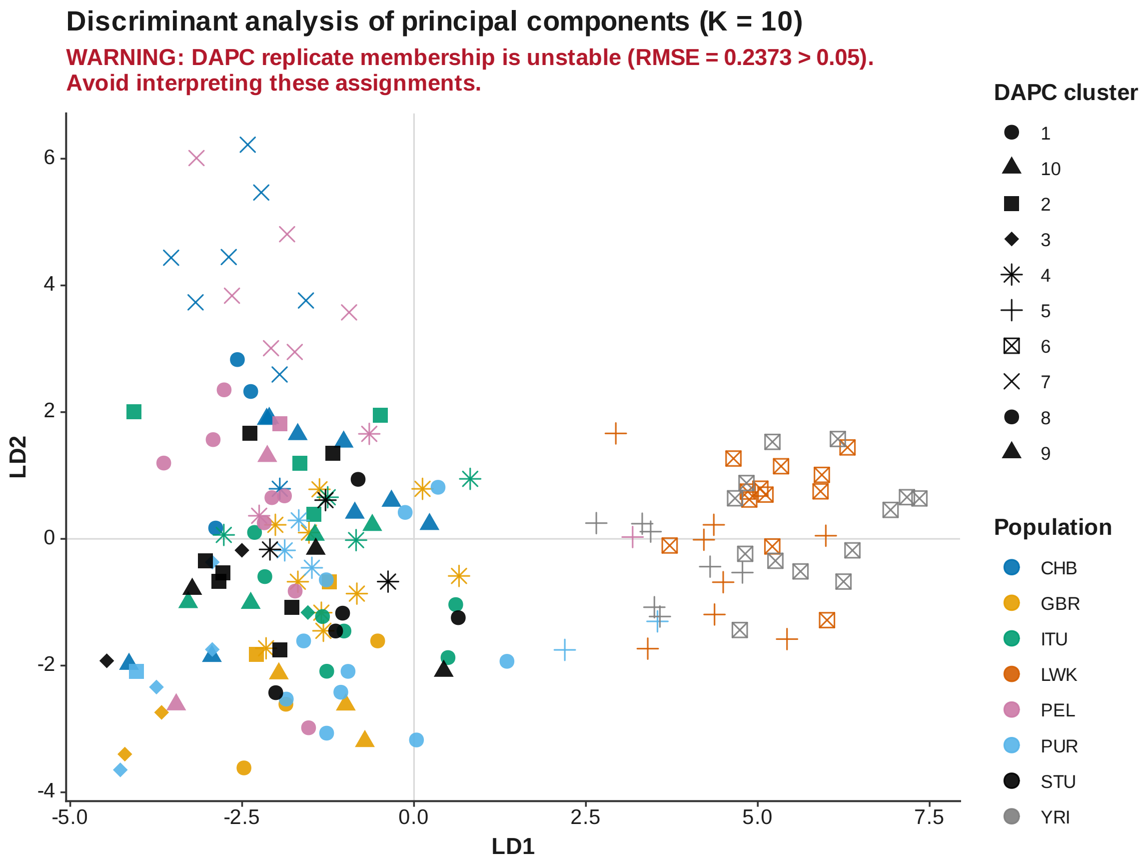

Figure 21: DAPC K 10

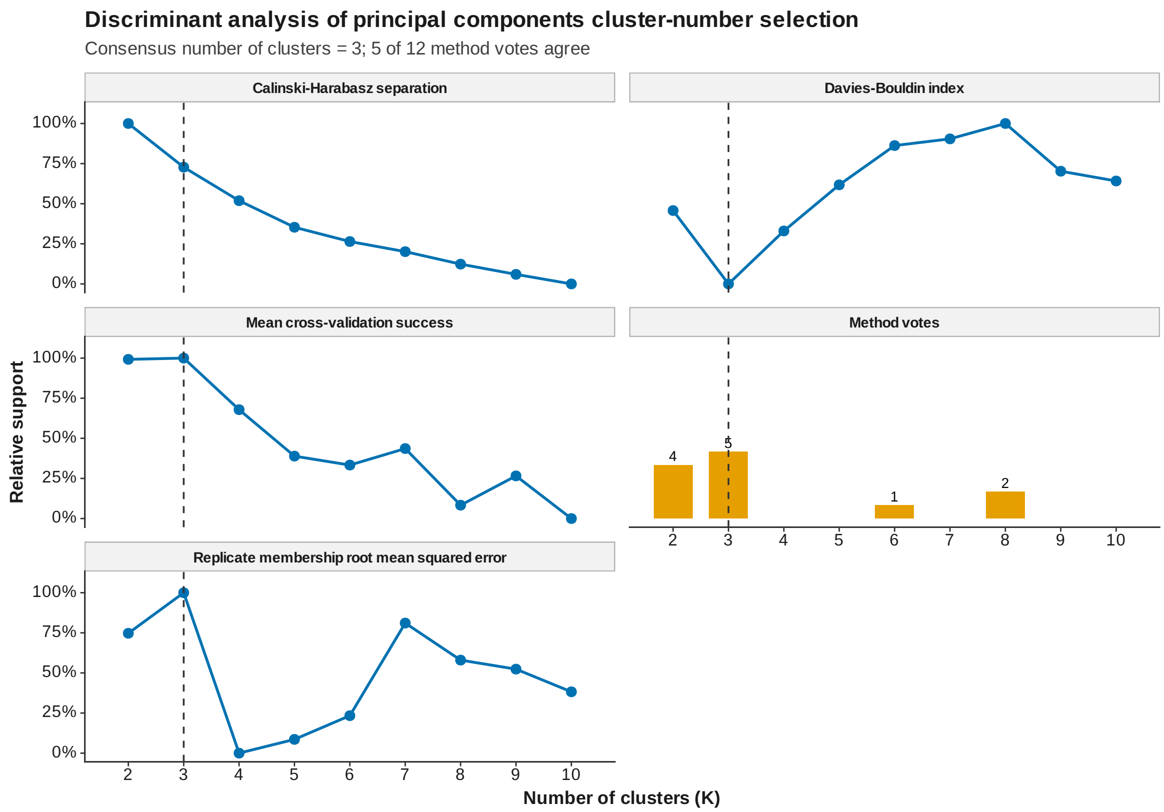

Figure 22: DAPC cluster number selection

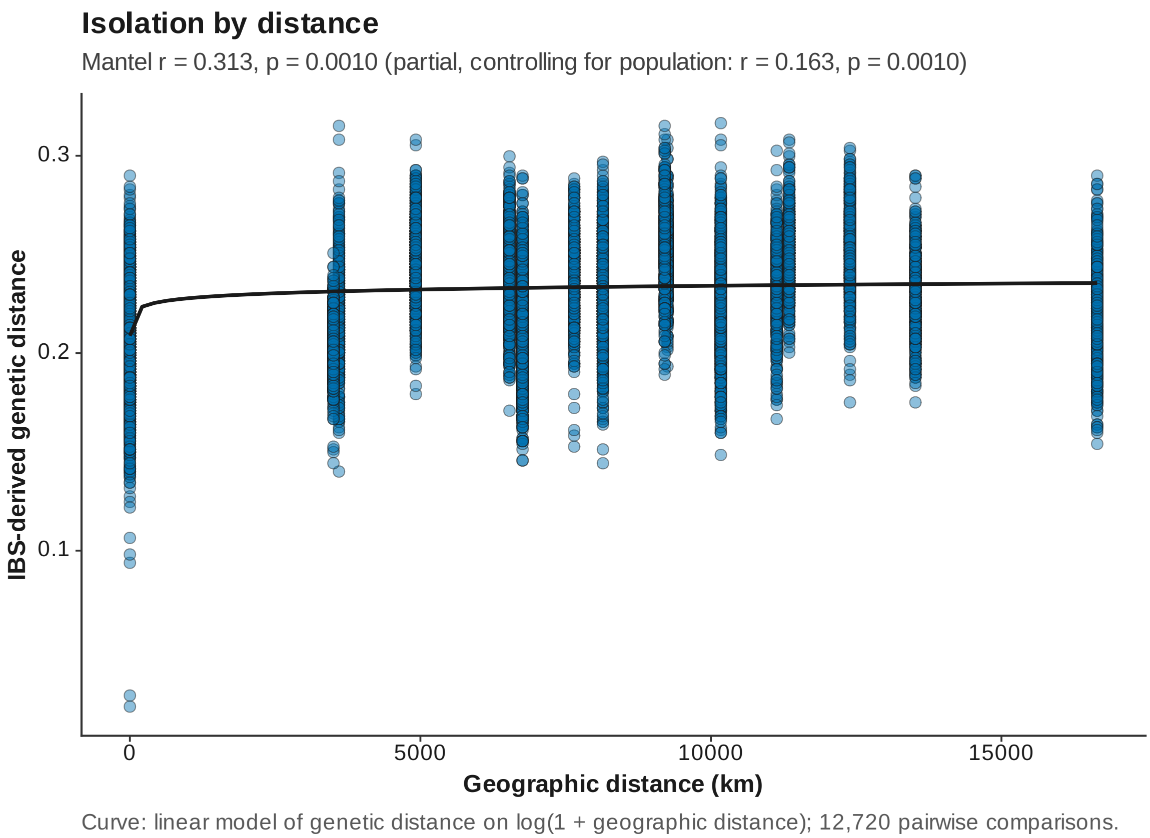

Figure 23: Isolation by distance

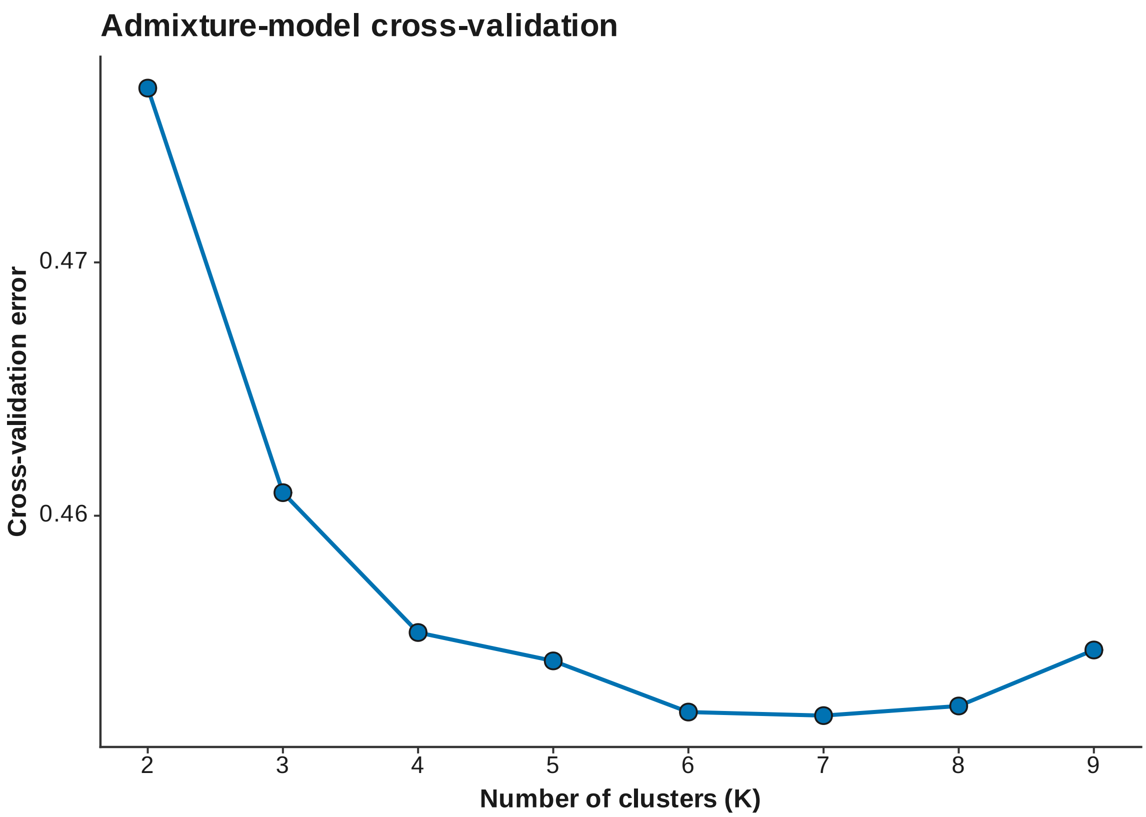

Figure 24: ADMIXTURE CV

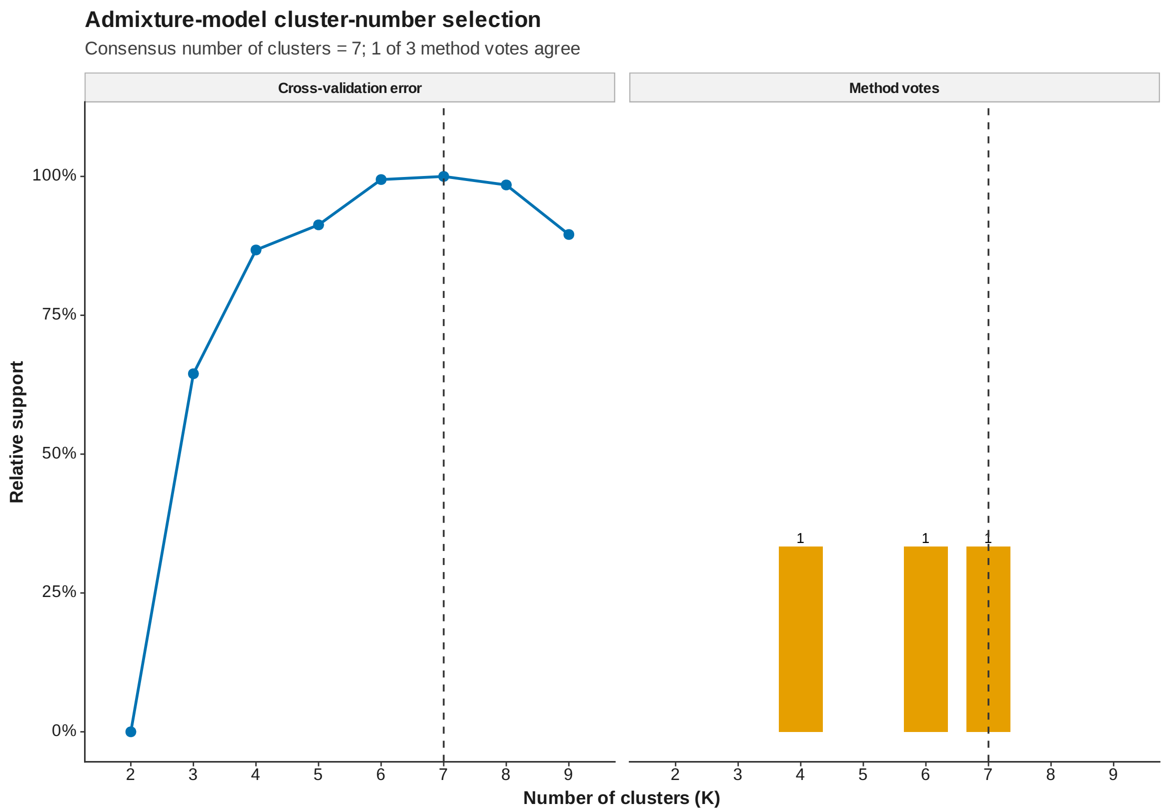

Figure 25: ADMIXTURE cluster number selection

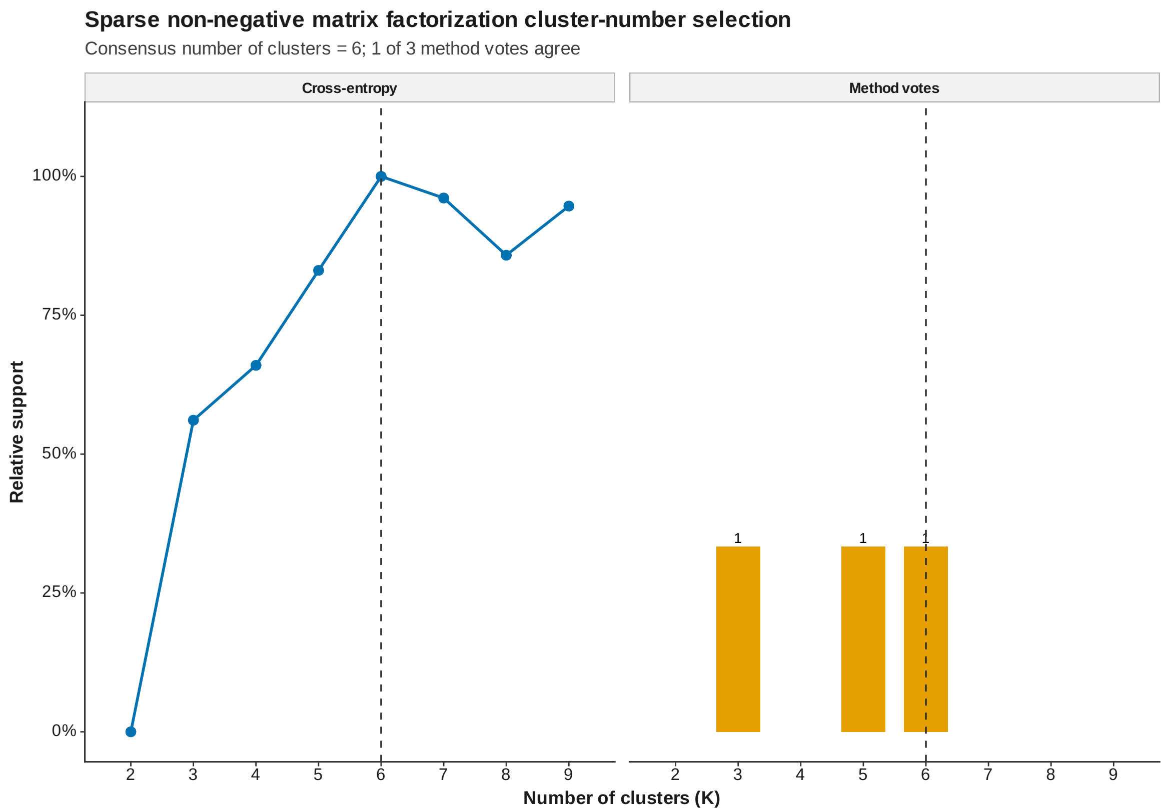

Figure 26: SNMF cluster number selection

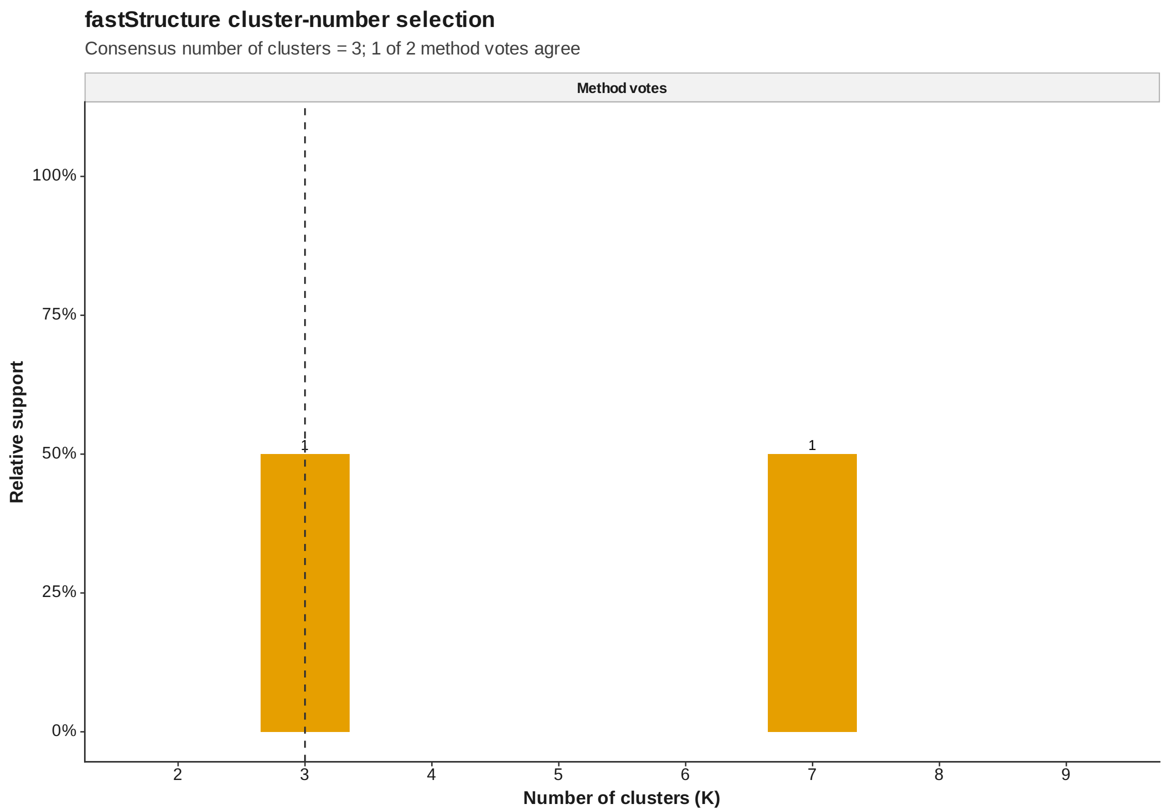

Figure 27: FastStructure cluster number selection

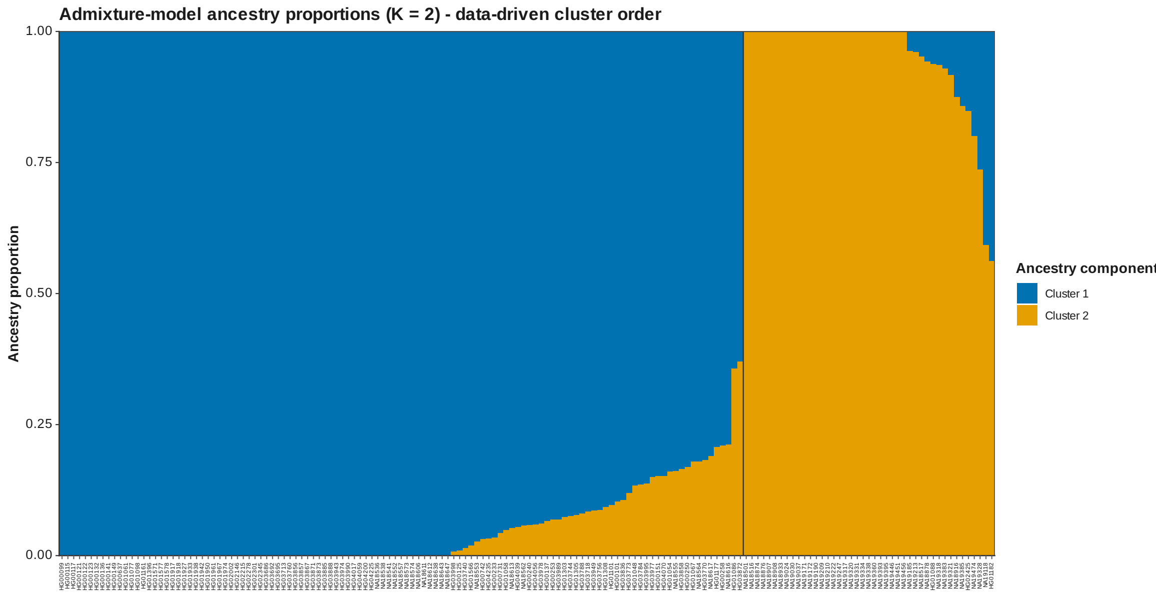

Figure 28: ADMIXTURE Q data driven K 2

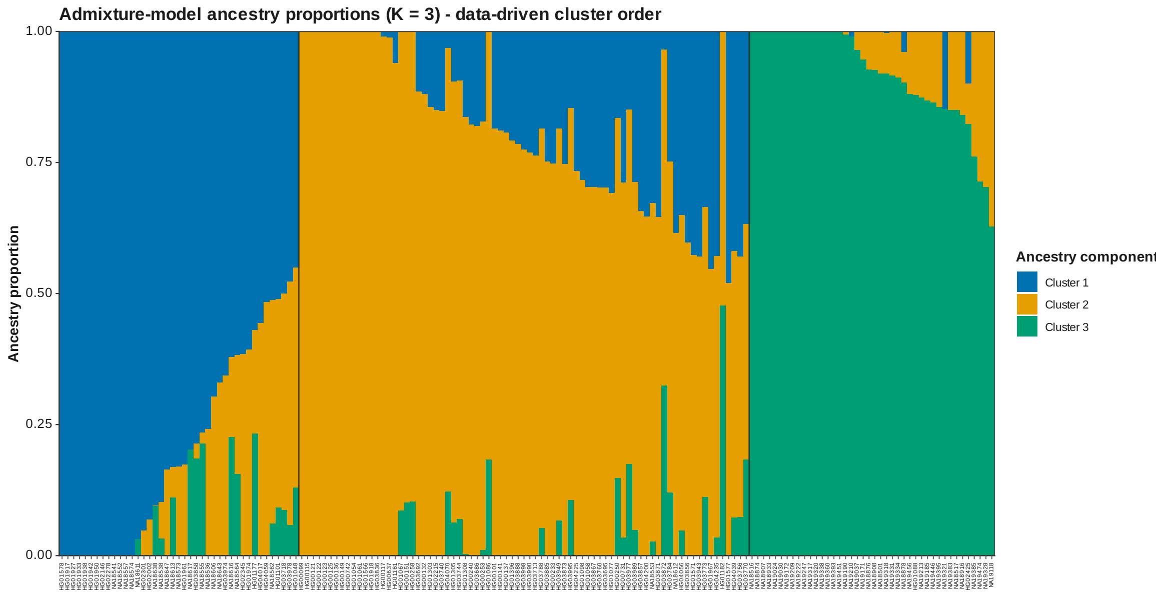

Figure 29: ADMIXTURE Q data driven K 3

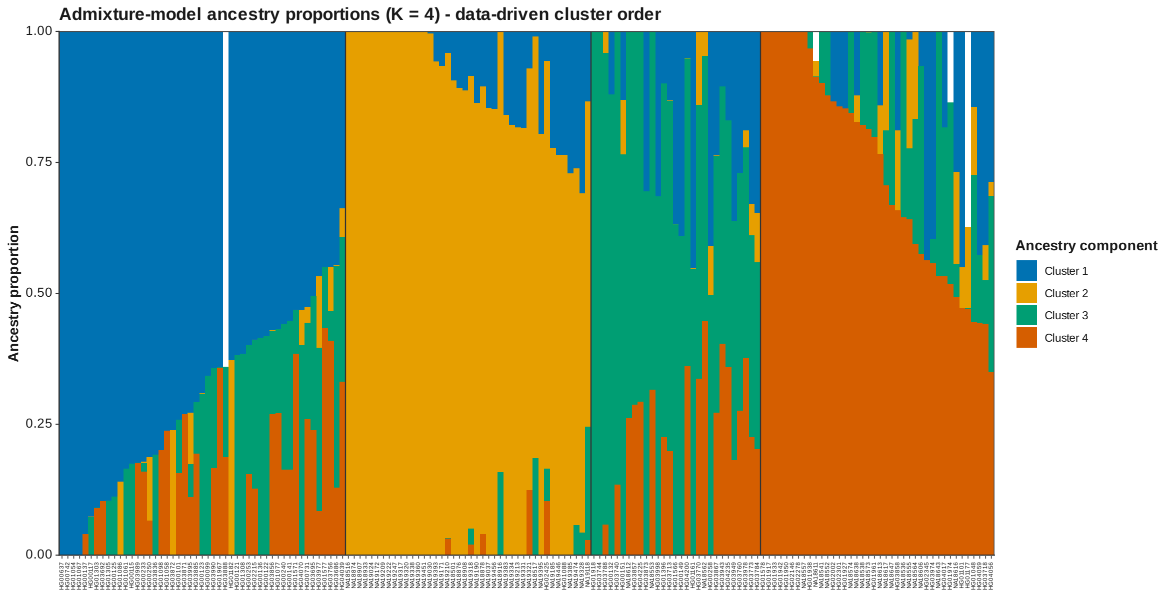

Figure 30: ADMIXTURE Q data driven K 4

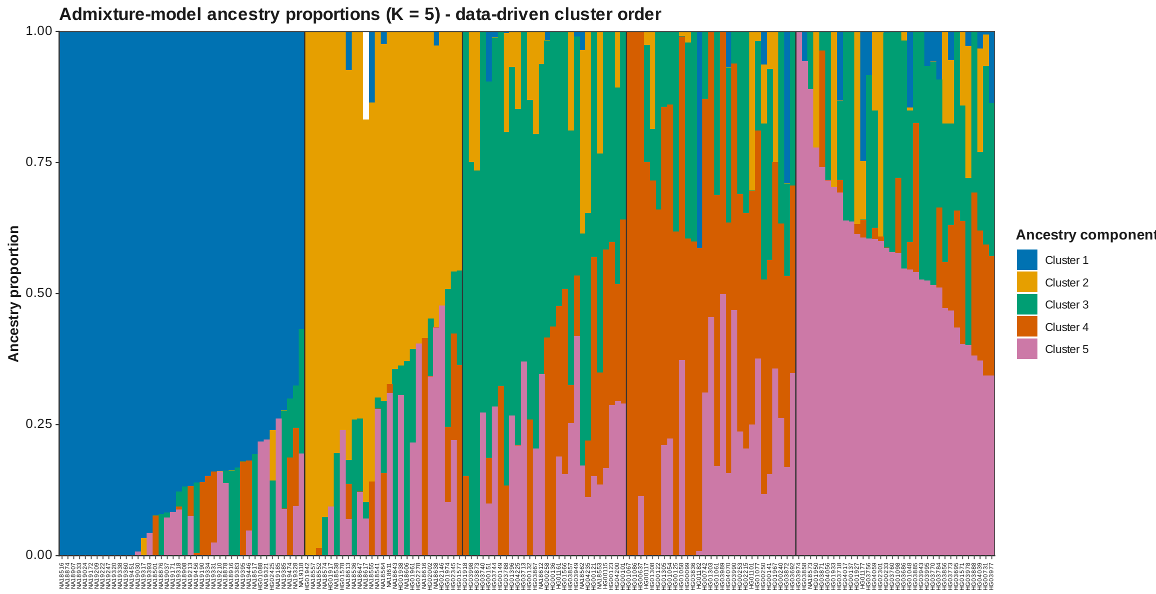

Figure 31: ADMIXTURE Q data driven K 5

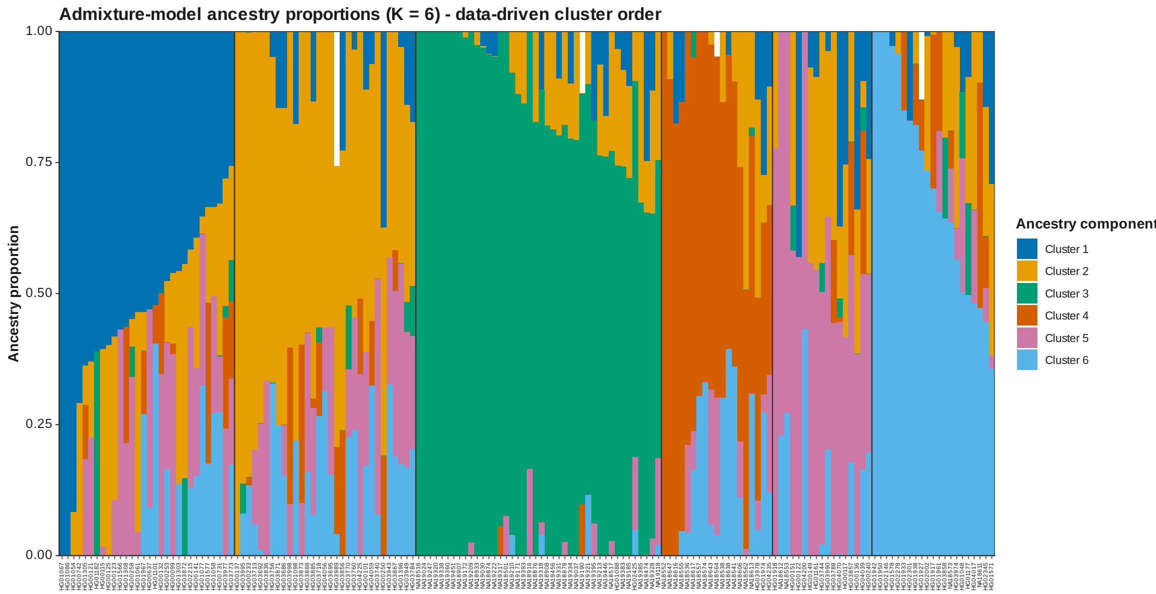

Figure 32: ADMIXTURE Q data driven K 6

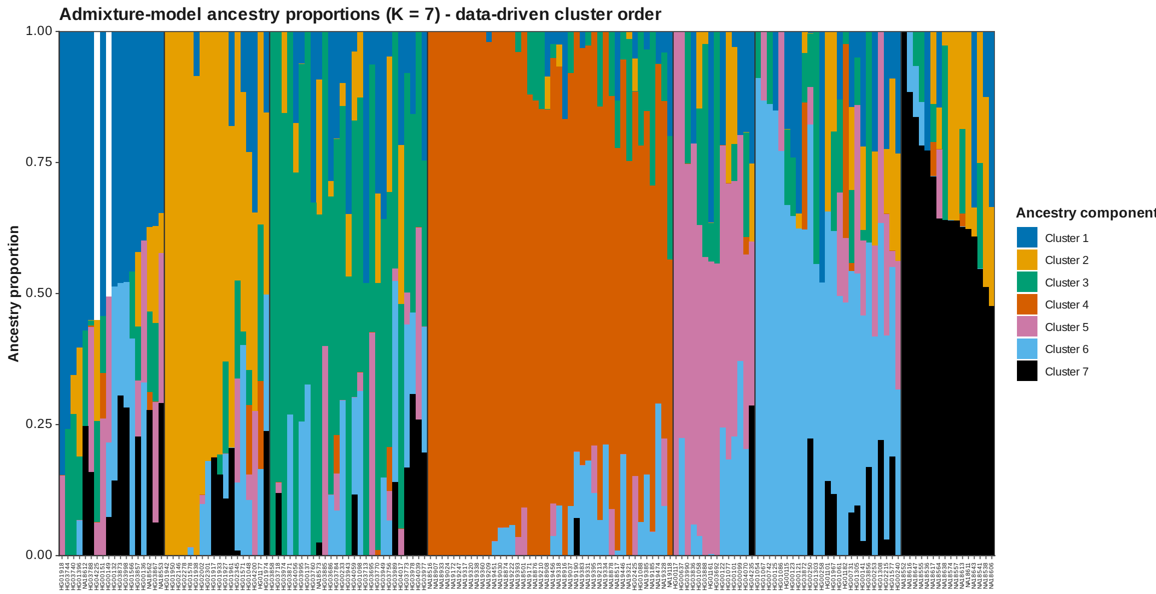

Figure 33: ADMIXTURE Q data driven K 7

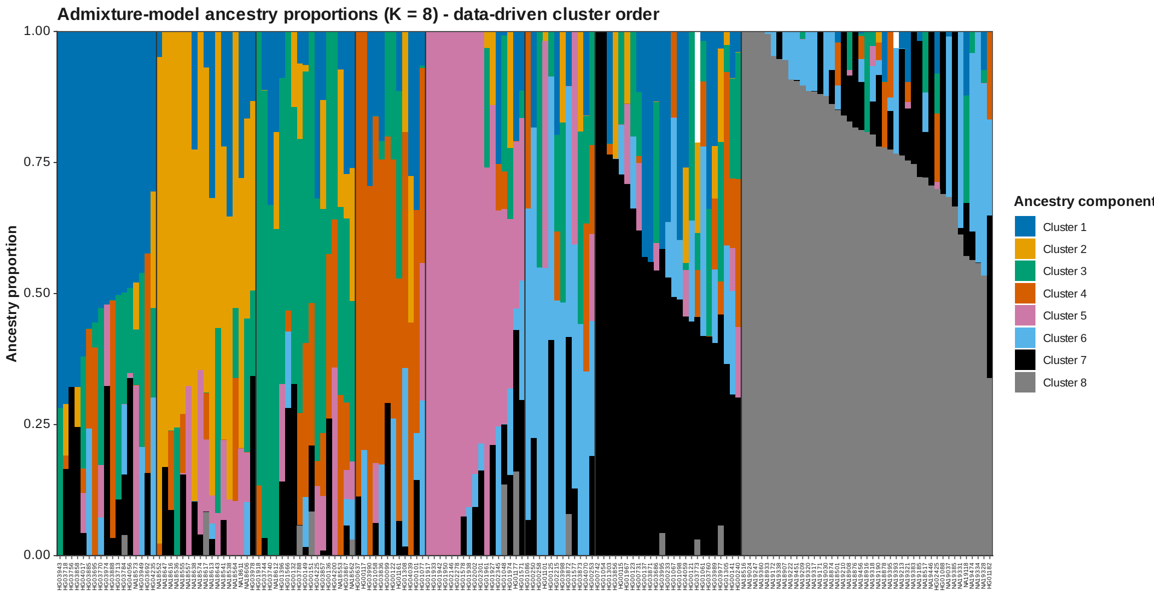

Figure 34: ADMIXTURE Q data driven K 8

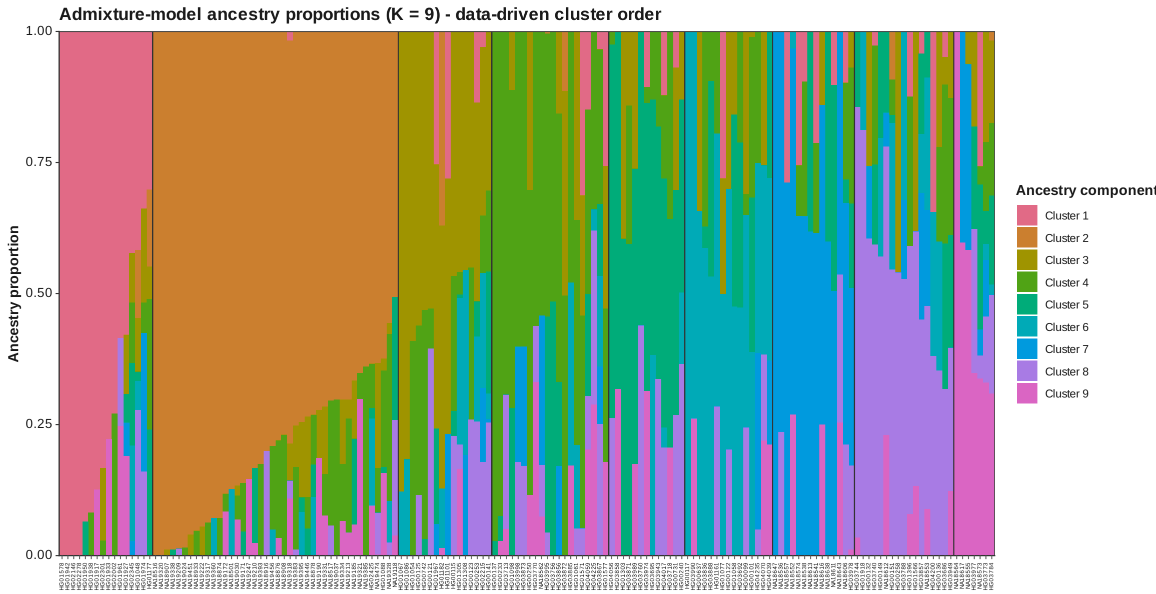

Figure 35: ADMIXTURE Q data driven K 9

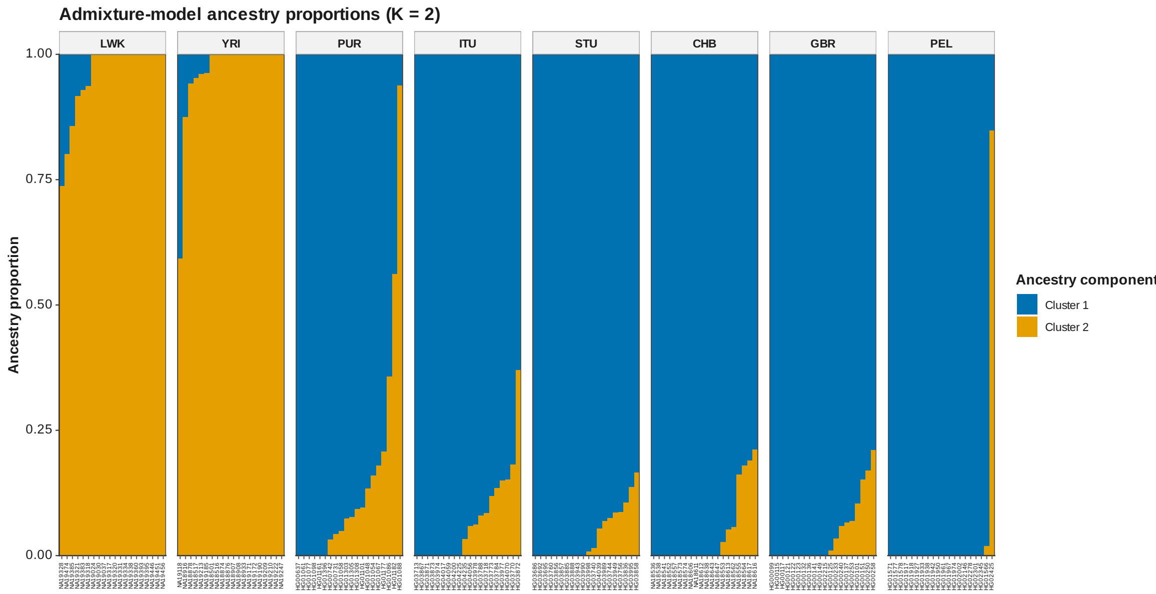

Figure 36: ADMIXTURE Q K 2

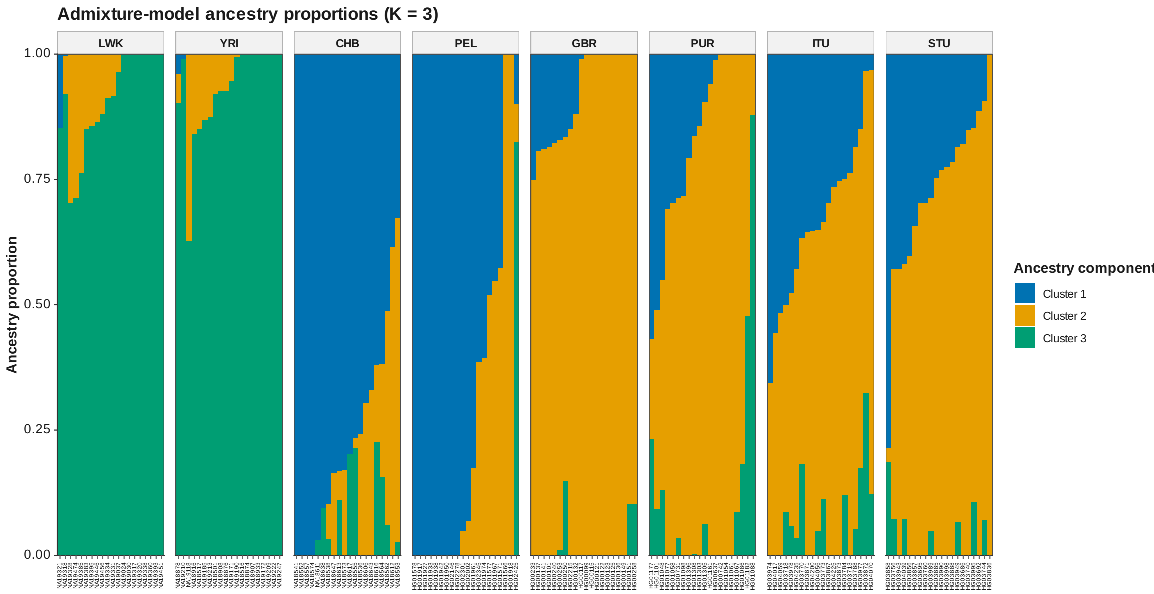

Figure 37: ADMIXTURE Q K 3

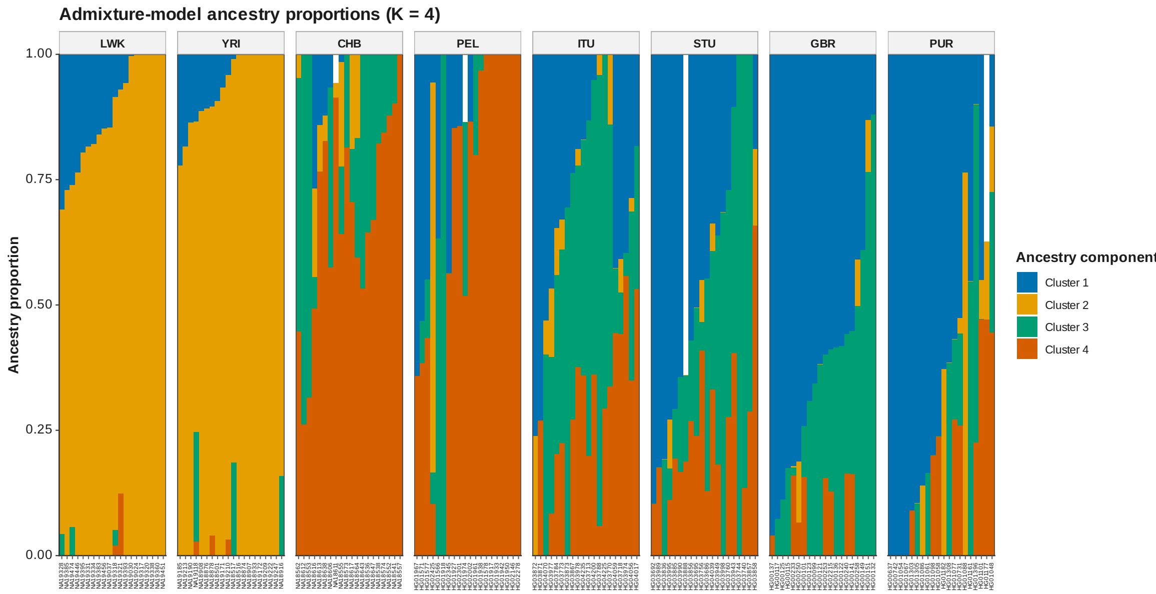

Figure 38: ADMIXTURE Q K 4

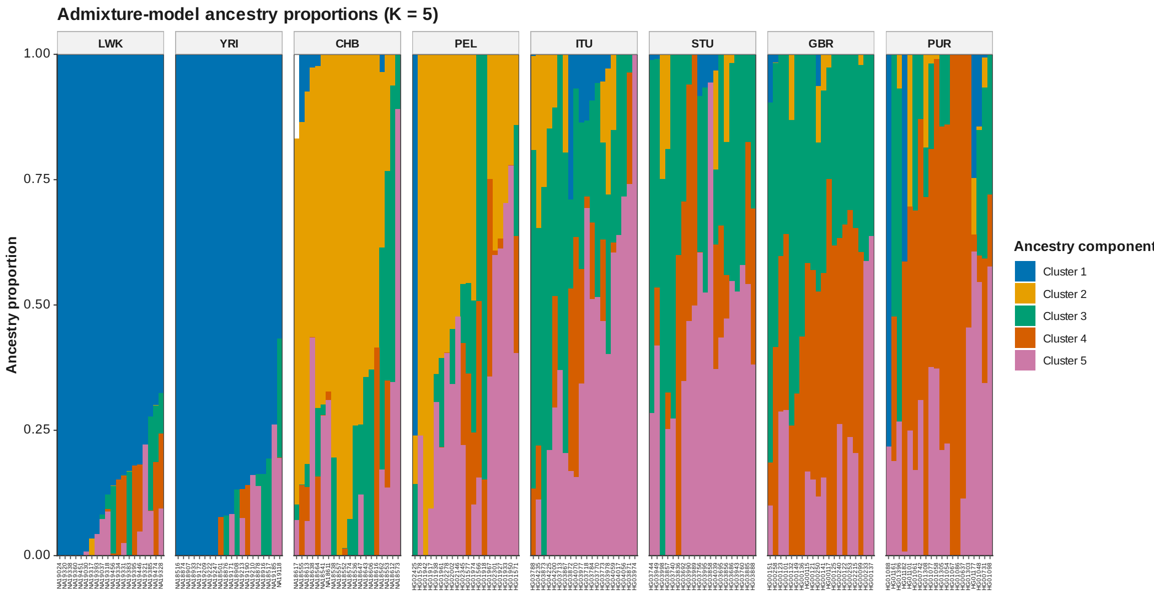

Figure 39: ADMIXTURE Q K 5

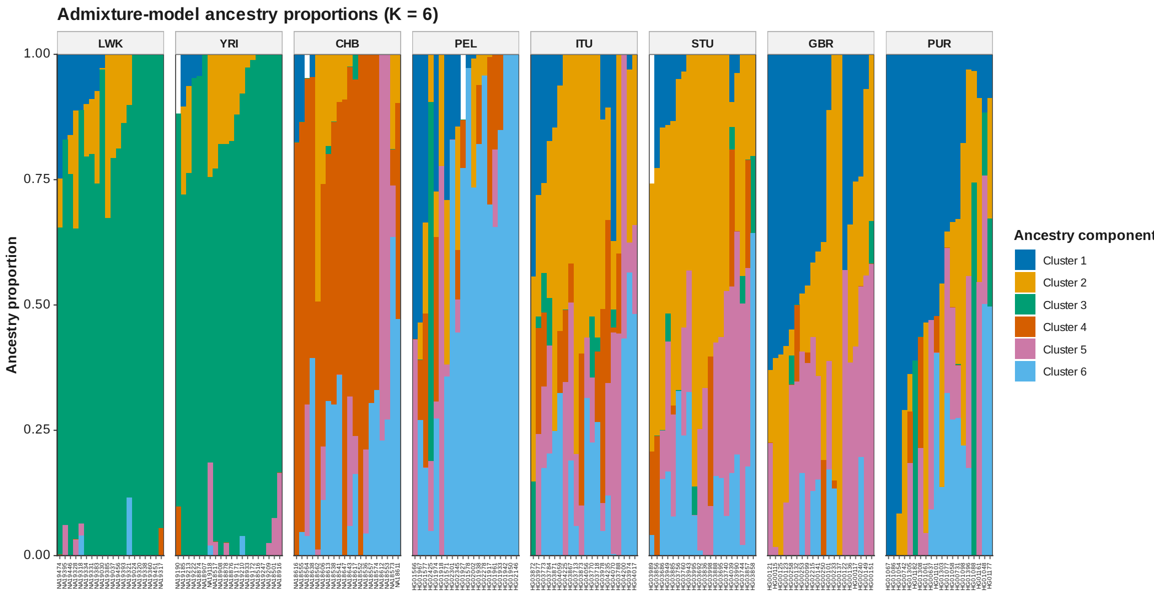

Figure 40: ADMIXTURE Q K 6

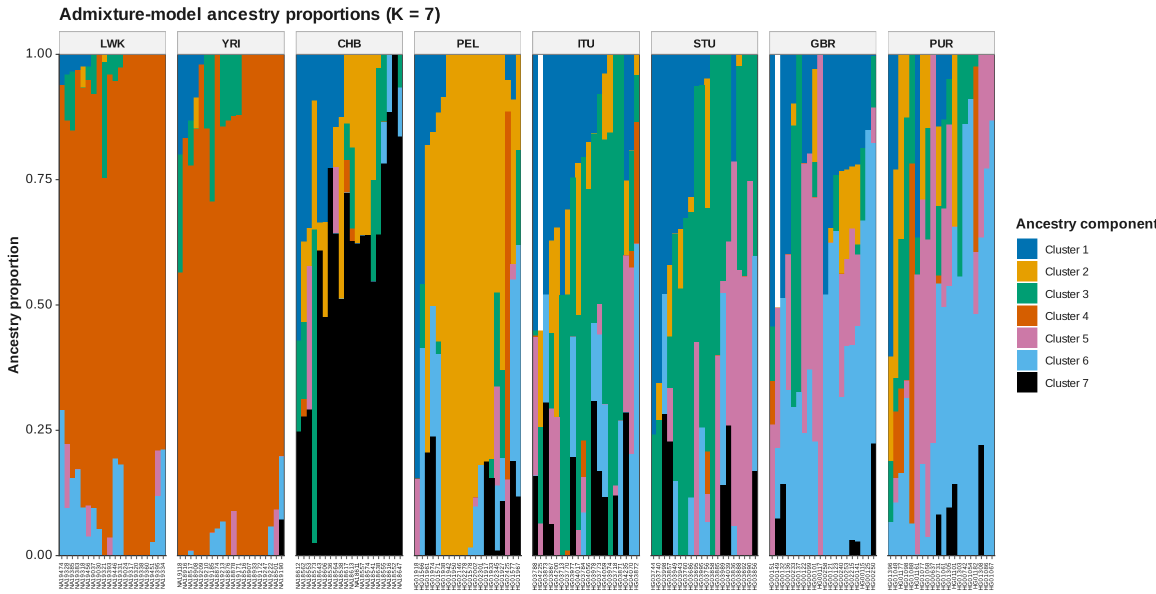

Figure 41: ADMIXTURE Q K 7

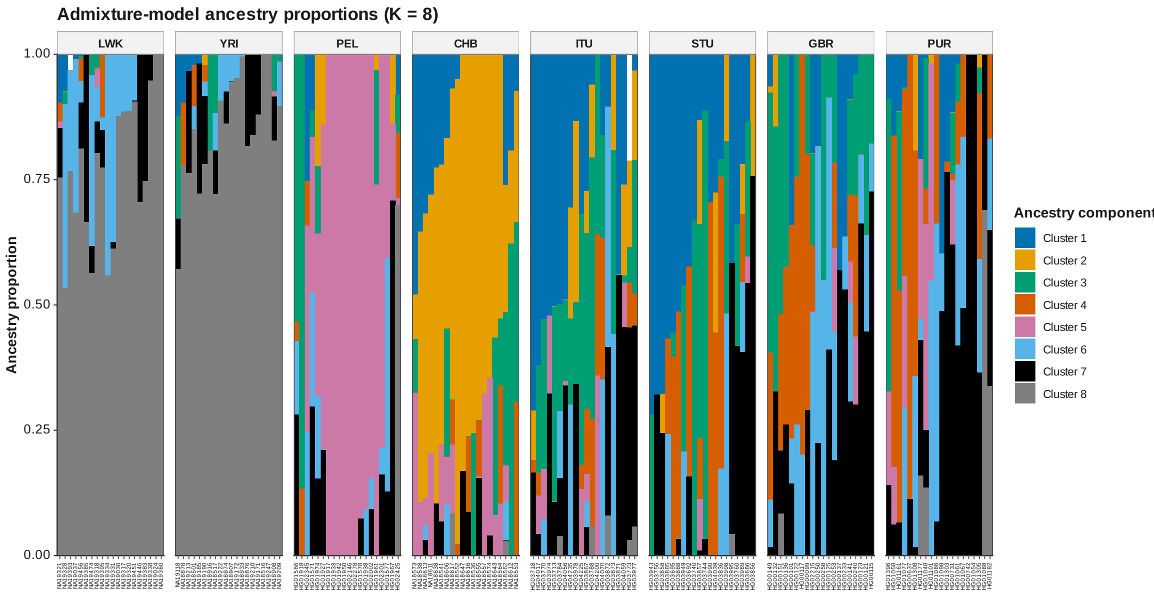

Figure 42: ADMIXTURE Q K 8

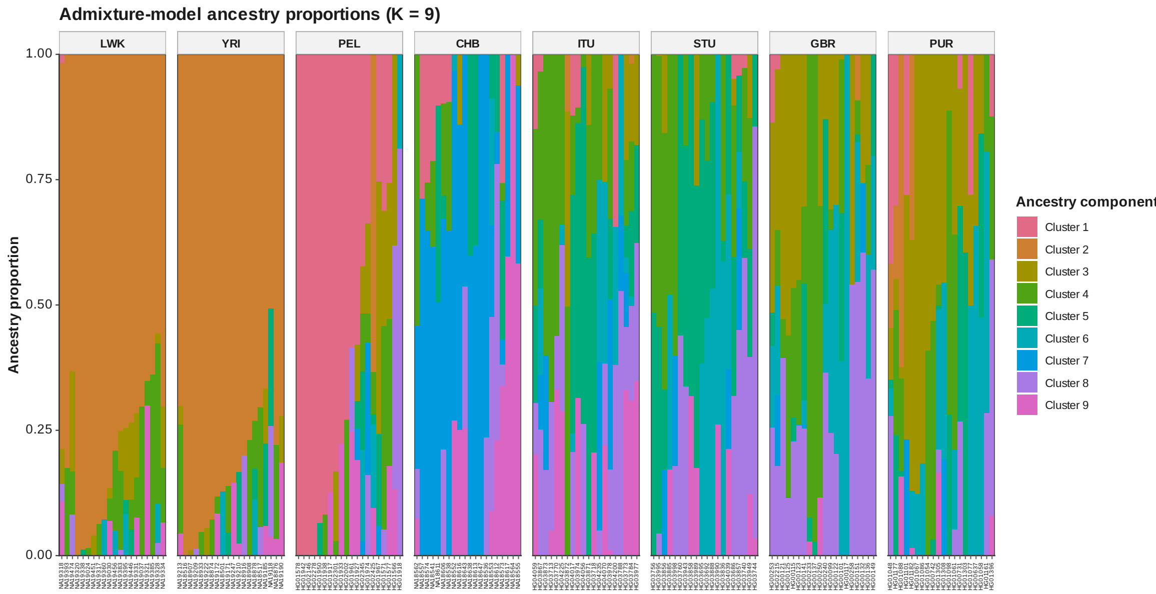

Figure 43: ADMIXTURE Q K 9

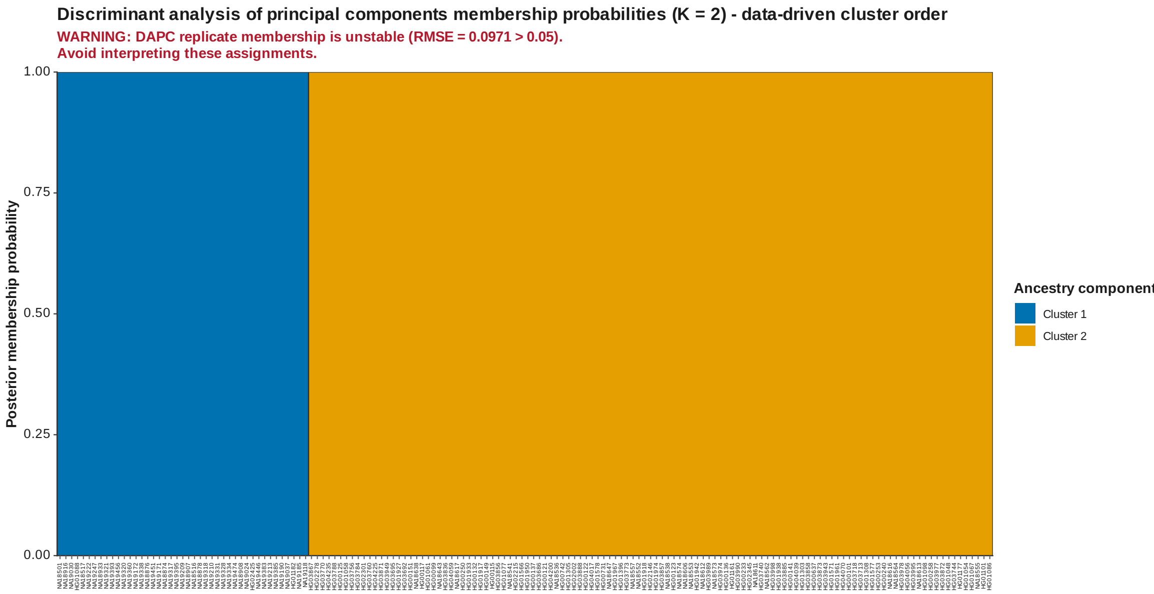

Figure 44: DAPC membership data driven K 2

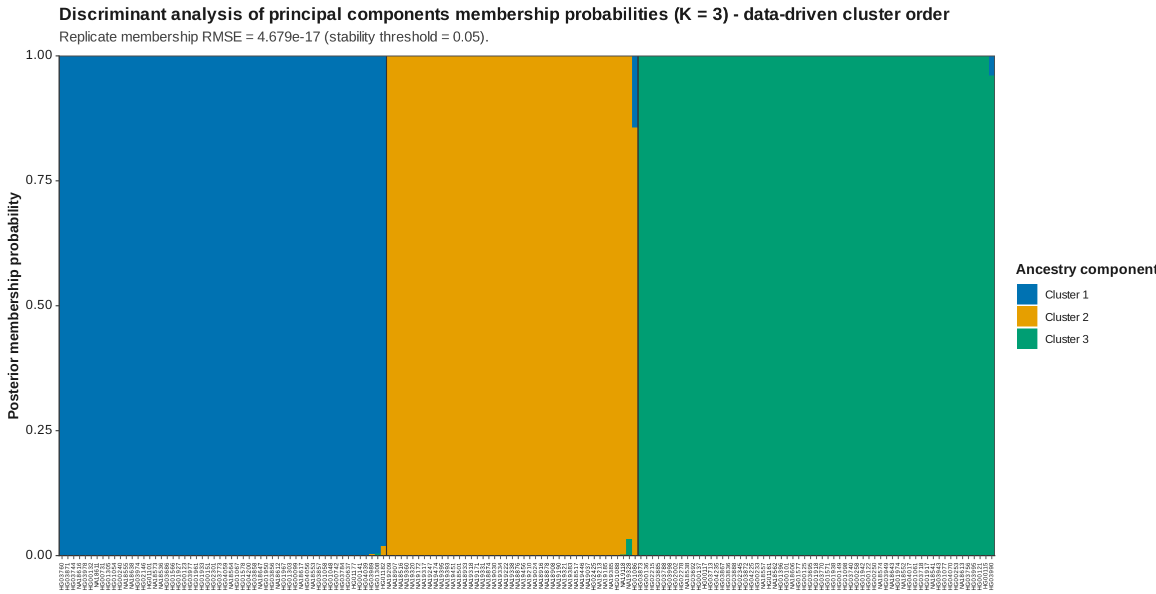

Figure 45: DAPC membership data driven K 3

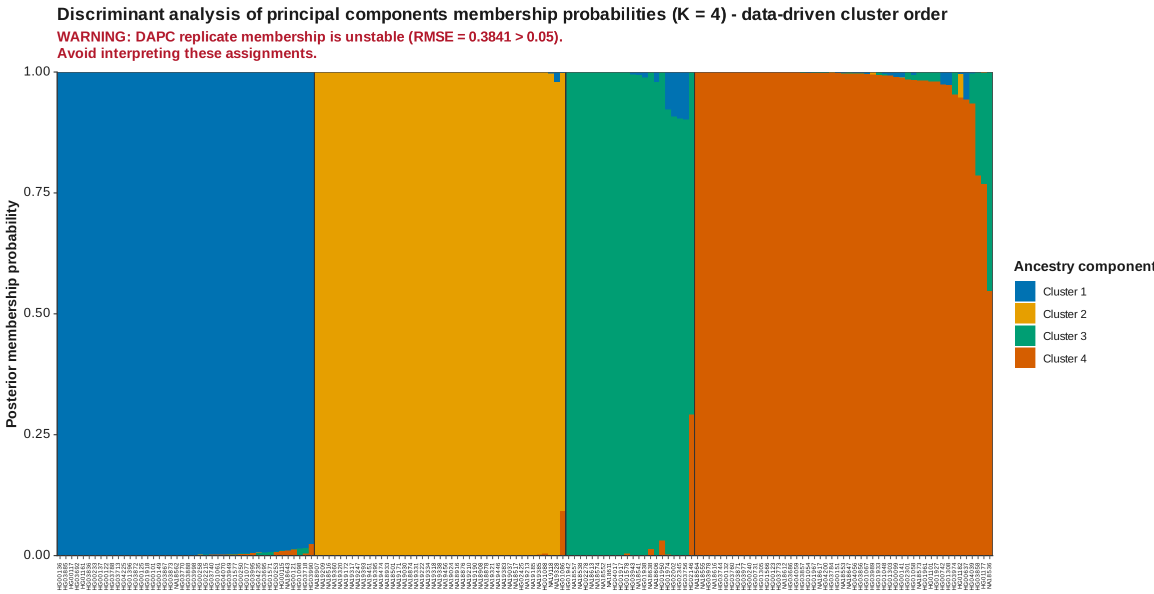

Figure 46: DAPC membership data driven K 4

Figure 47: DAPC membership data driven K 5

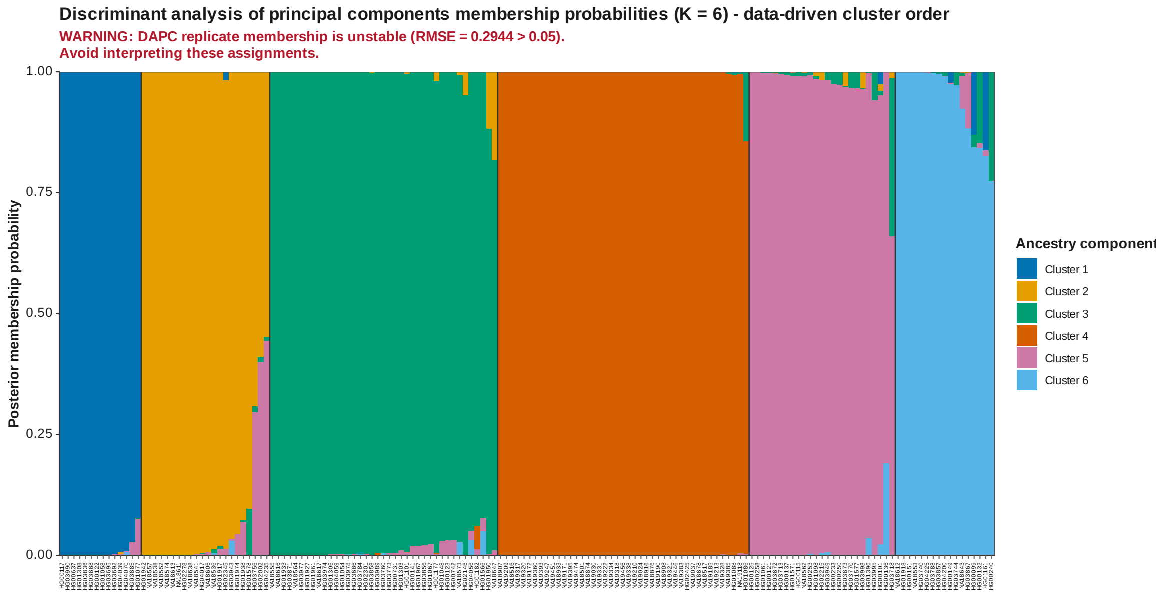

Figure 48: DAPC membership data driven K 6

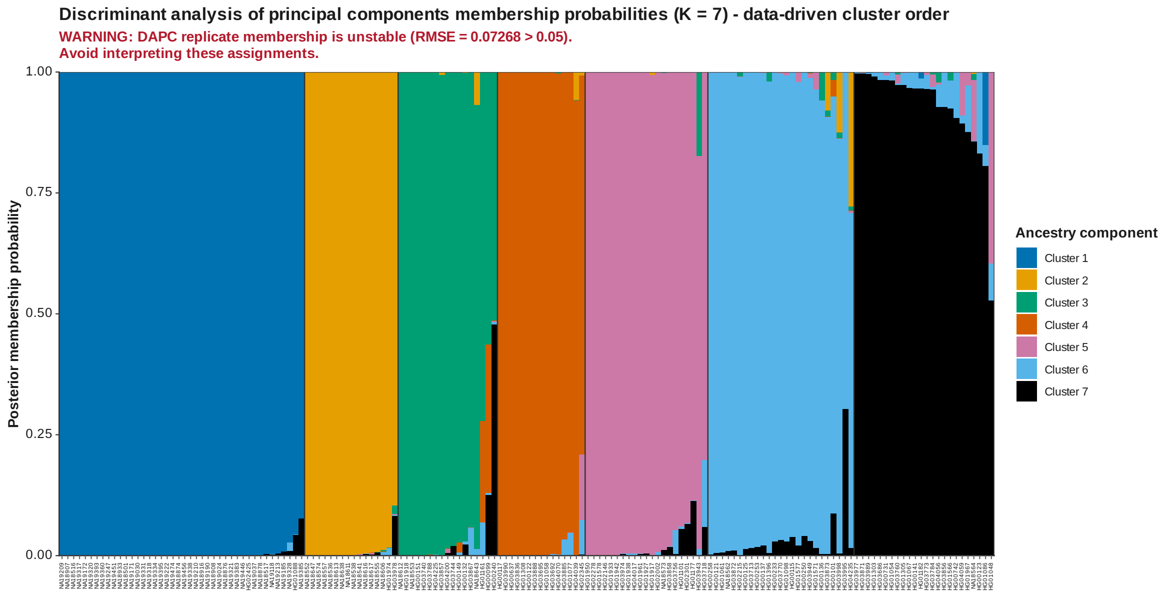

Figure 49: DAPC membership data driven K 7

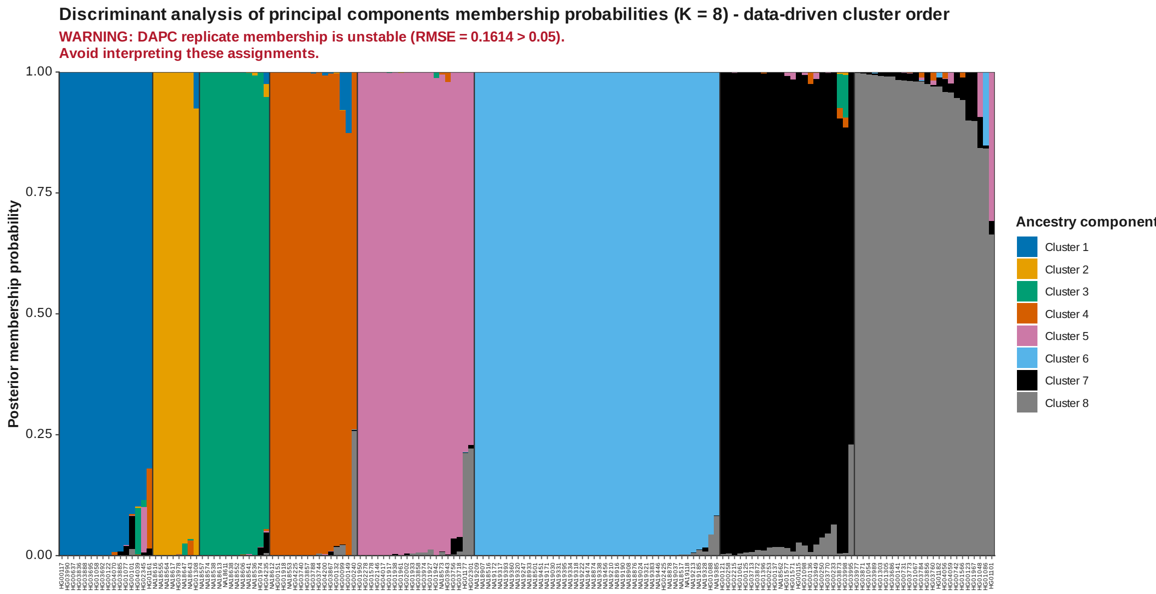

Figure 50: DAPC membership data driven K 8

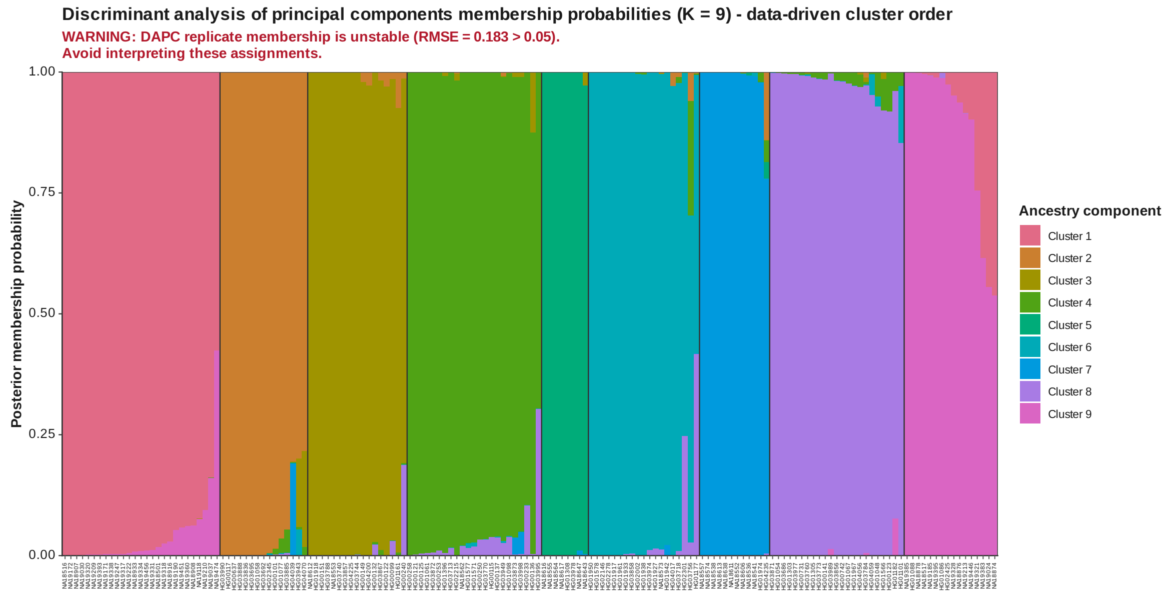

Figure 51: DAPC membership data driven K 9

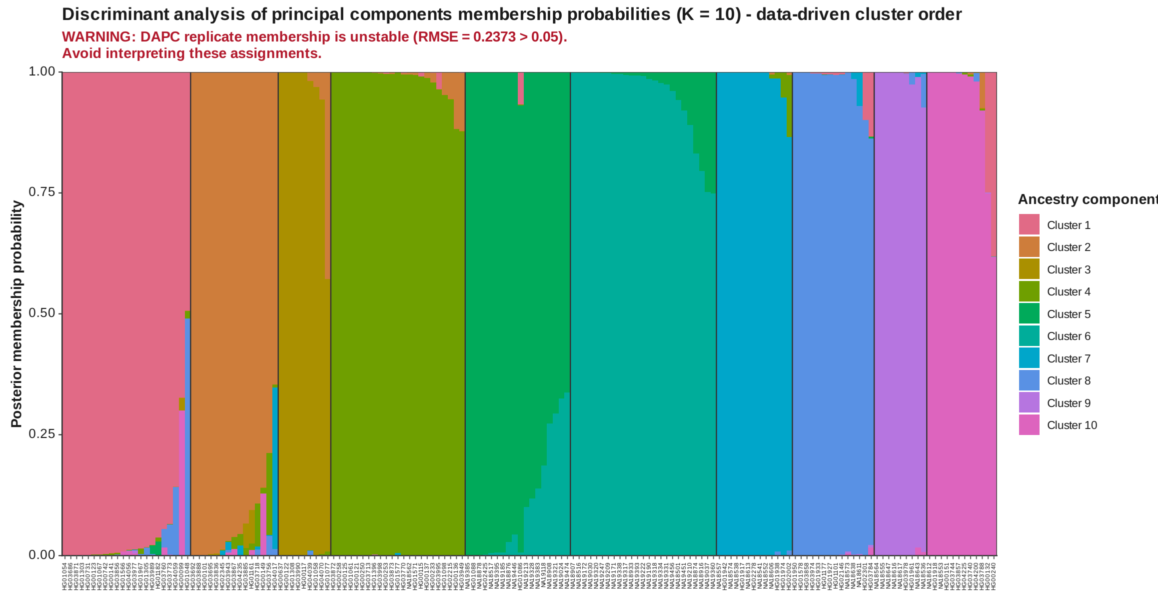

Figure 52: DAPC membership data driven K 10

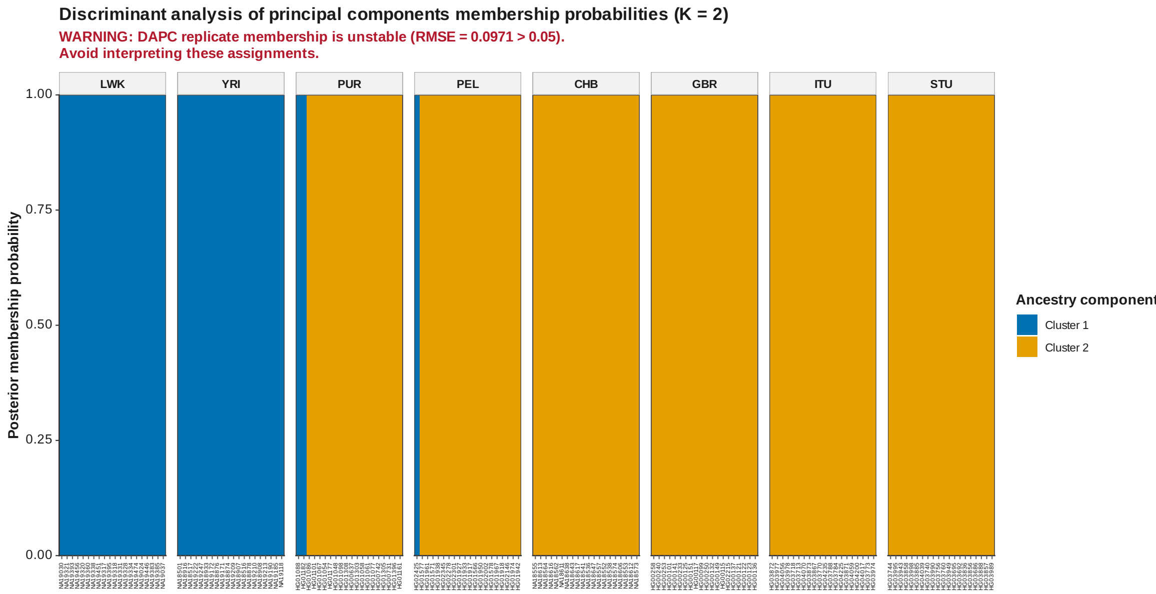

Figure 53: DAPC membership K 2

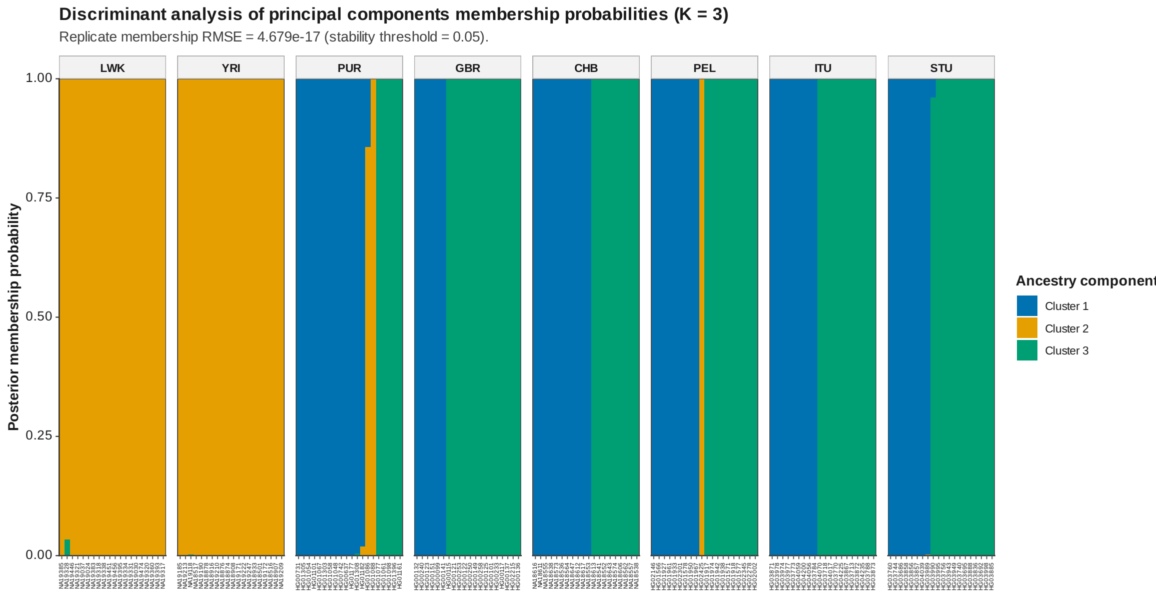

Figure 54: DAPC membership K 3

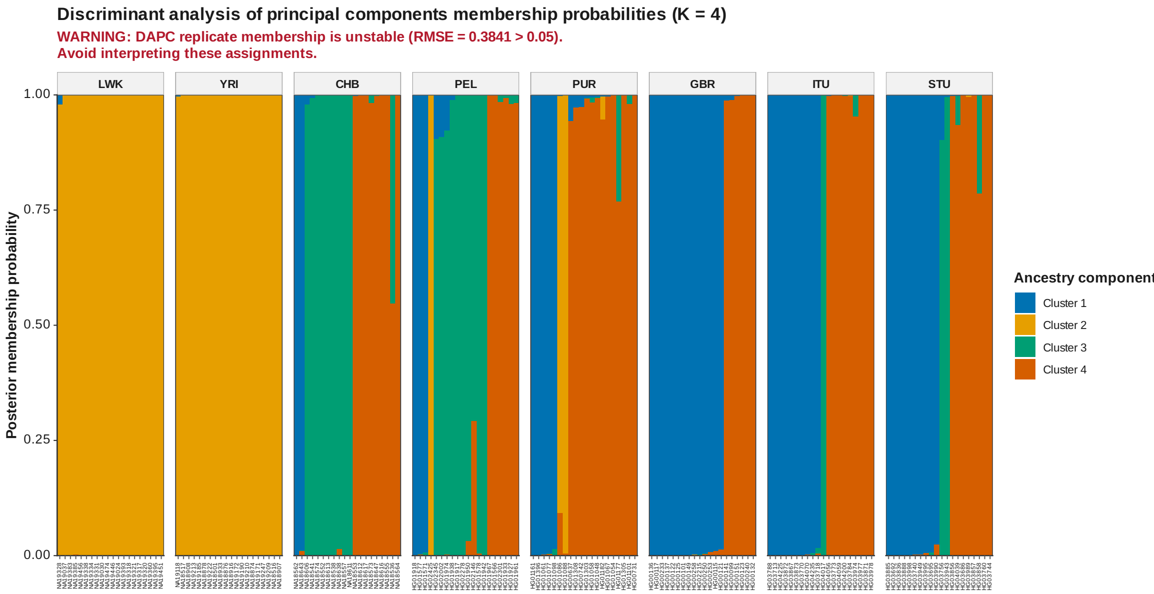

Figure 55: DAPC membership K 4

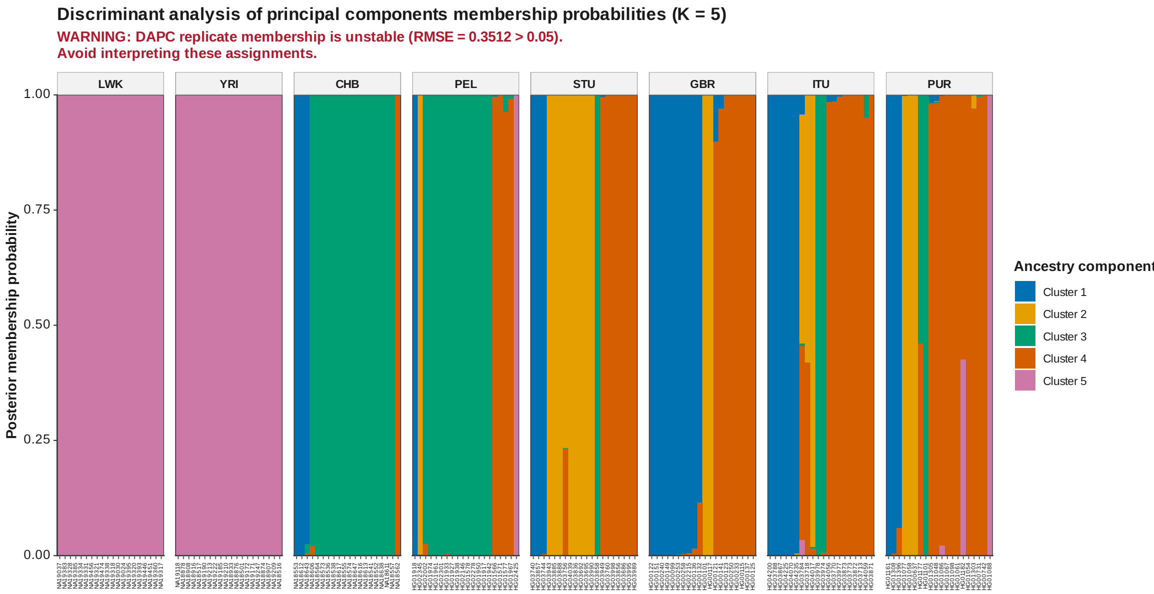

Figure 56: DAPC membership K 5

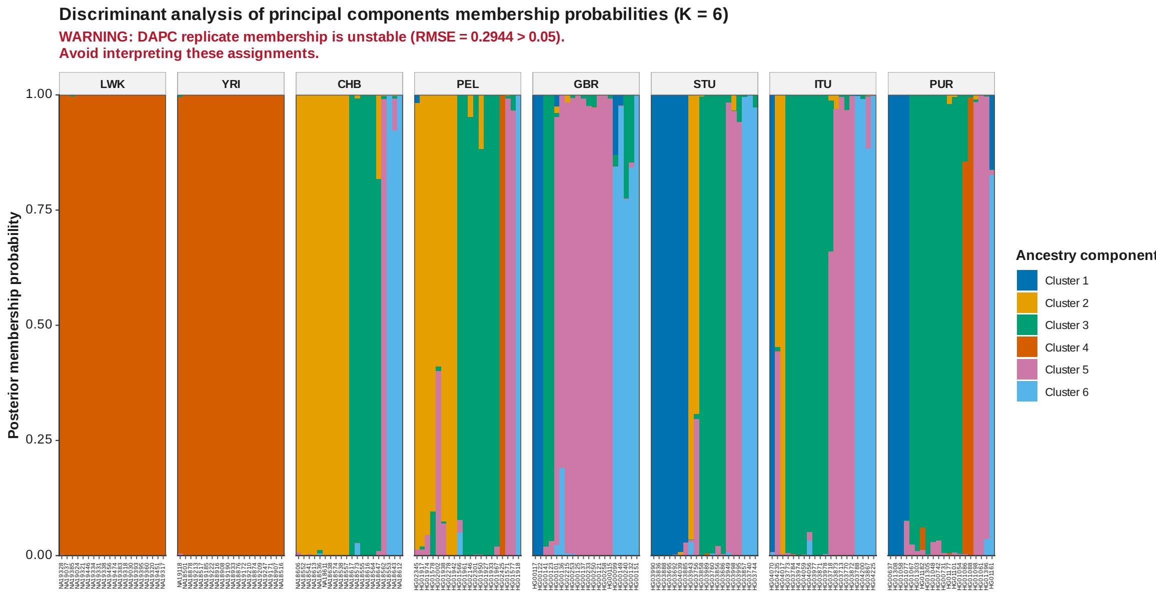

Figure 57: DAPC membership K 6

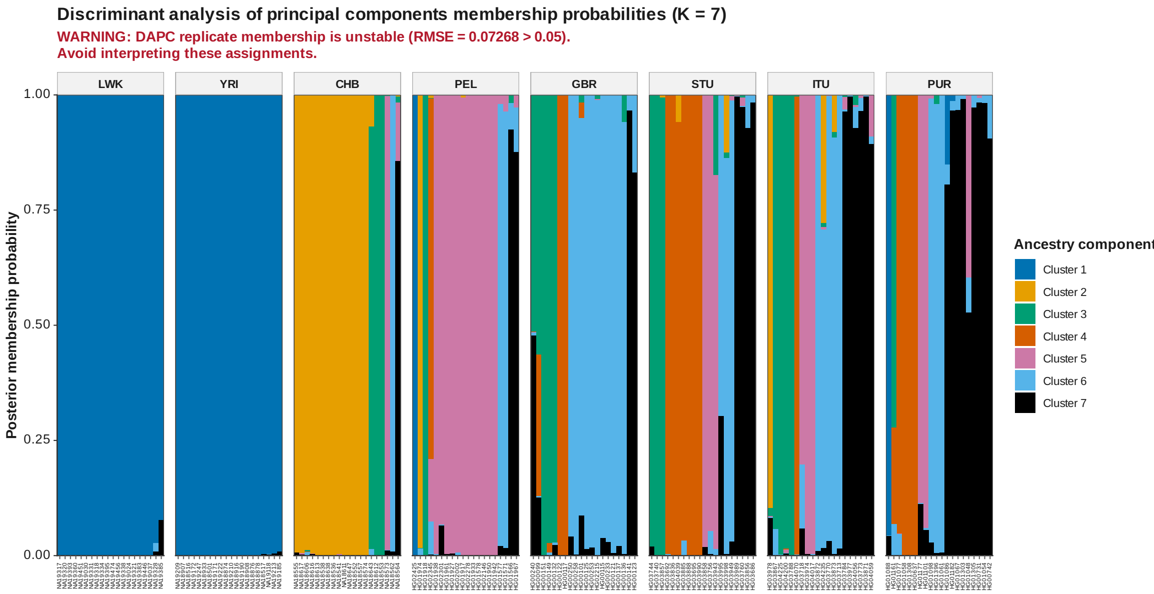

Figure 58: DAPC membership K 7

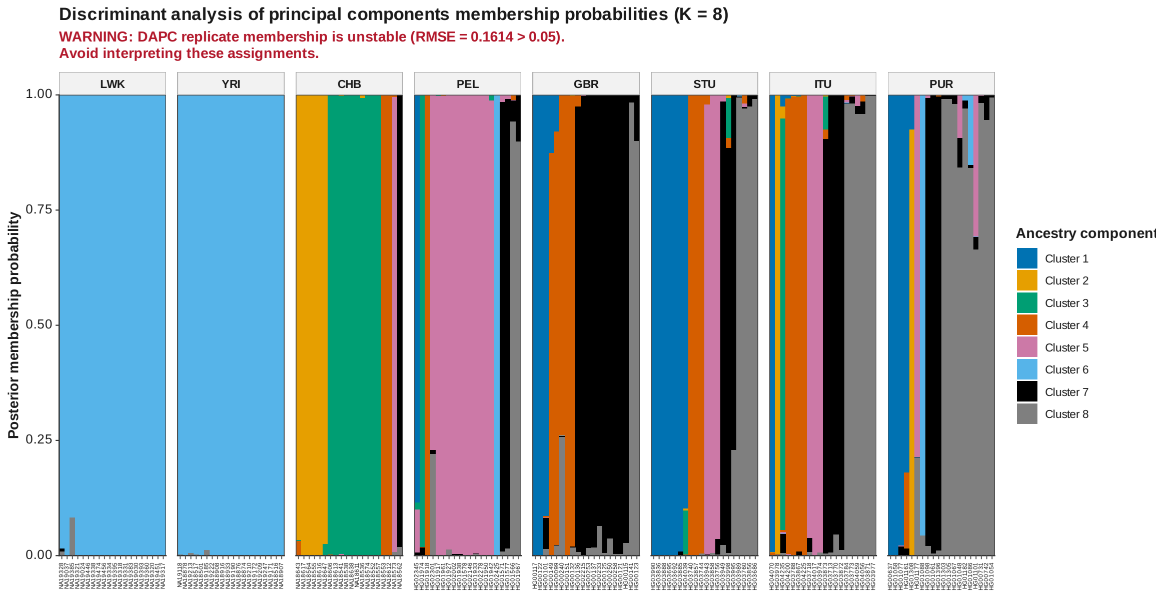

Figure 59: DAPC membership K 8

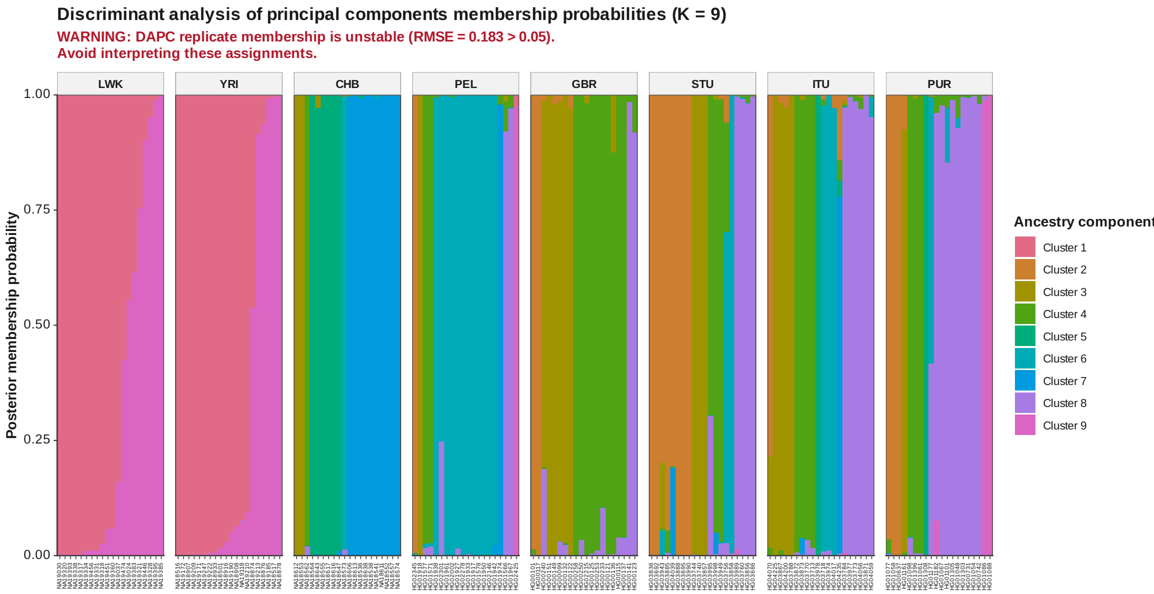

Figure 60: DAPC membership K 9

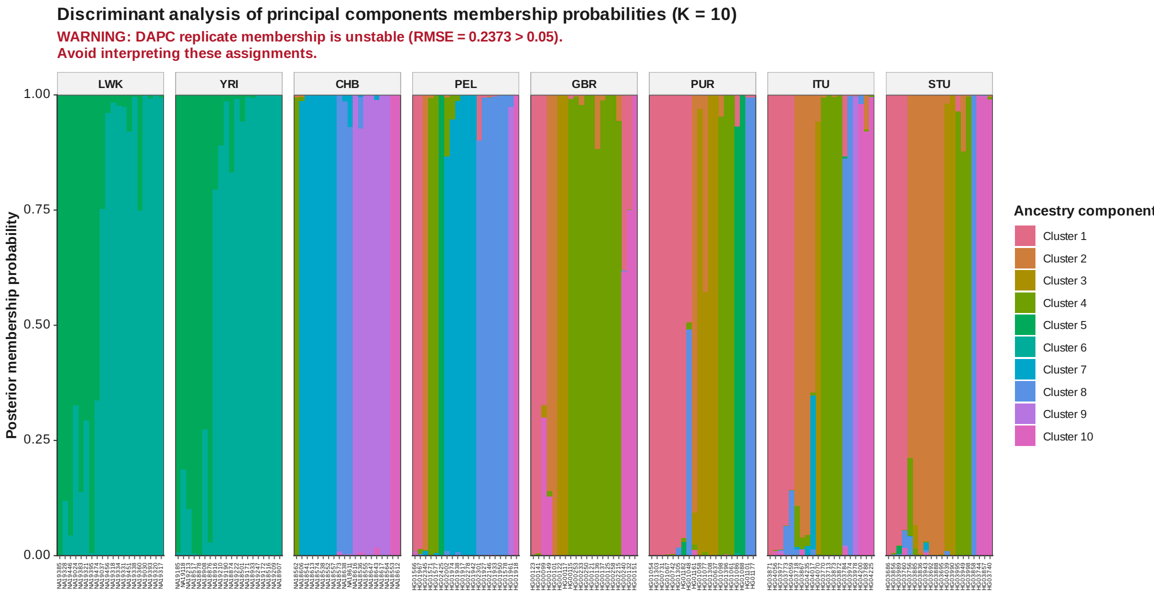

Figure 61: DAPC membership K 10

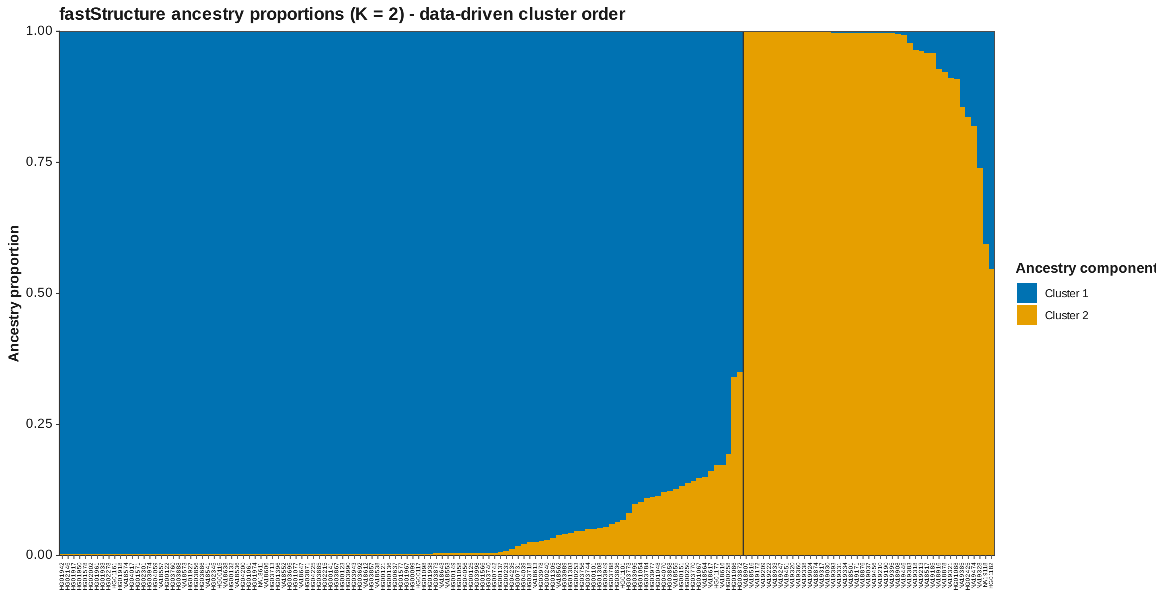

Figure 62: FastStructure Q data driven K 2

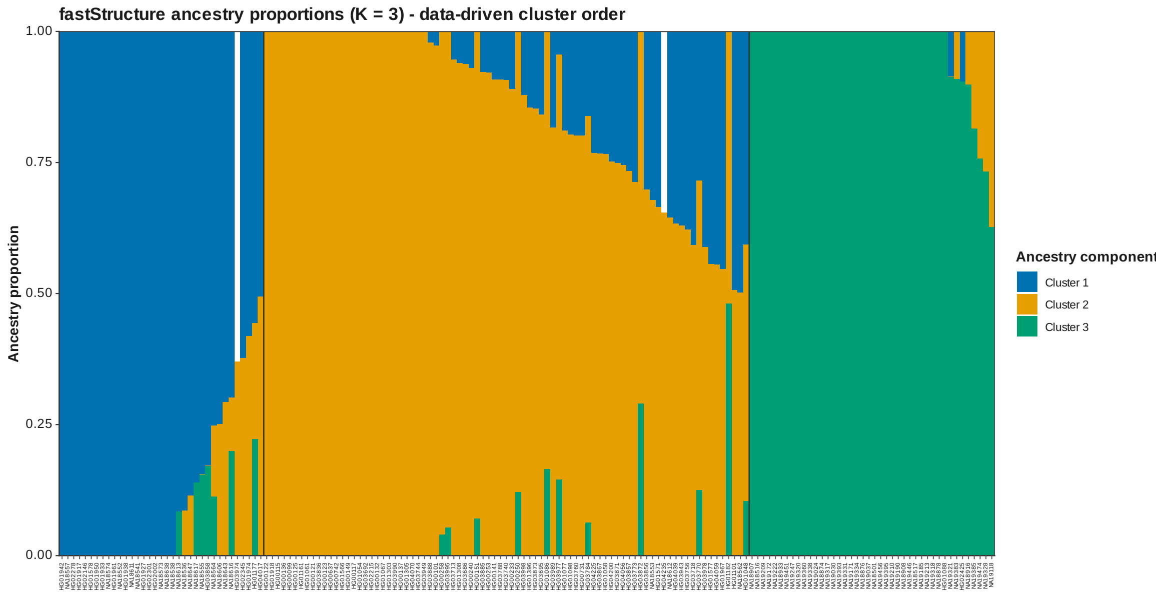

Figure 63: FastStructure Q data driven K 3

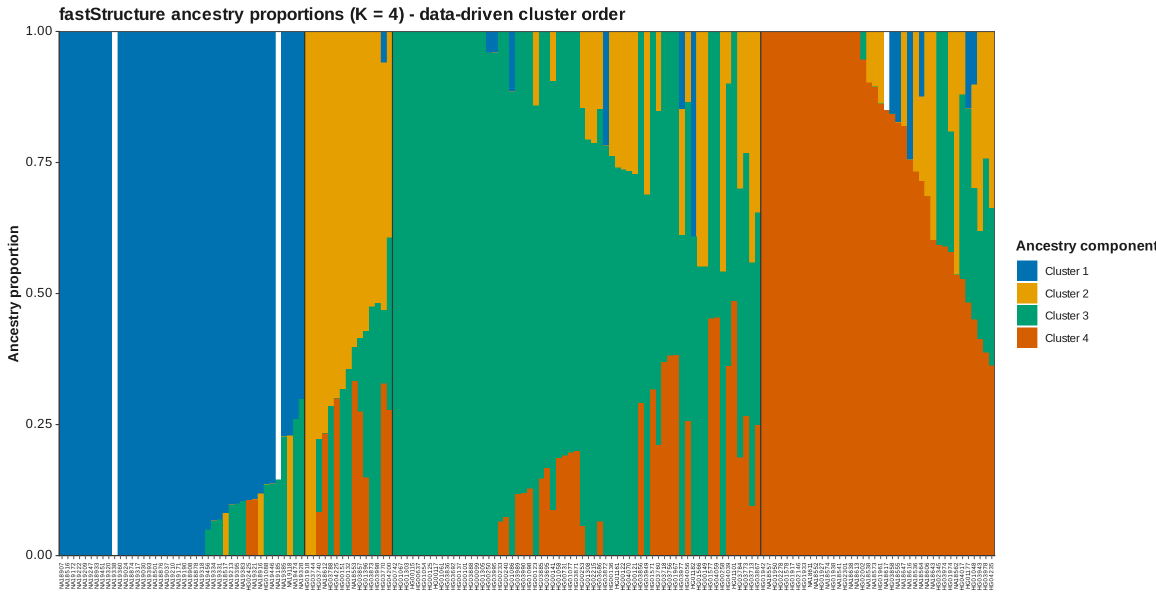

Figure 64: FastStructure Q data driven K 4

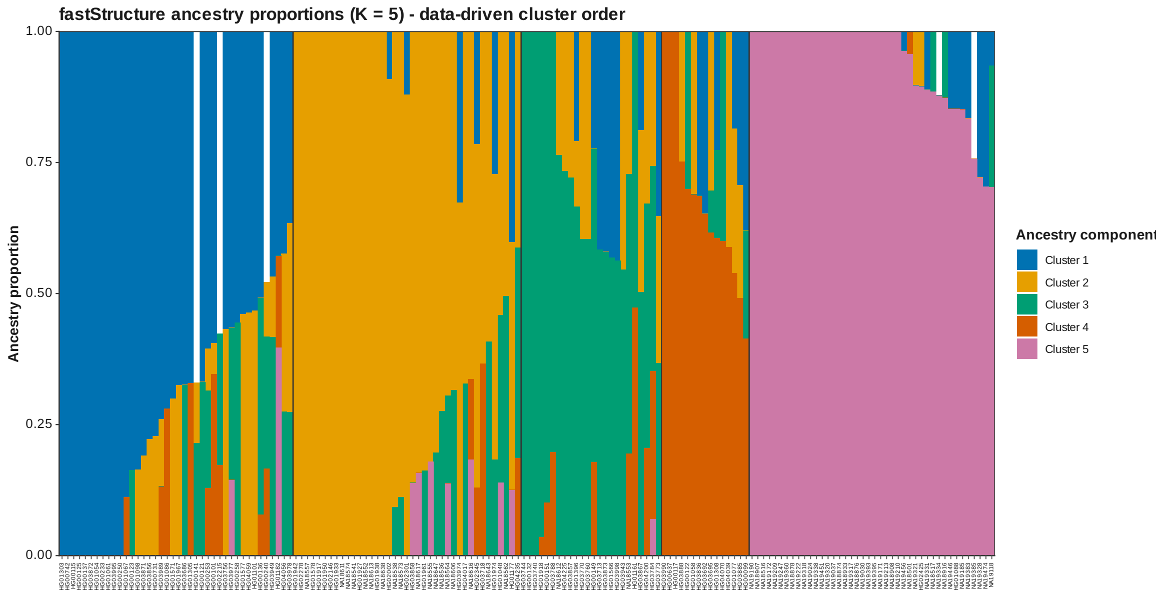

Figure 65: FastStructure Q data driven K 5

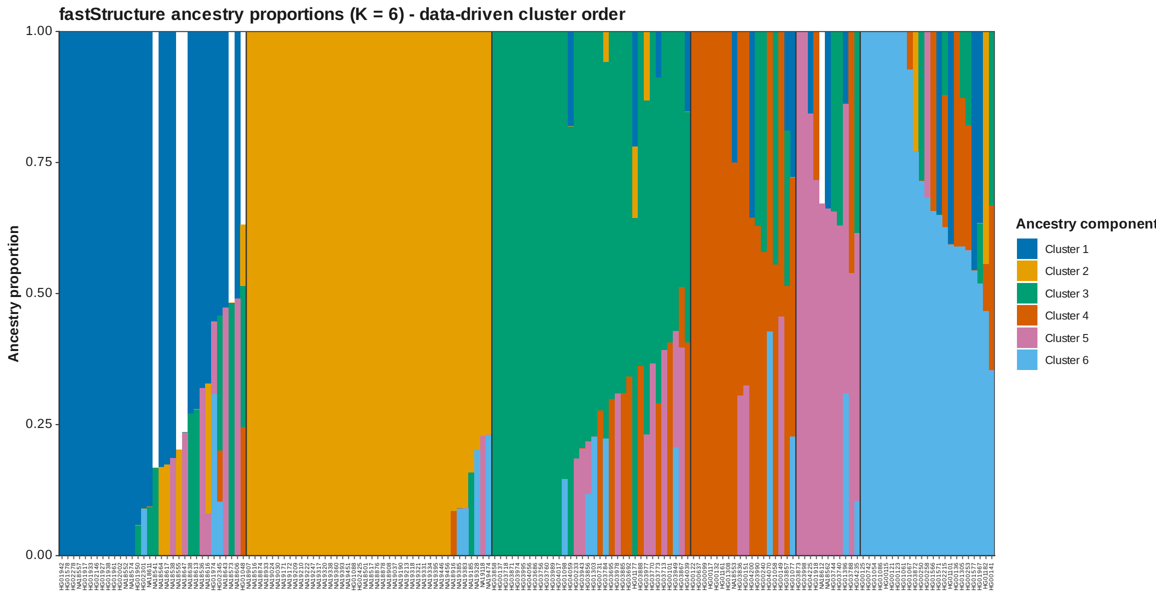

Figure 66: FastStructure Q data driven K 6

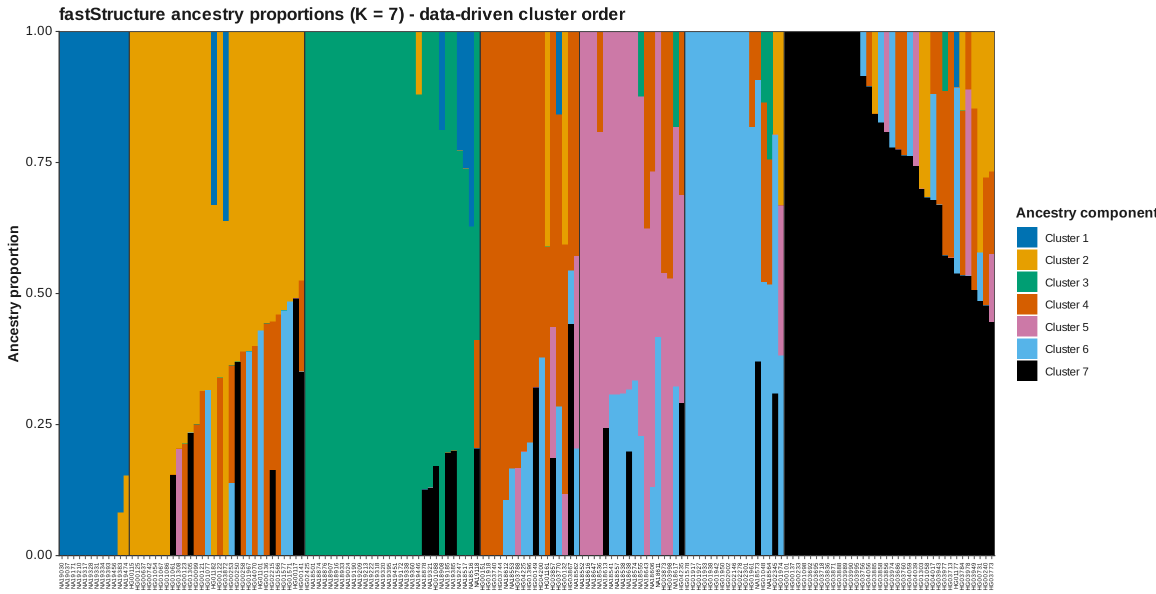

Figure 67: FastStructure Q data driven K 7

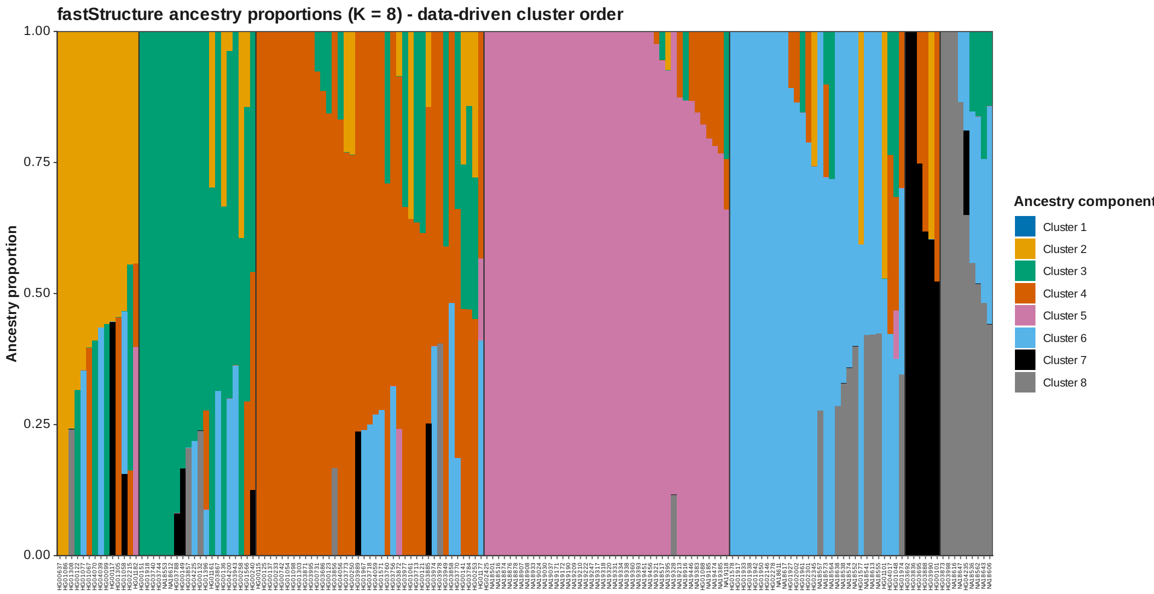

Figure 68: FastStructure Q data driven K 8

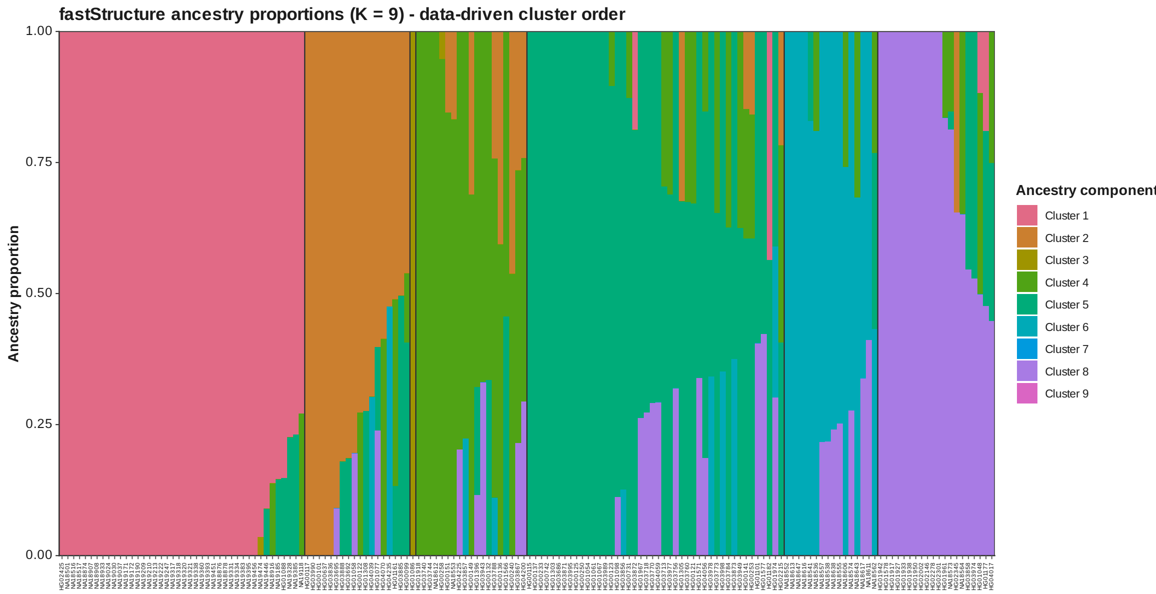

Figure 69: FastStructure Q data driven K 9

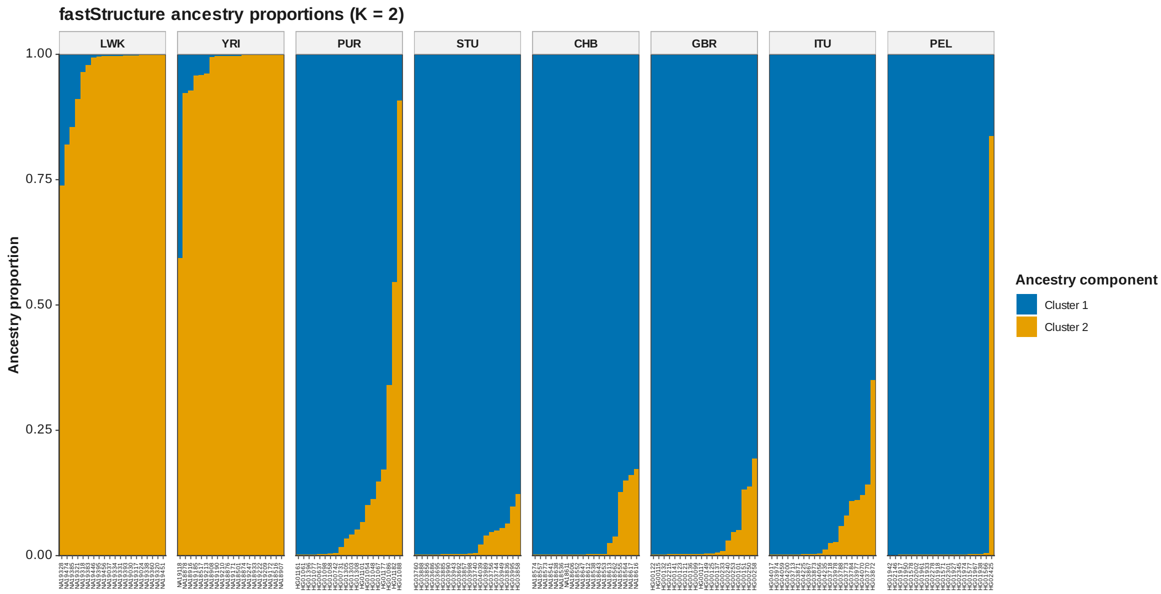

Figure 70: FastStructure Q K 2

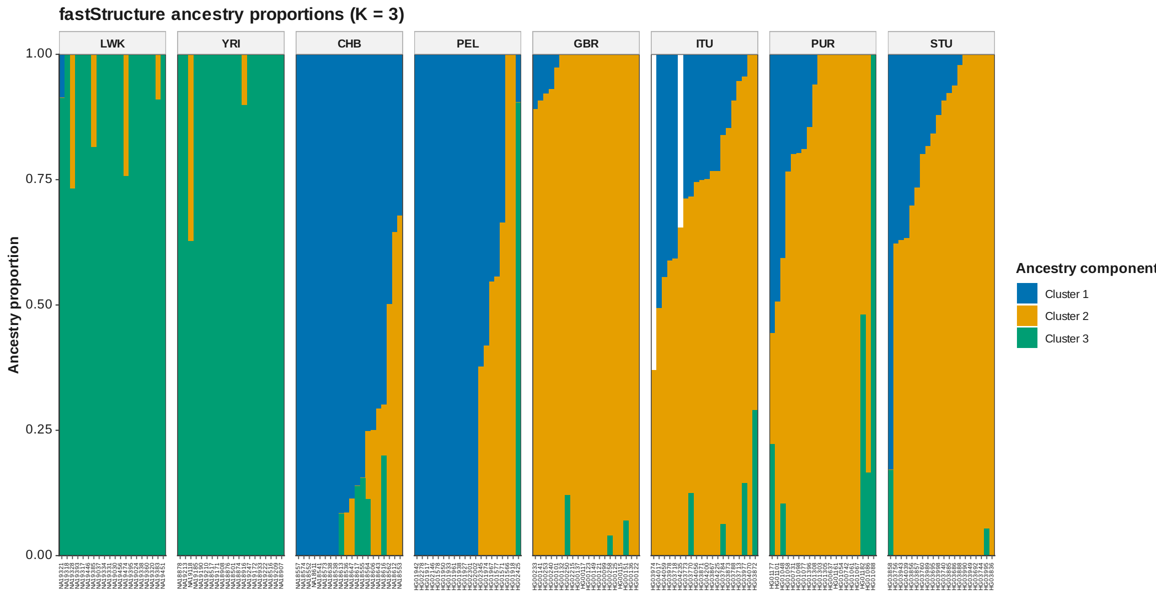

Figure 71: FastStructure Q K 3

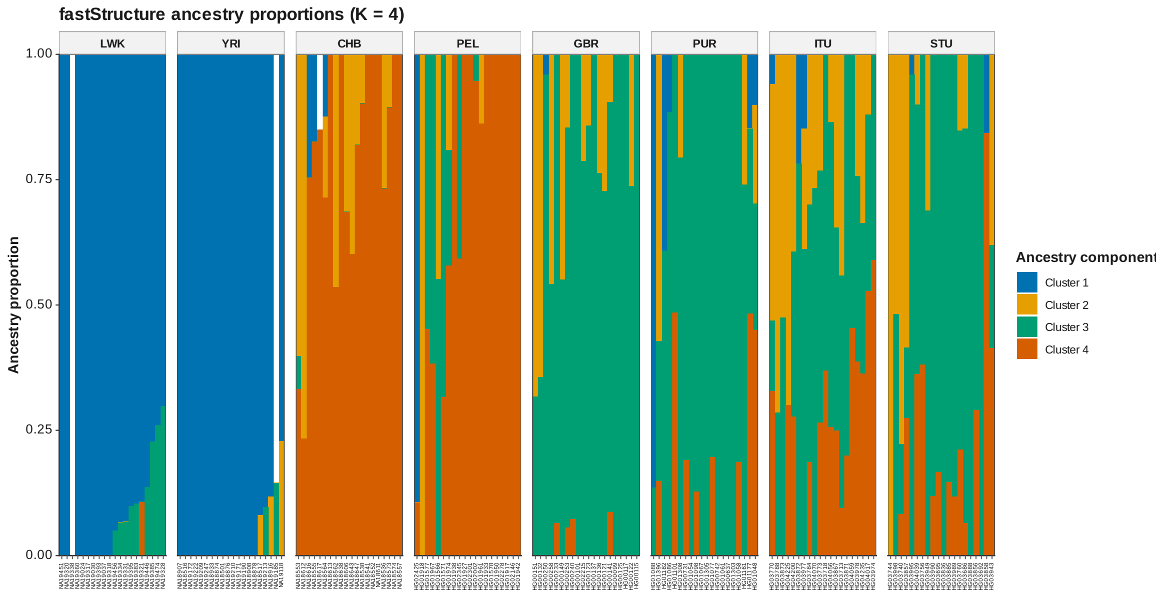

Figure 72: FastStructure Q K 4

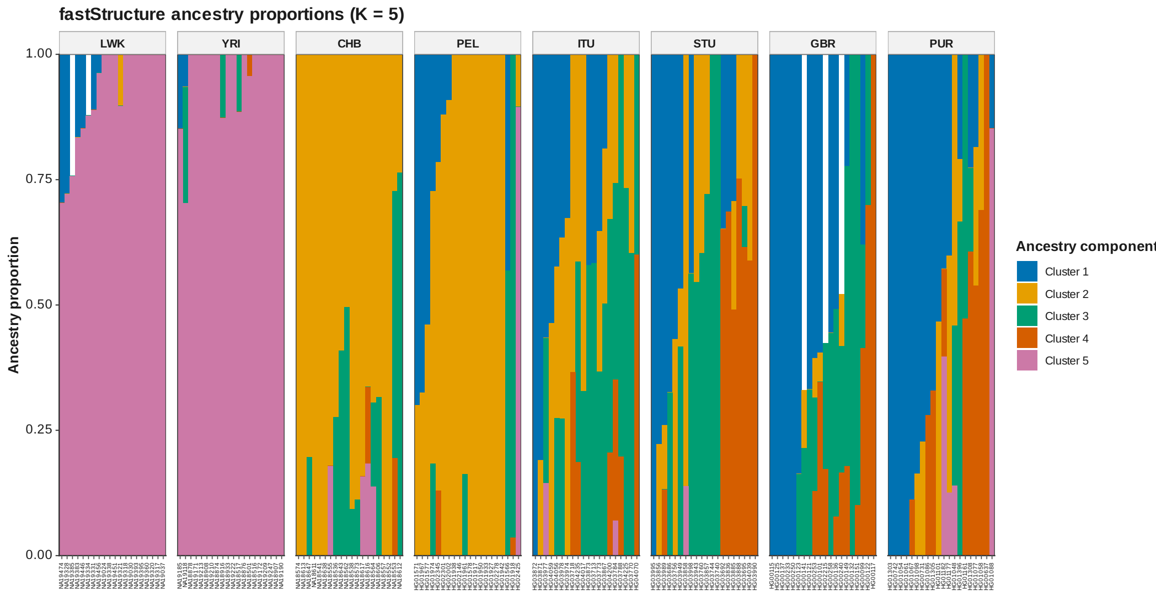

Figure 73: FastStructure Q K 5

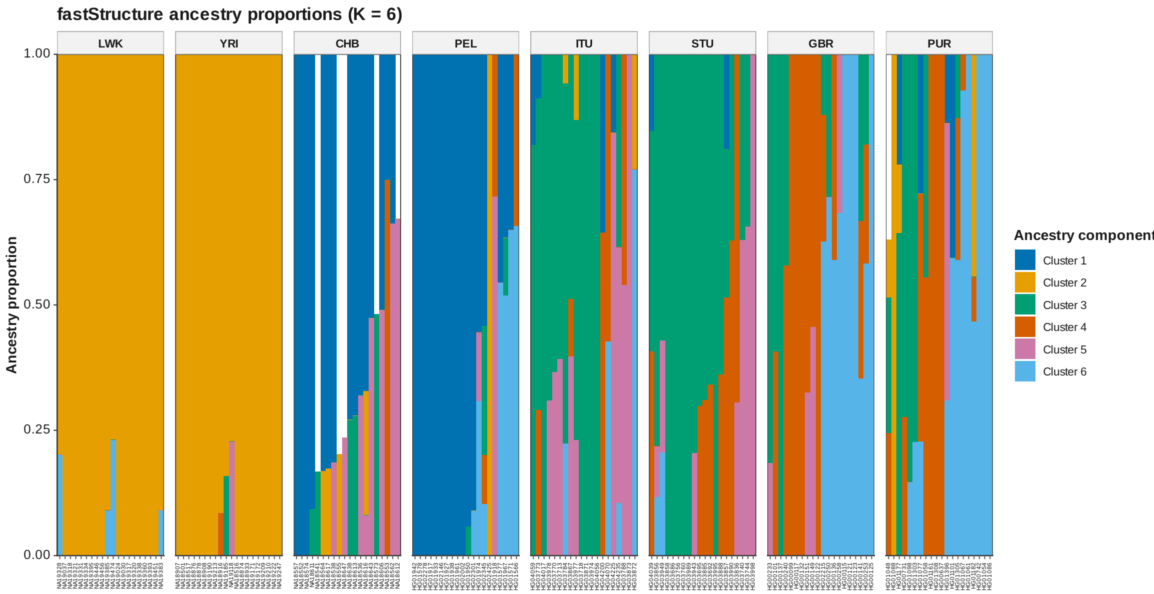

Figure 74: FastStructure Q K 6

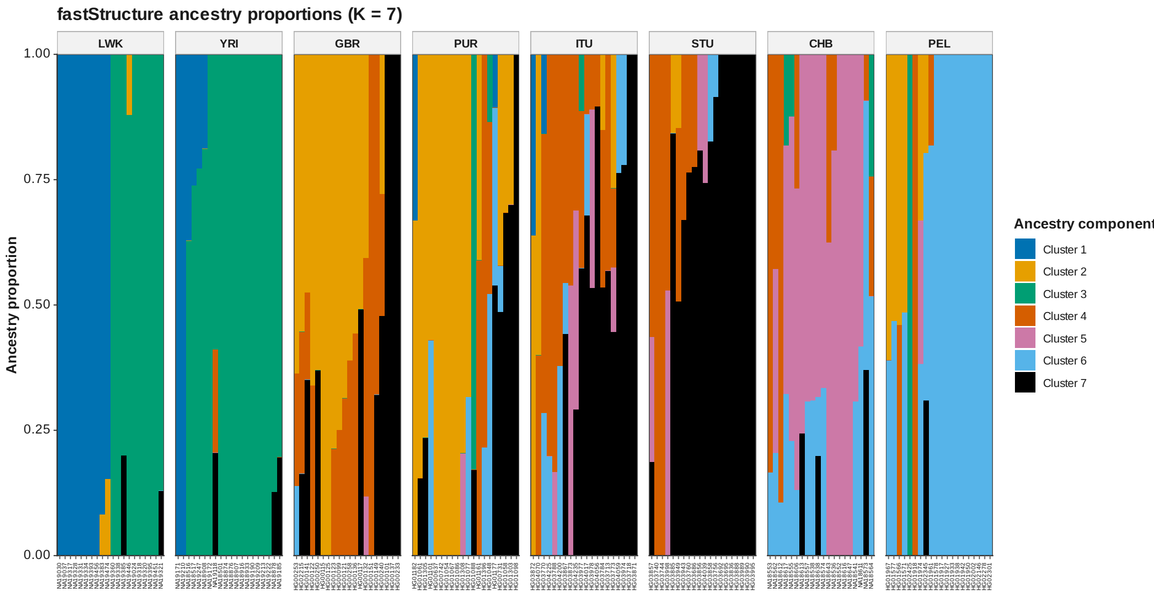

Figure 75: FastStructure Q K 7

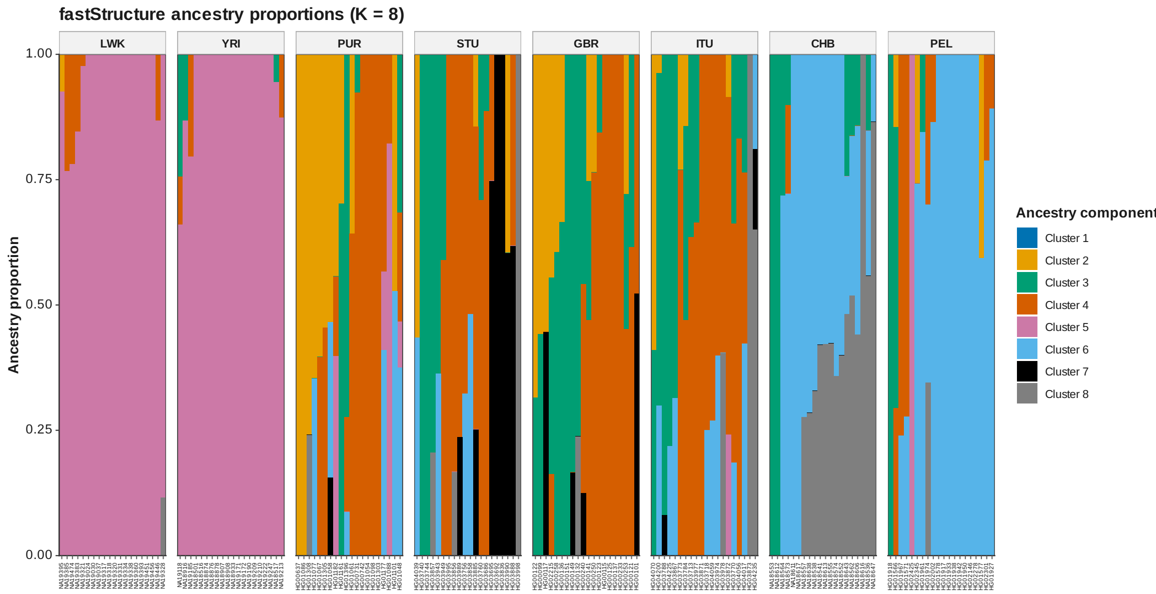

Figure 76: FastStructure Q K 8

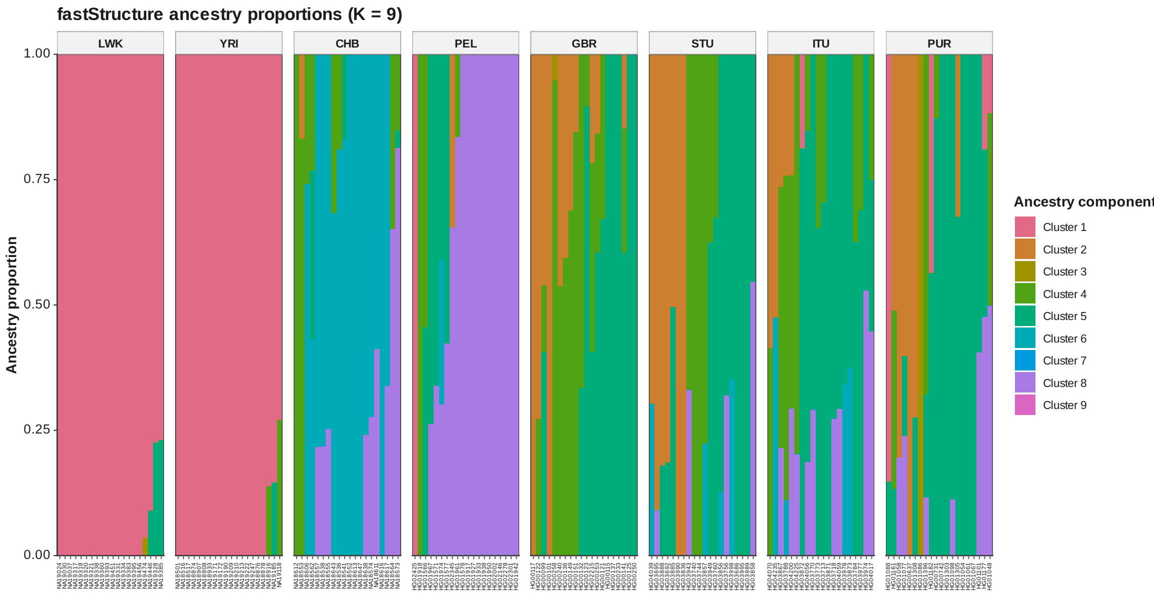

Figure 77: FastStructure Q K 9

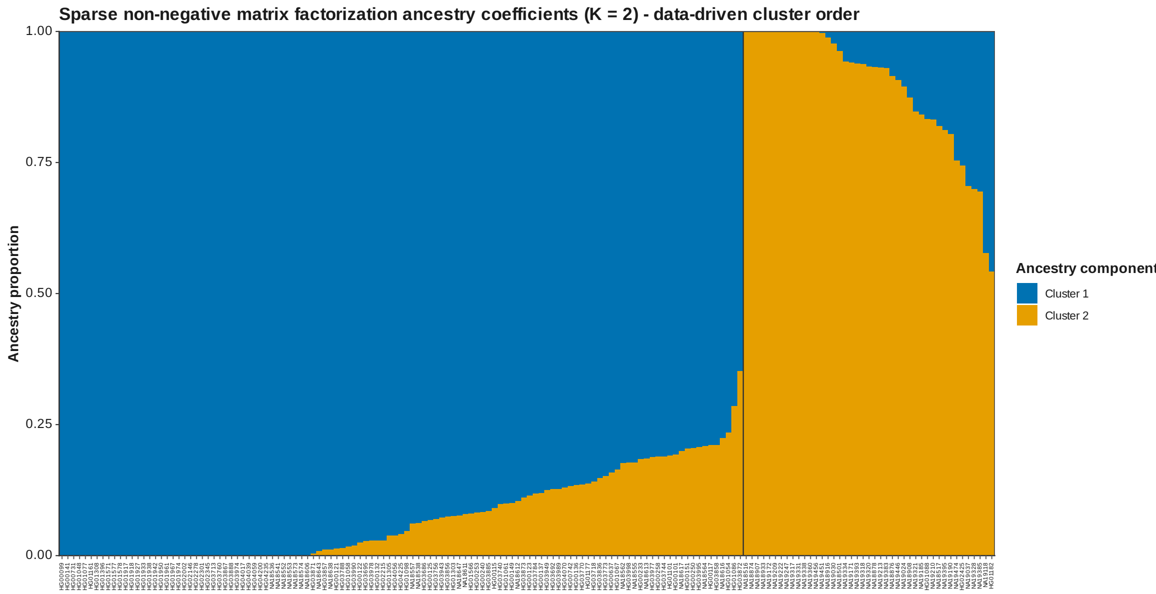

Figure 78: SNMF Q data driven K 2

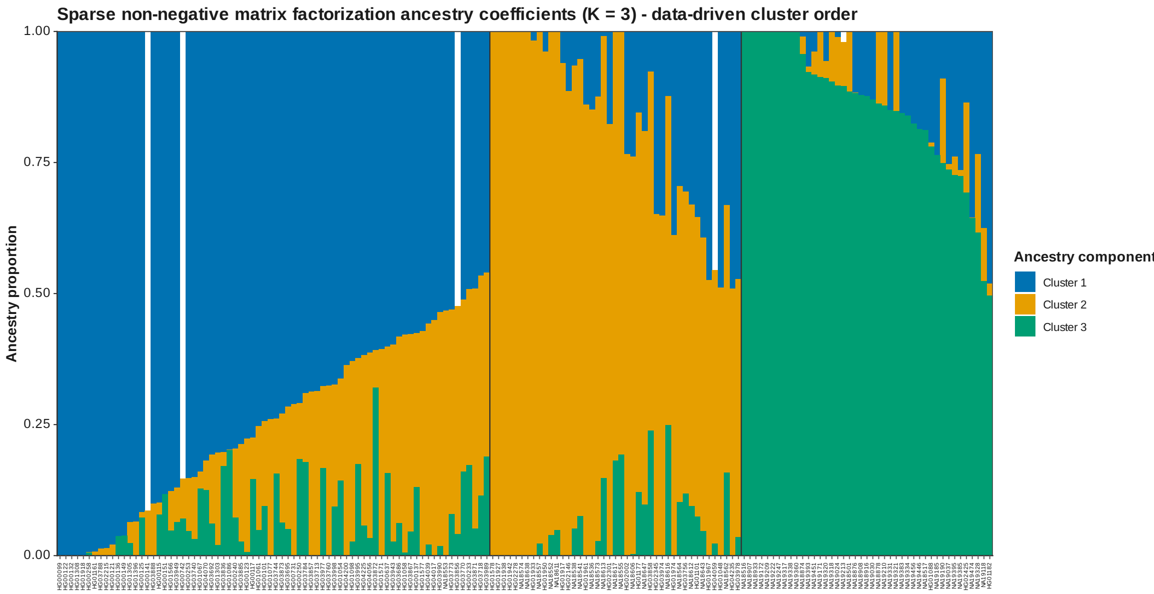

Figure 79: SNMF Q data driven K 3

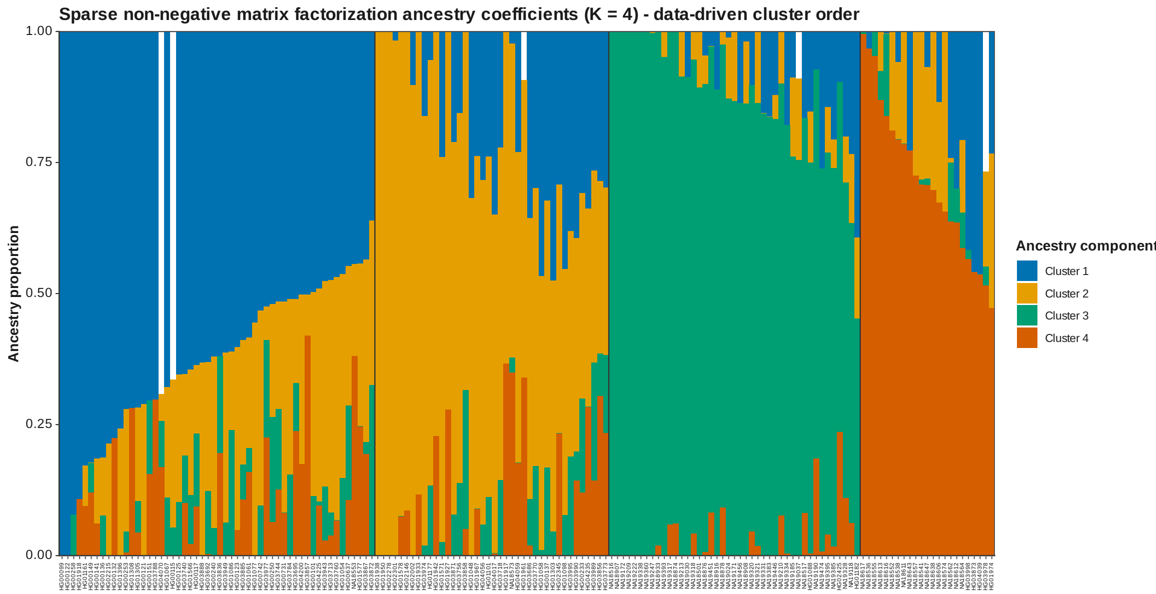

Figure 80: SNMF Q data driven K 4

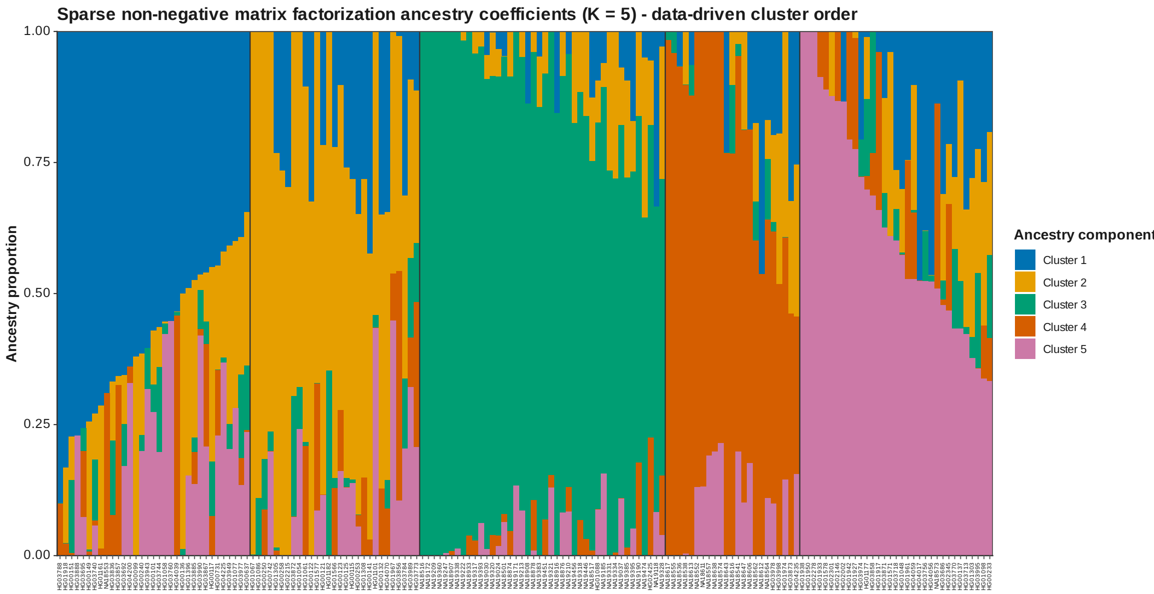

Figure 81: SNMF Q data driven K 5

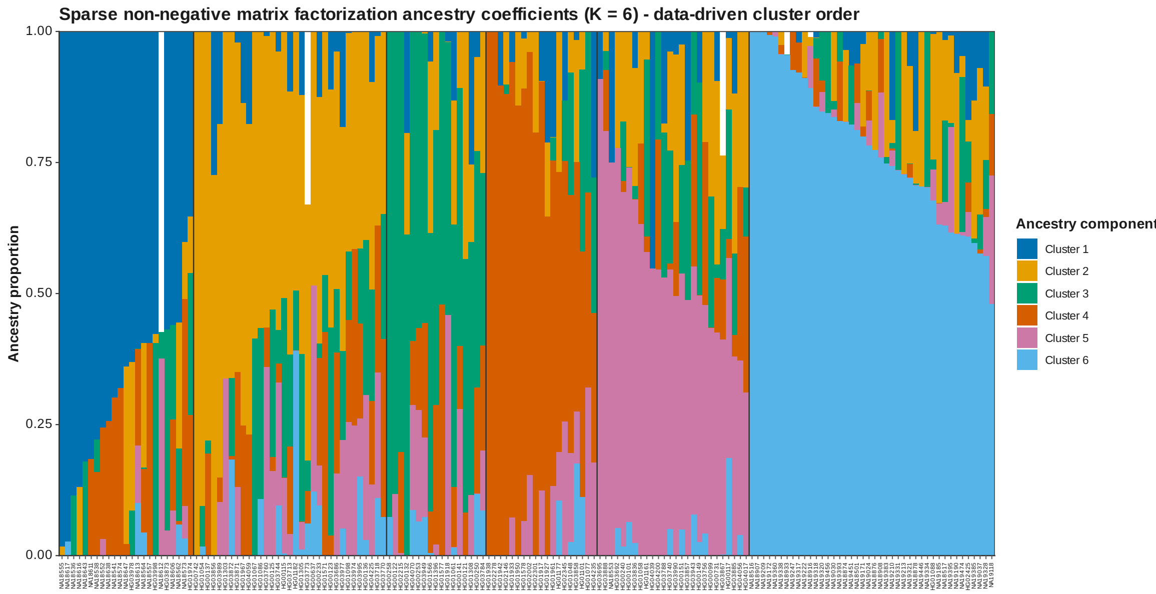

Figure 82: SNMF Q data driven K 6

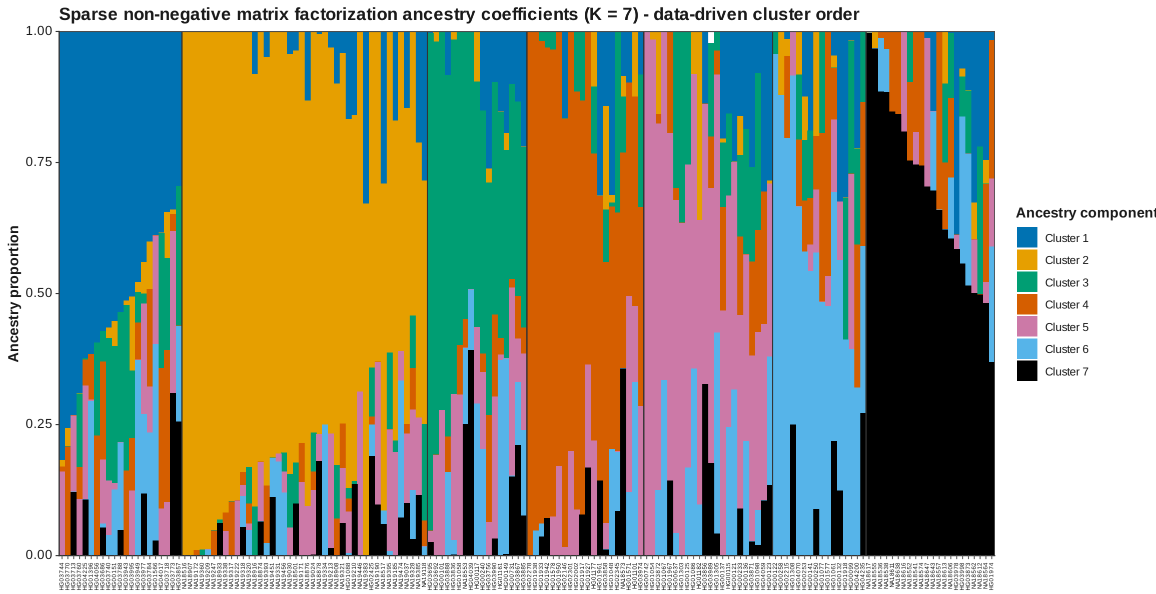

Figure 83: SNMF Q data driven K 7

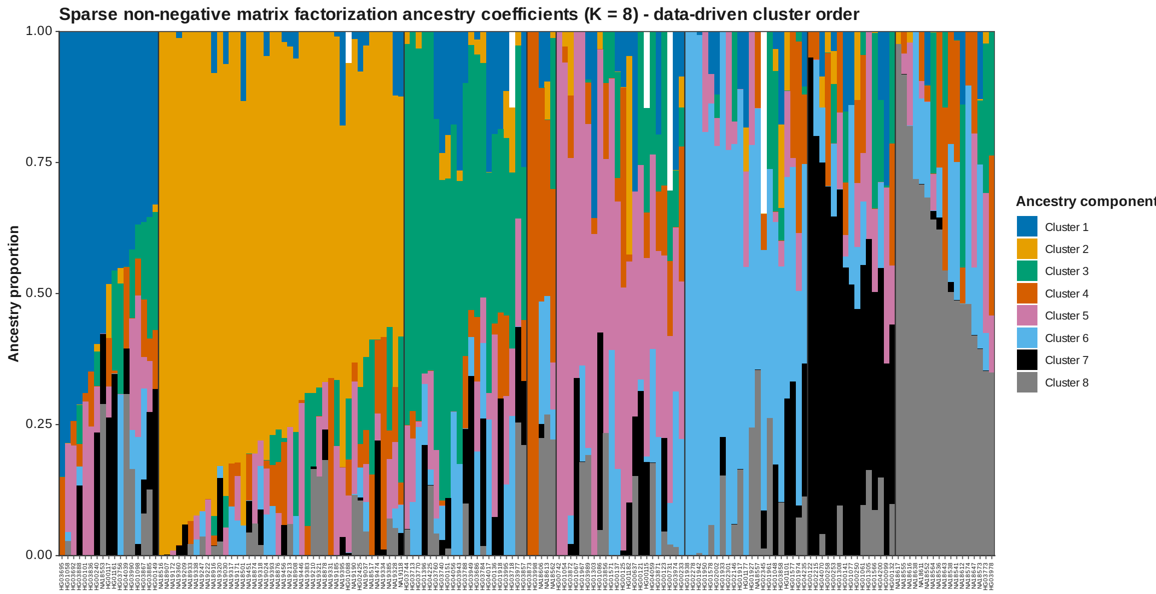

Figure 84: SNMF Q data driven K 8

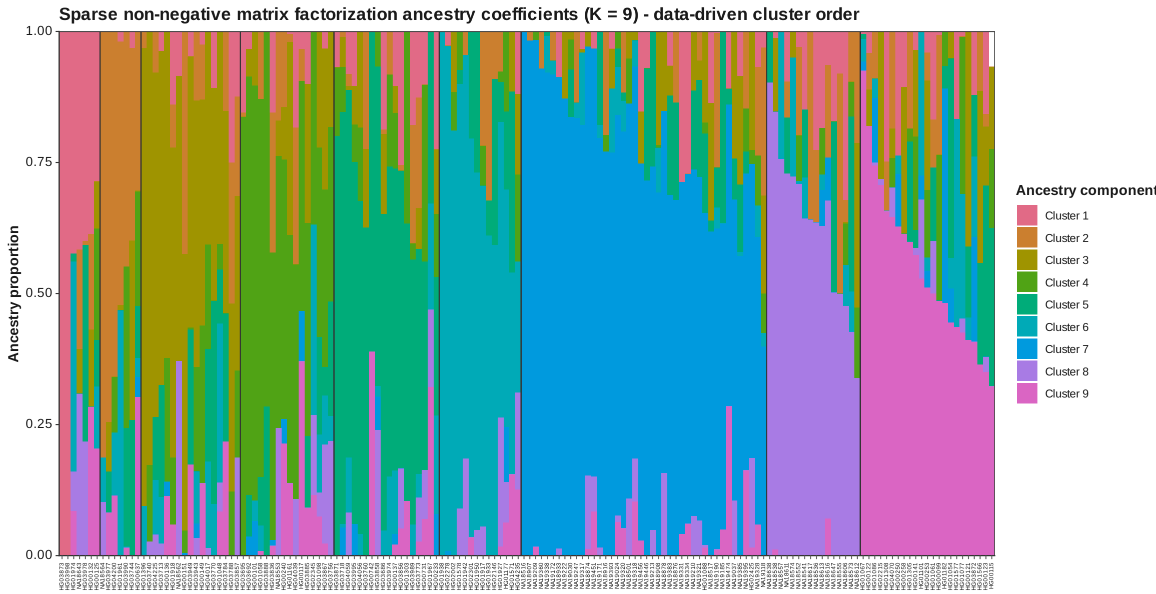

Figure 85: SNMF Q data driven K 9

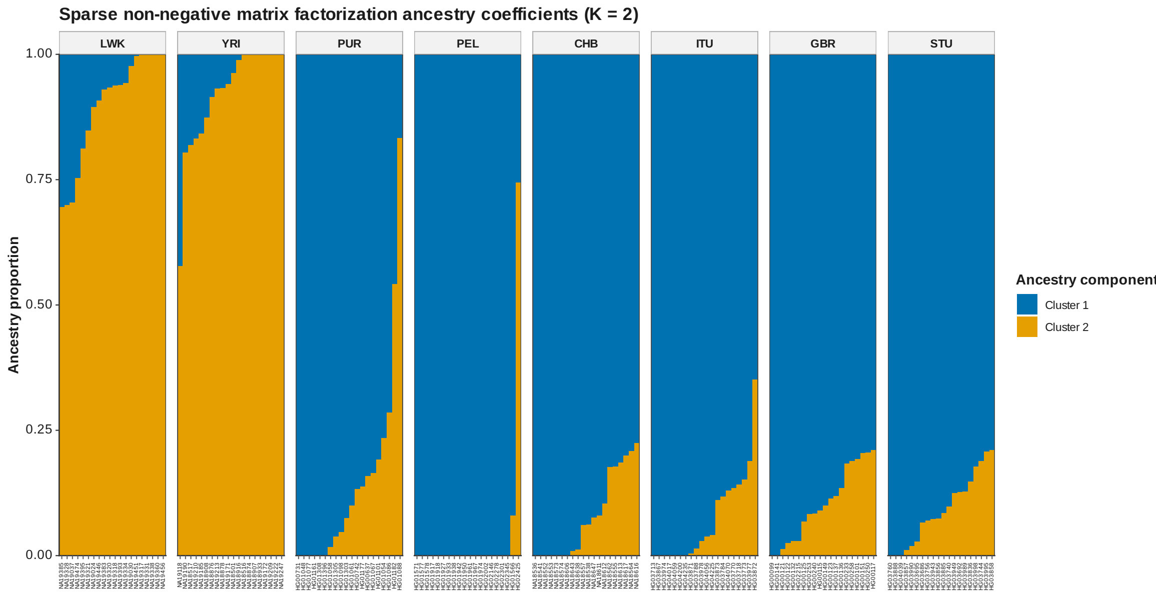

Figure 86: SNMF Q K 2

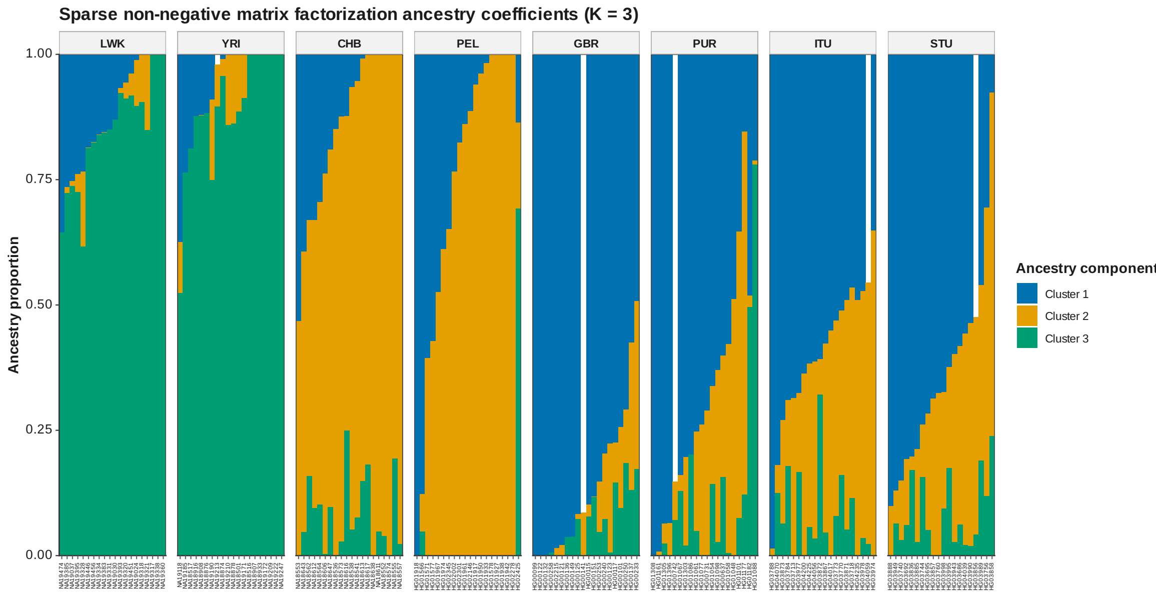

Figure 87: SNMF Q K 3

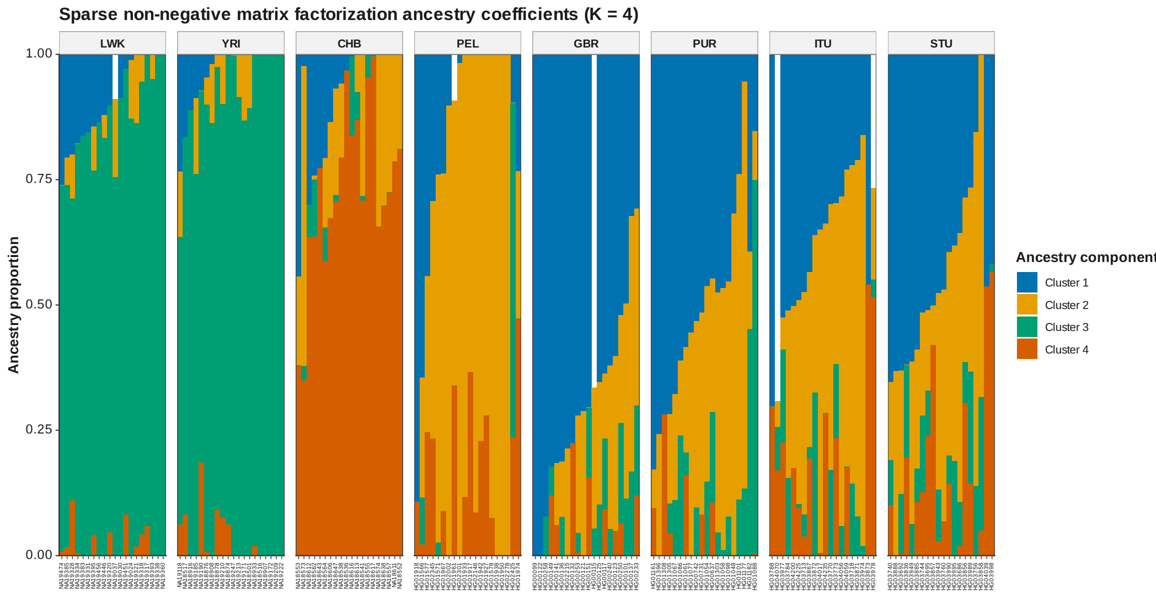

Figure 88: SNMF Q K 4

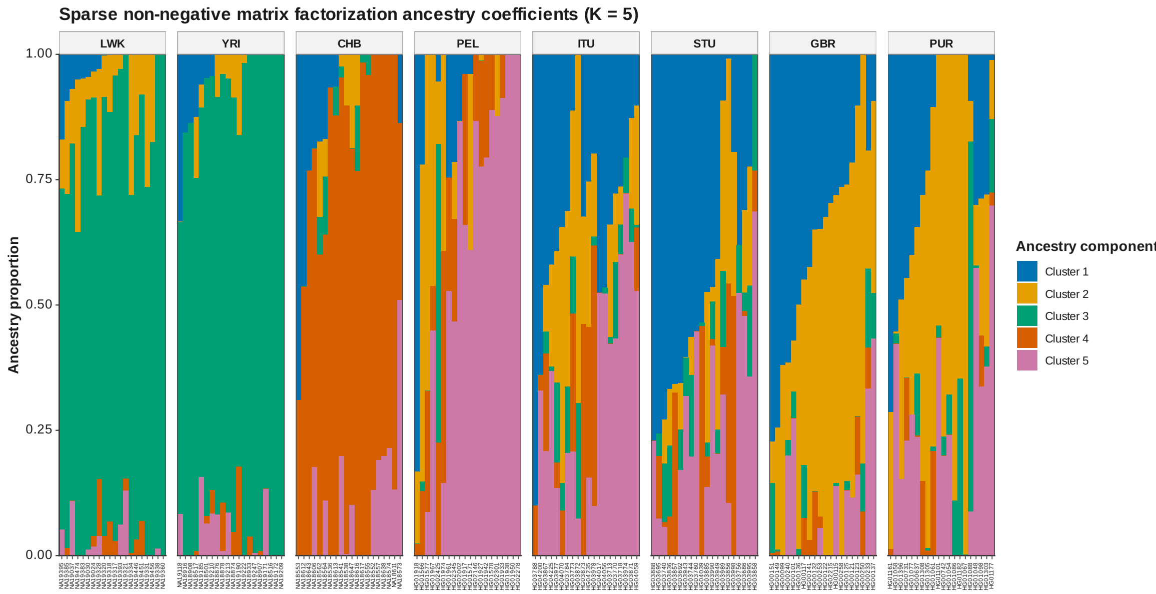

Figure 89: SNMF Q K 5

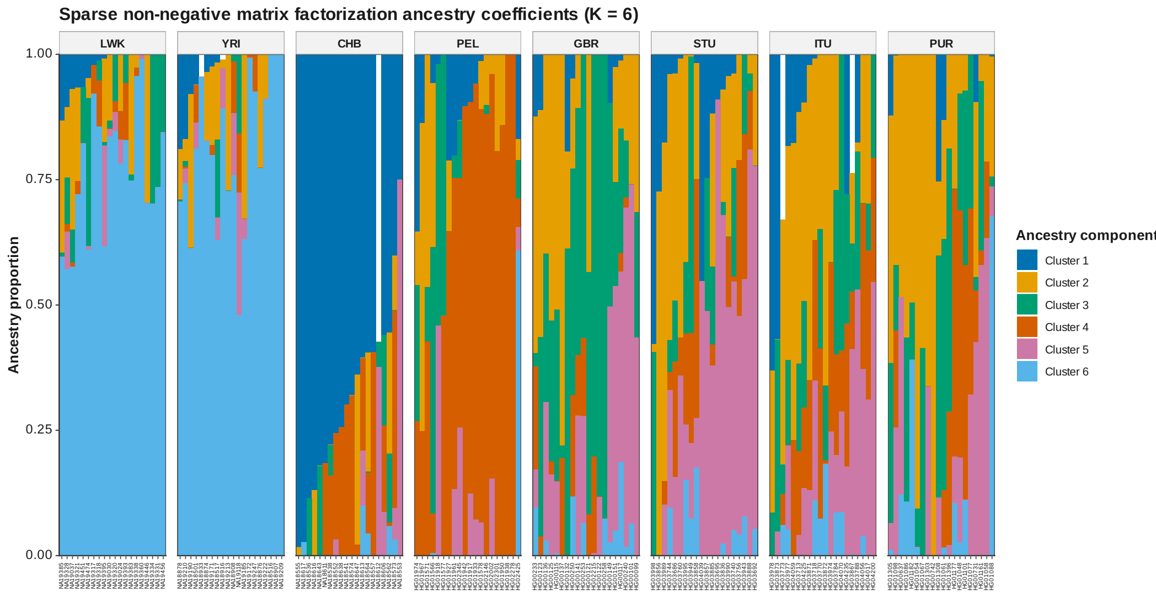

Figure 90: SNMF Q K 6

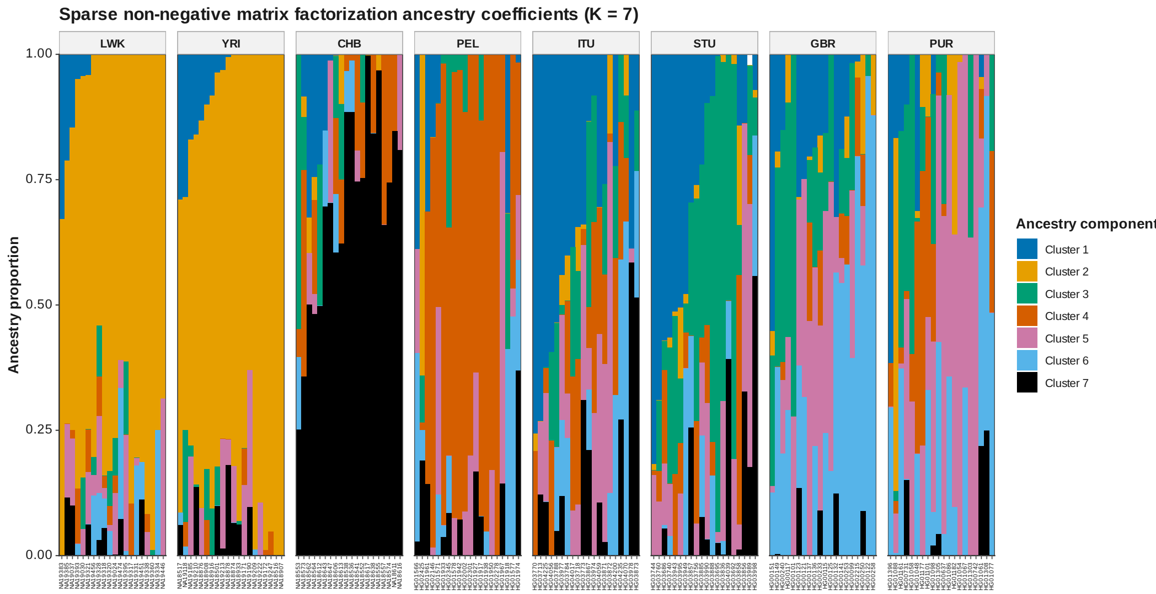

Figure 91: SNMF Q K 7

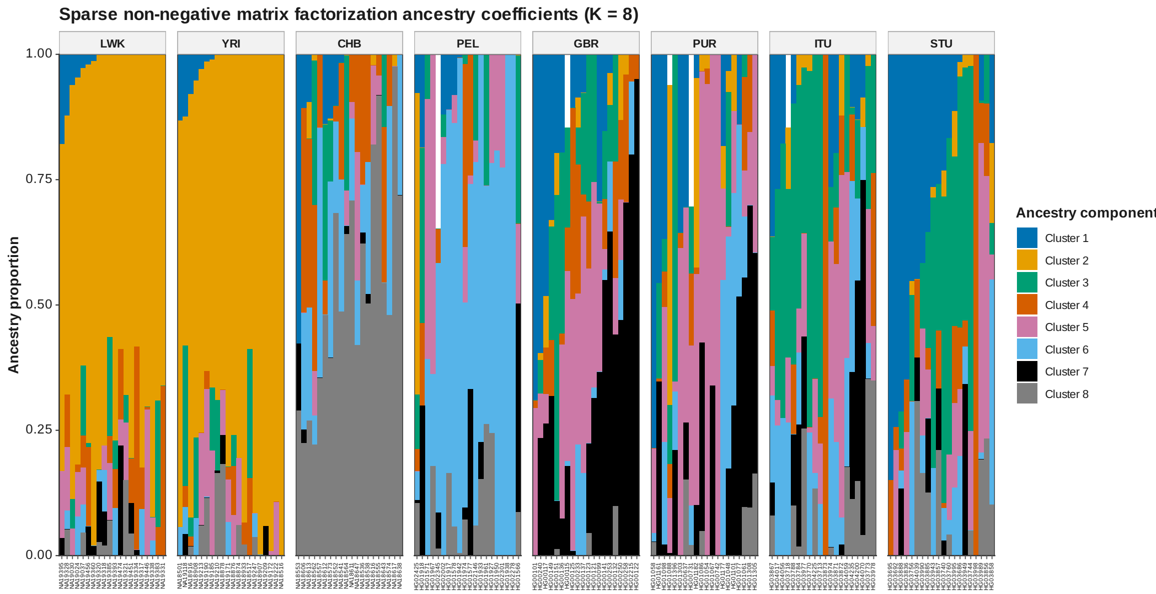

Figure 92: SNMF Q K 8

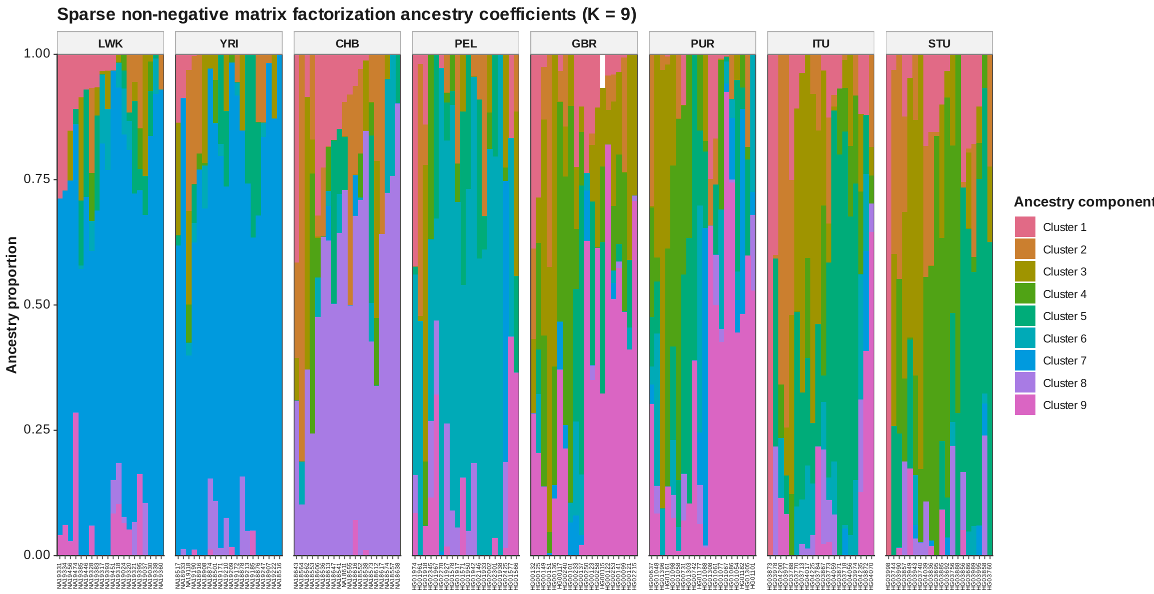

Figure 93: SNMF Q K 9

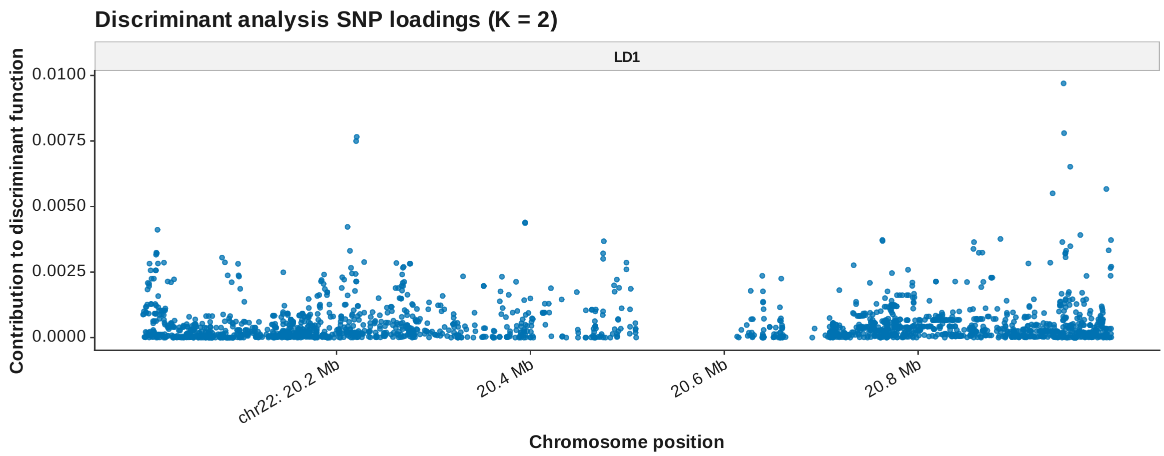

Figure 94: DAPC loadings manhattan K 2

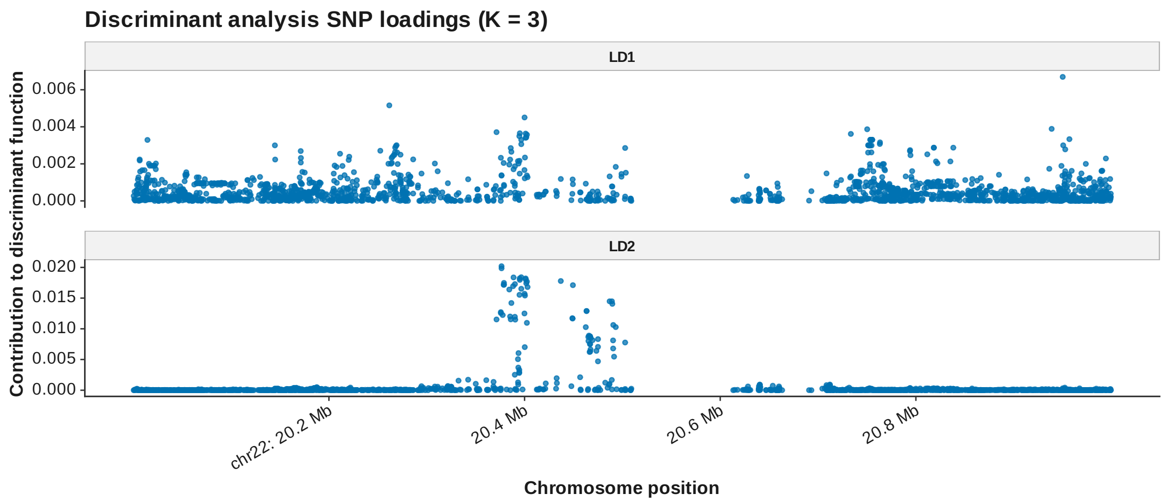

Figure 95: DAPC loadings manhattan K 3

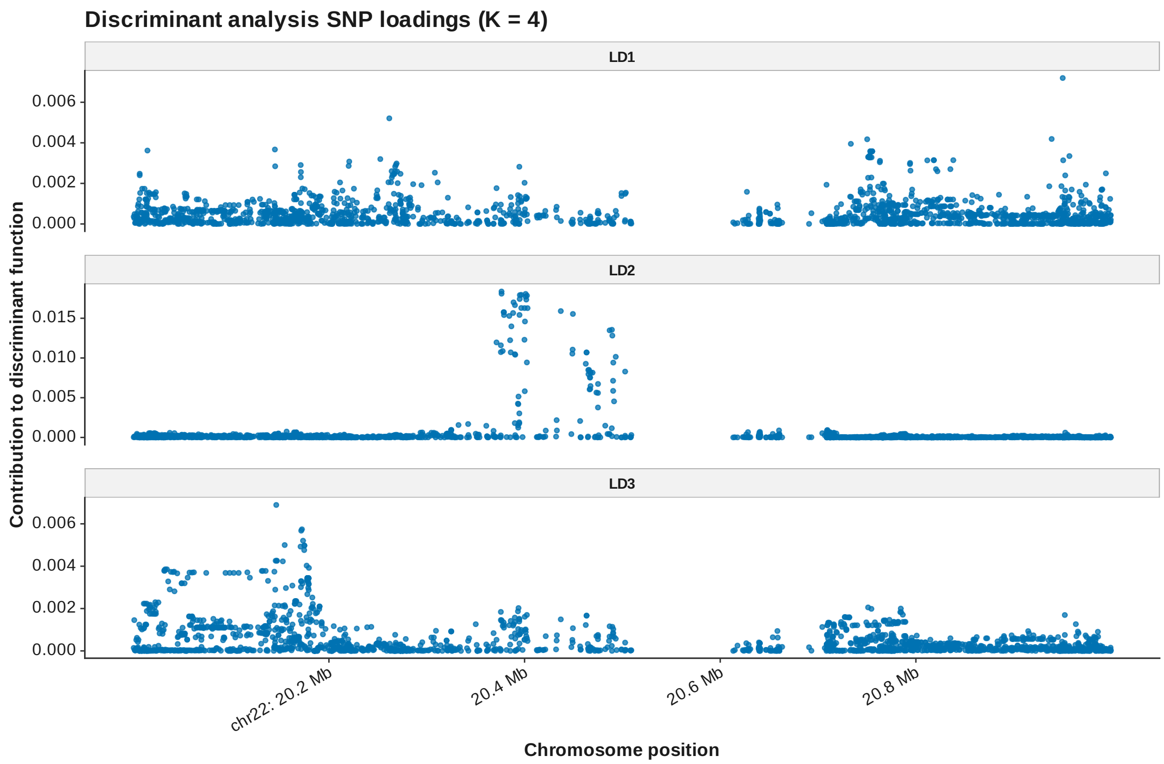

Figure 96: DAPC loadings manhattan K 4

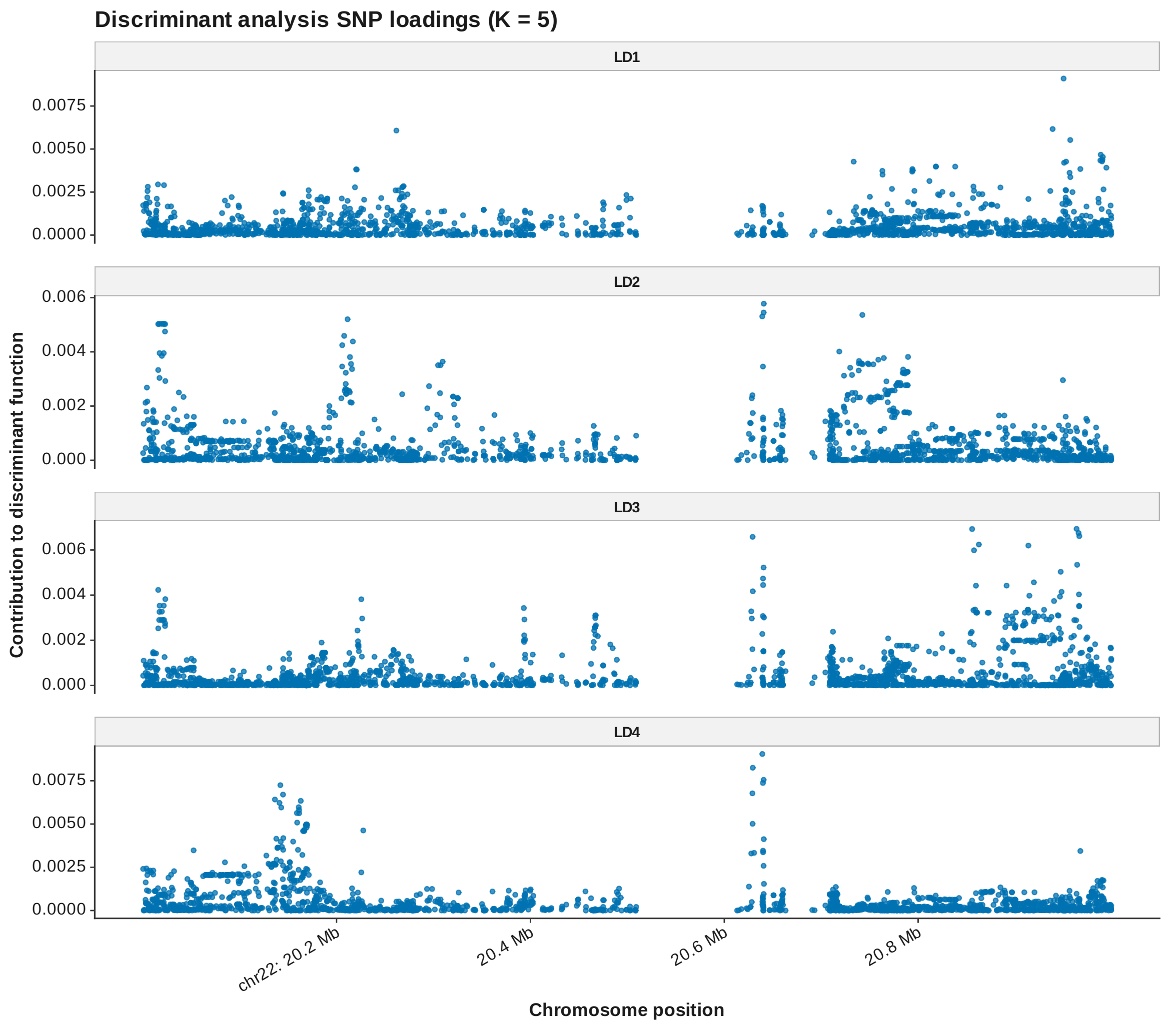

Figure 97: DAPC loadings manhattan K 5

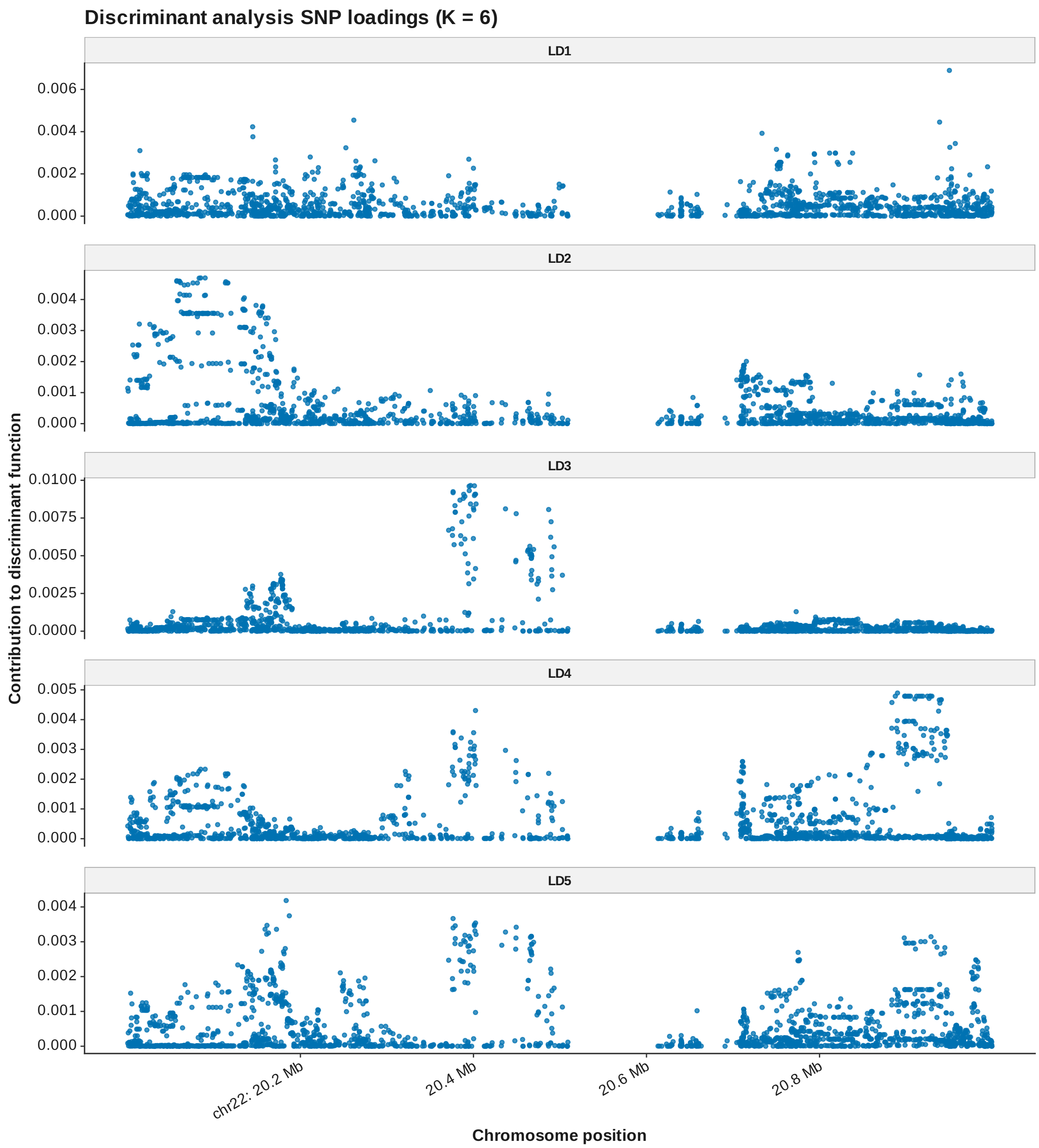

Figure 98: DAPC loadings manhattan K 6

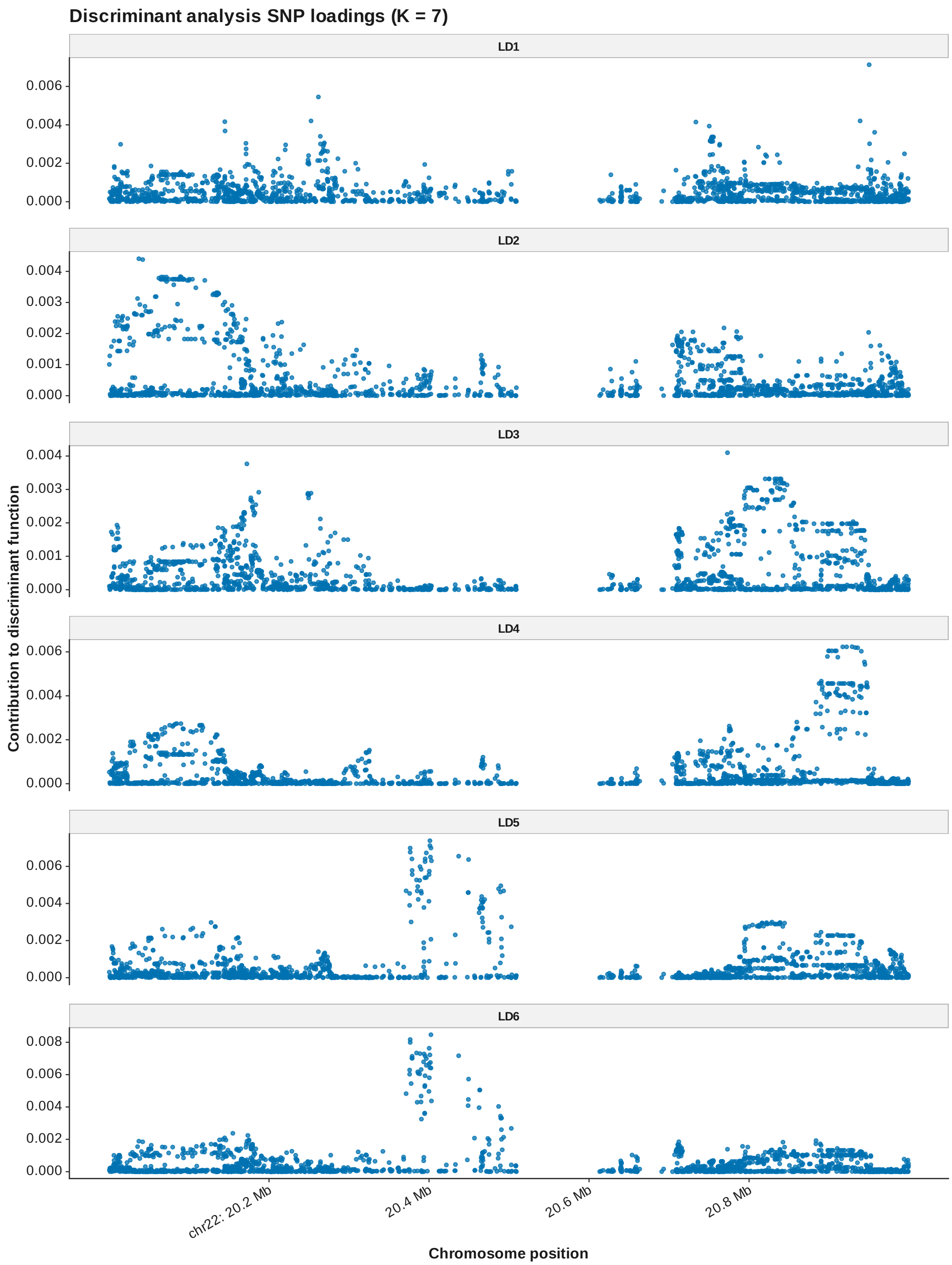

Figure 99: DAPC loadings manhattan K 7

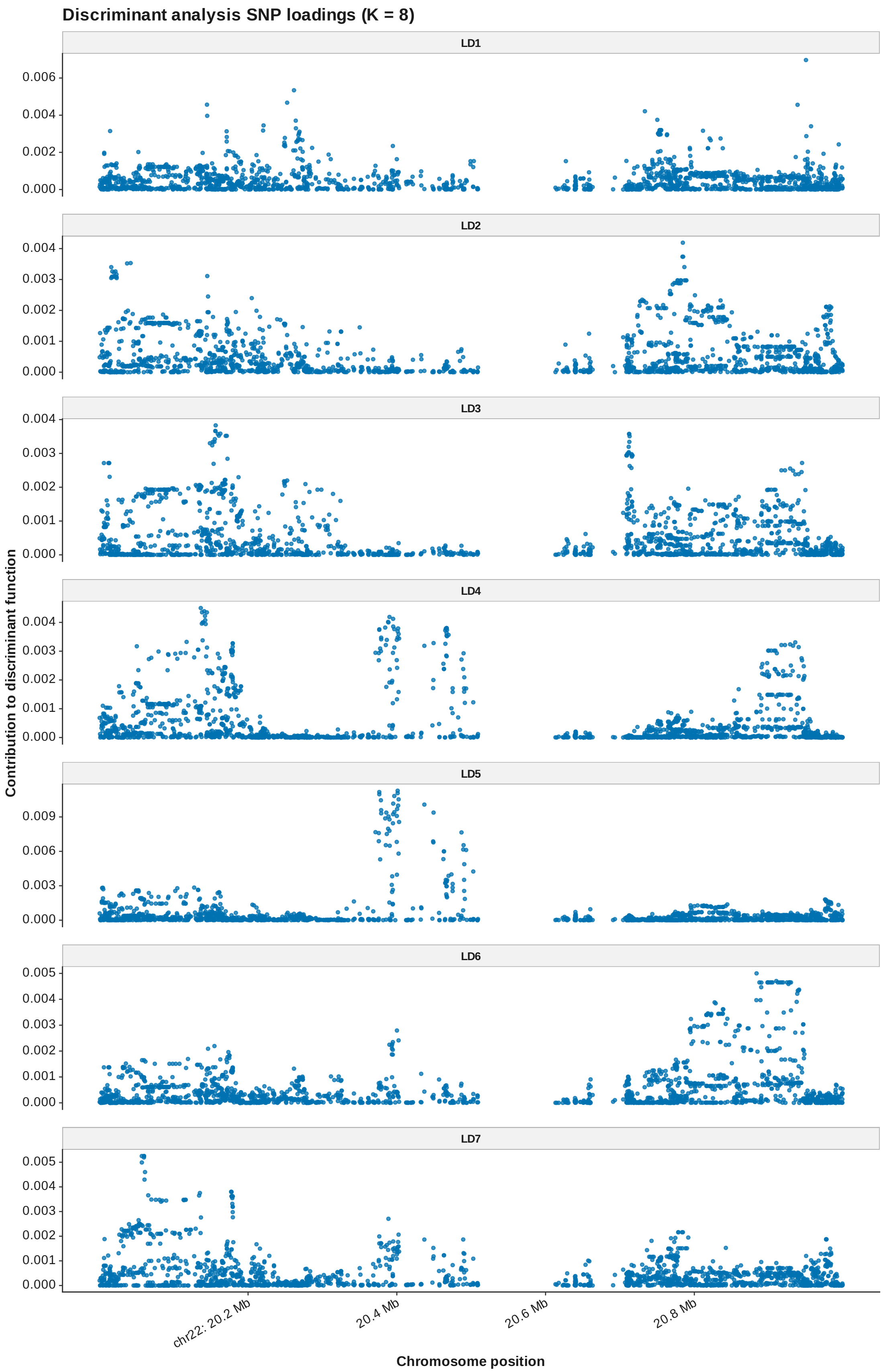

Figure 100: DAPC loadings manhattan K 8

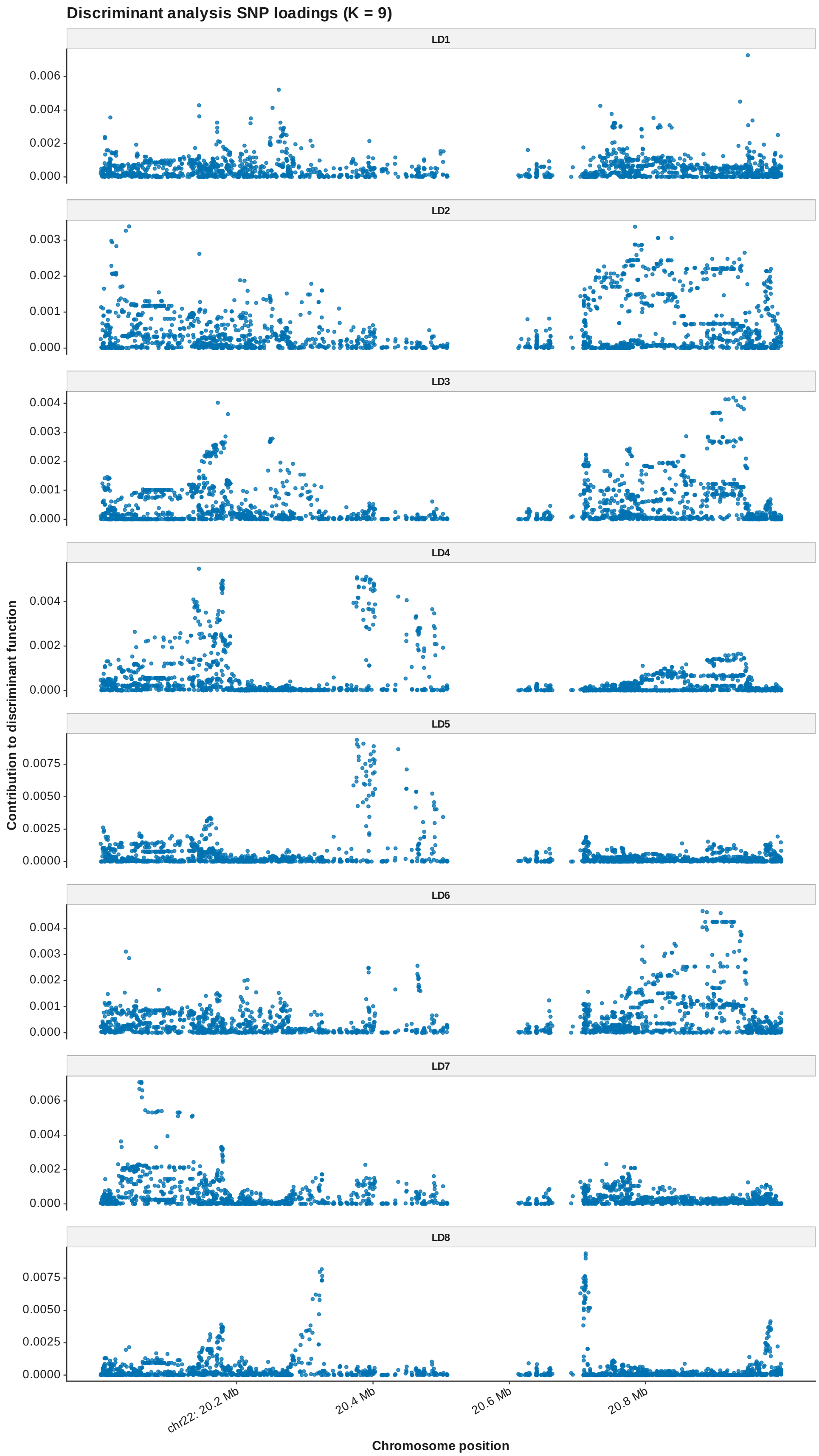

Figure 101: DAPC loadings manhattan K 9

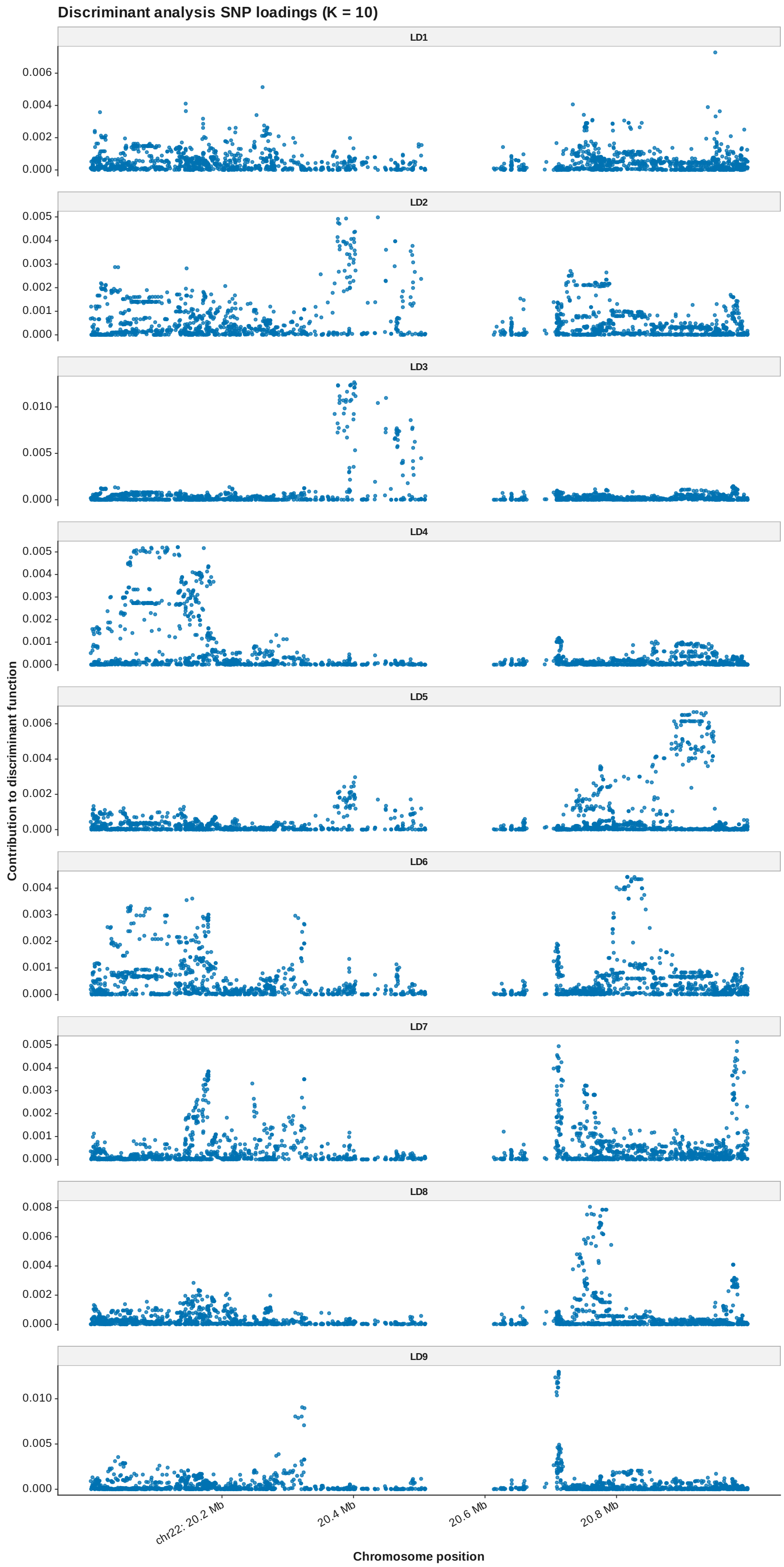

Figure 102: DAPC loadings manhattan K 10

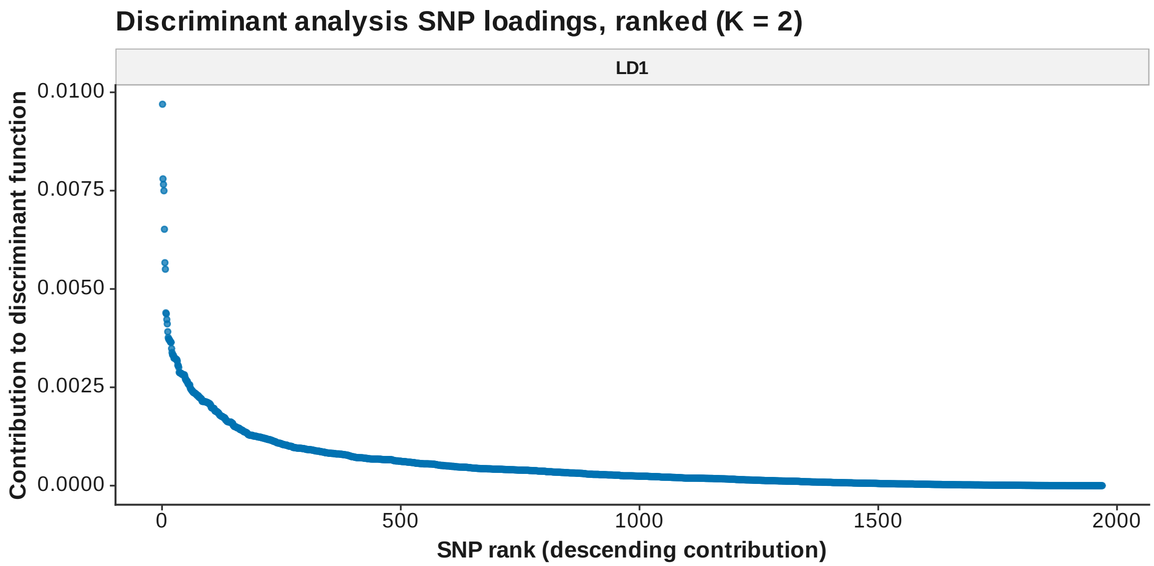

Figure 103: DAPC loadings ranked K 2

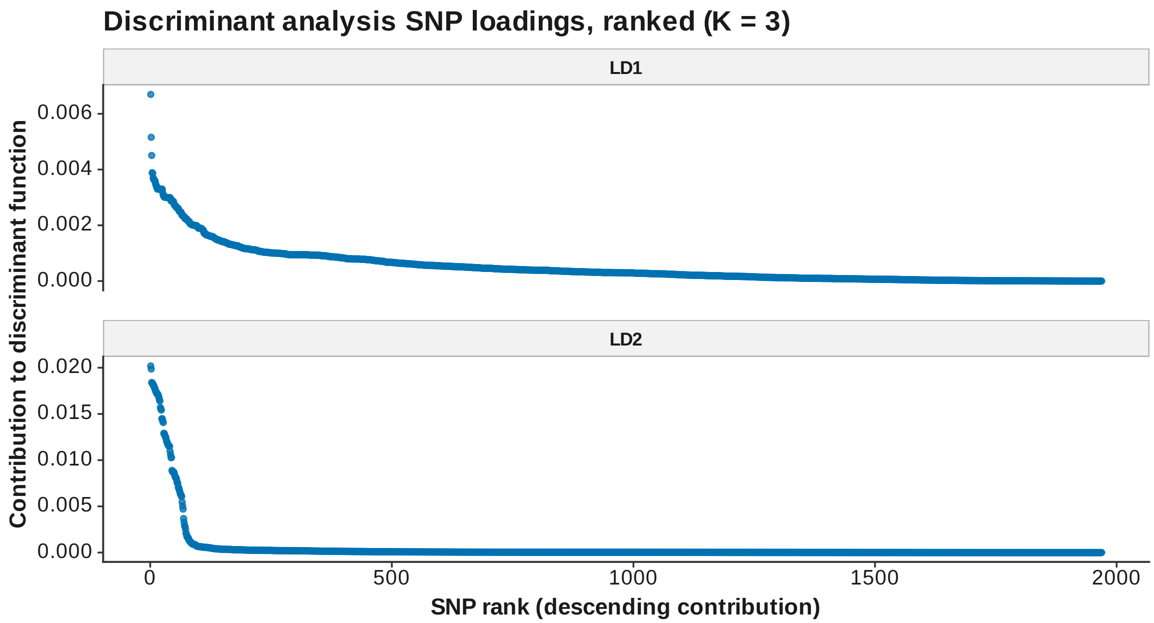

Figure 104: DAPC loadings ranked K 3

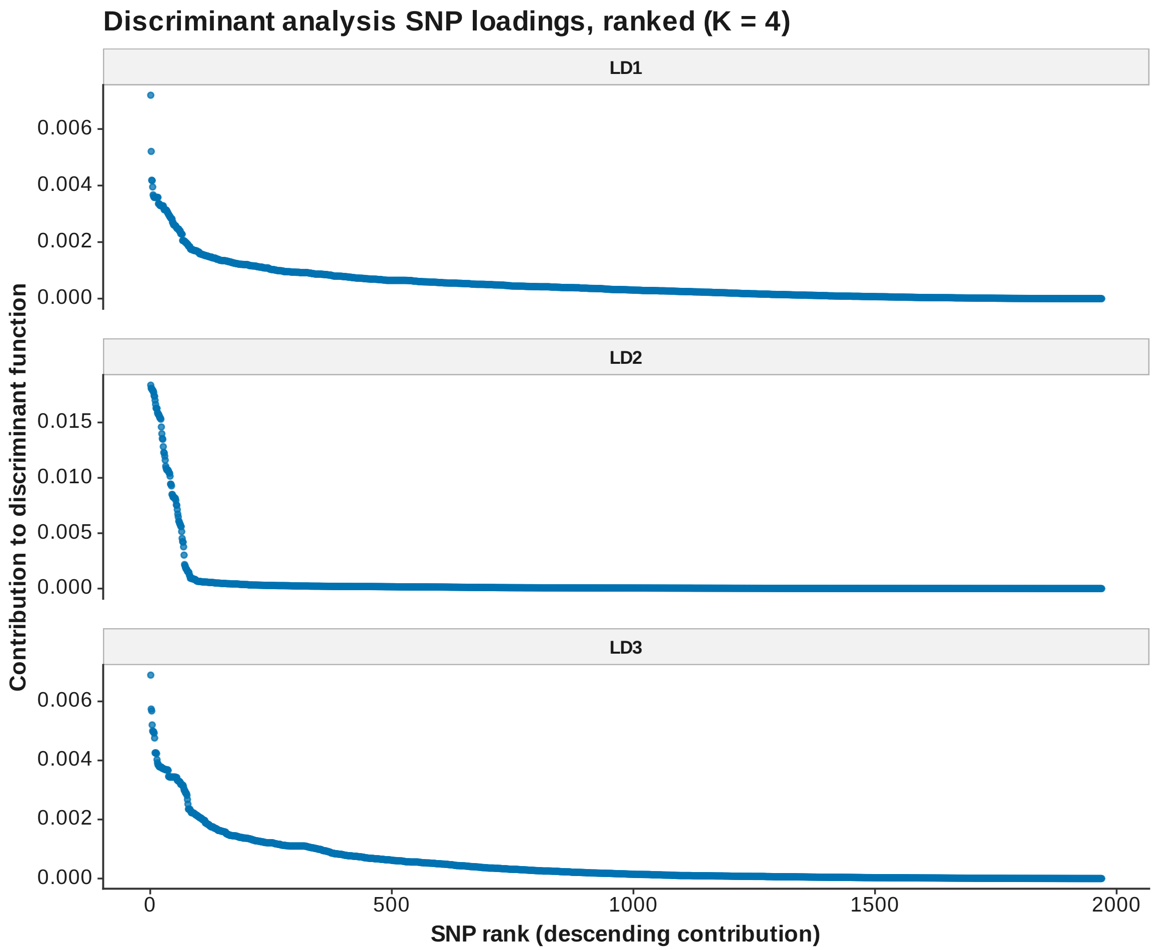

Figure 105: DAPC loadings ranked K 4

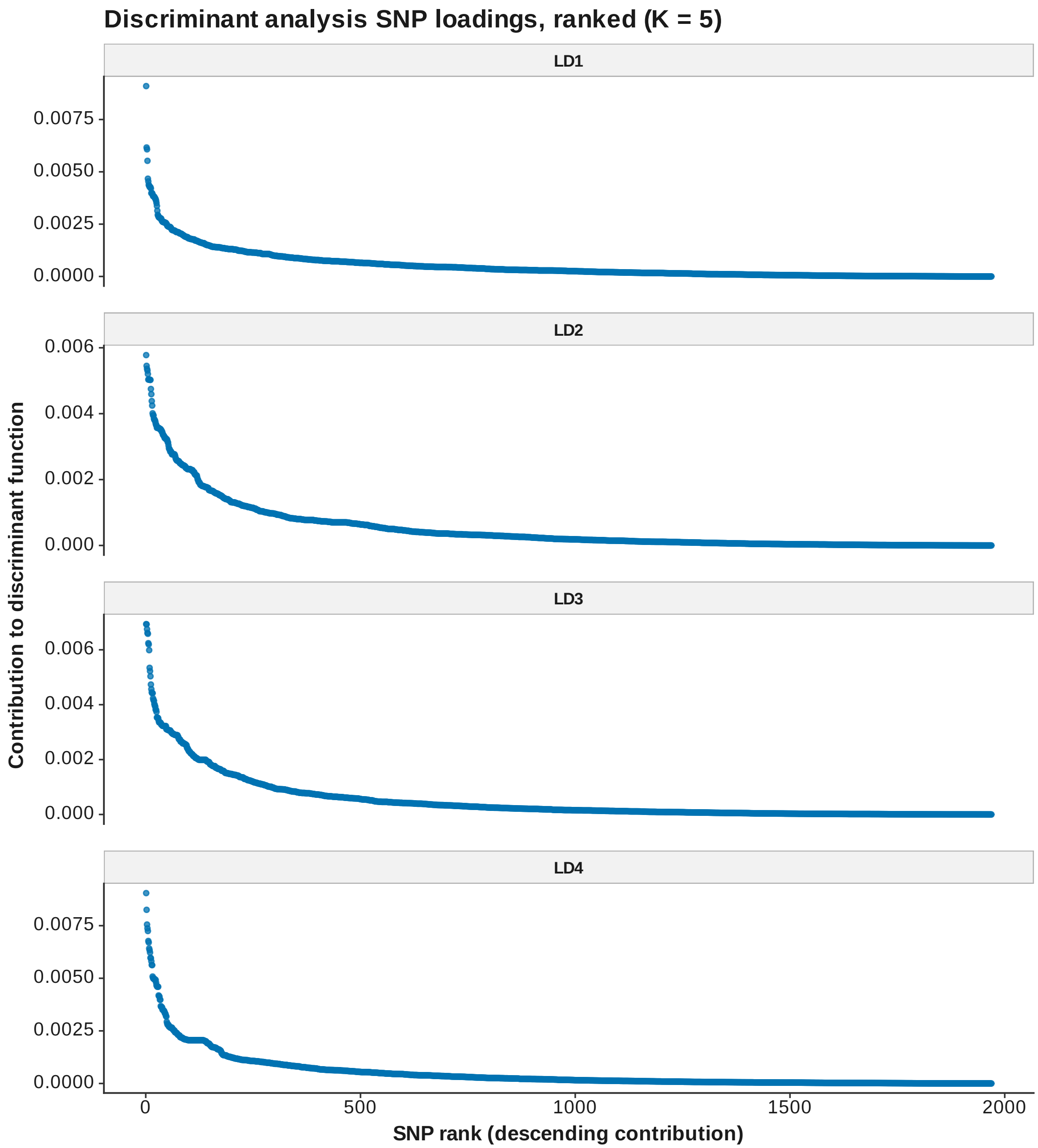

Figure 106: DAPC loadings ranked K 5

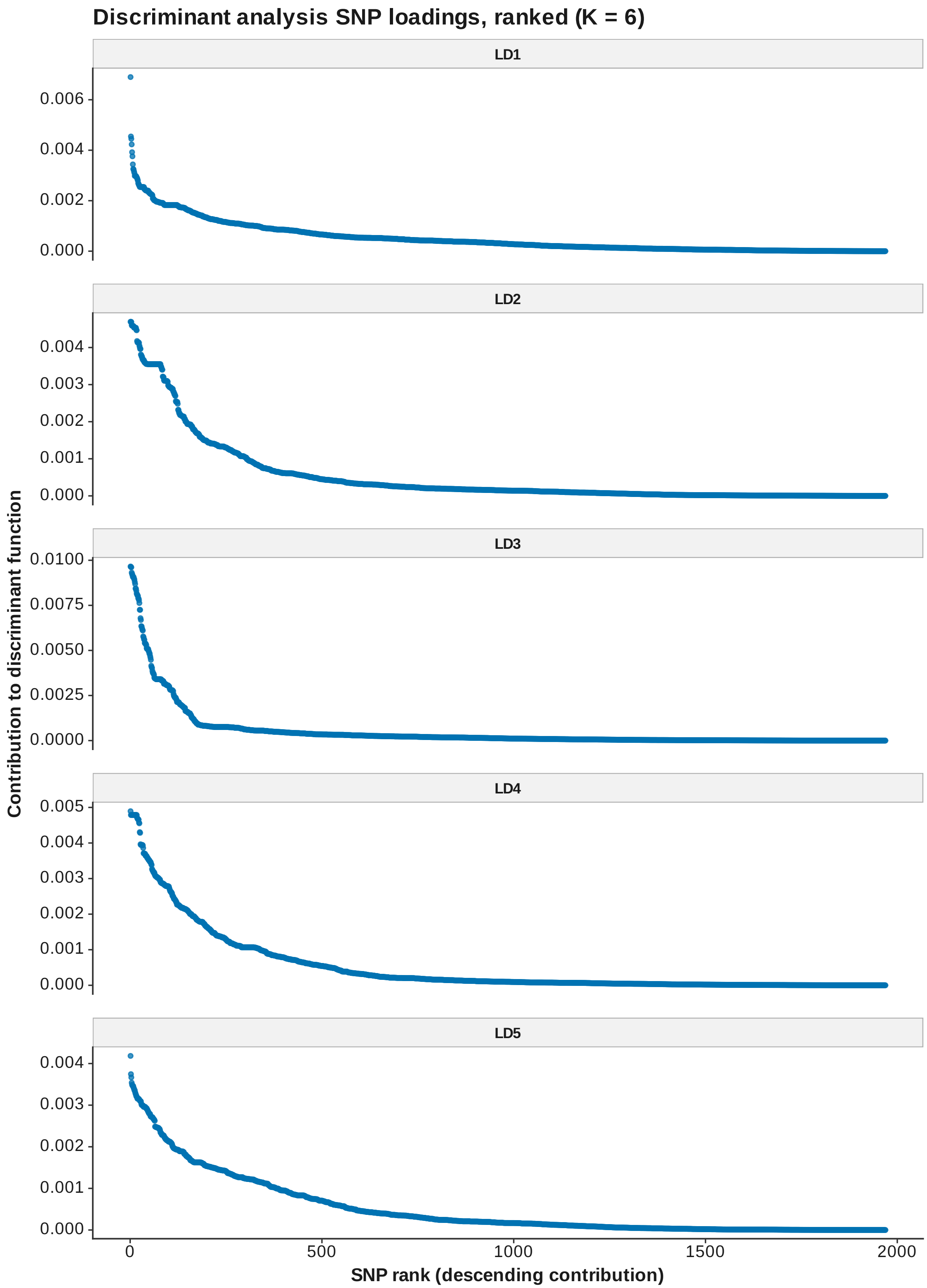

Figure 107: DAPC loadings ranked K 6

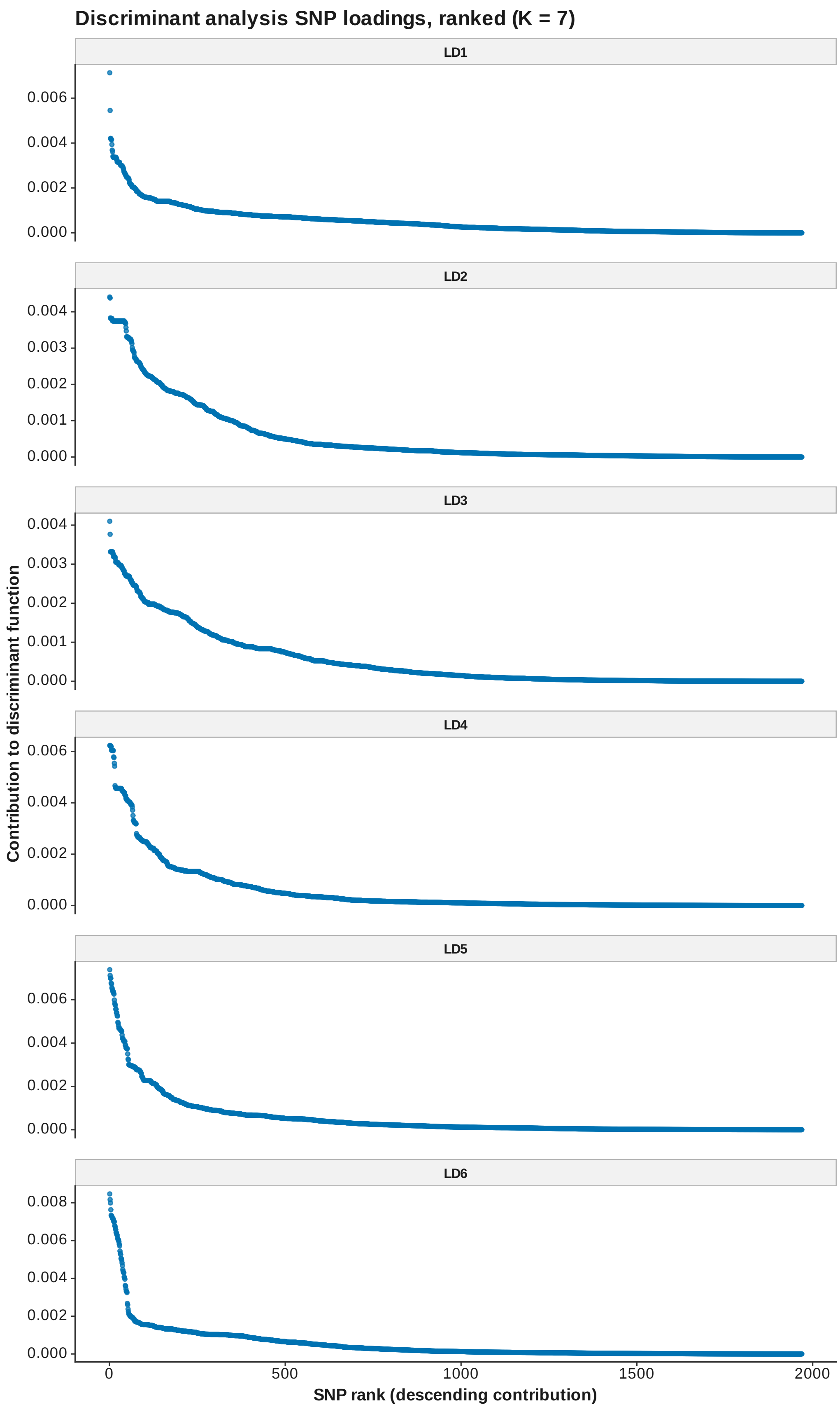

Figure 108: DAPC loadings ranked K 7

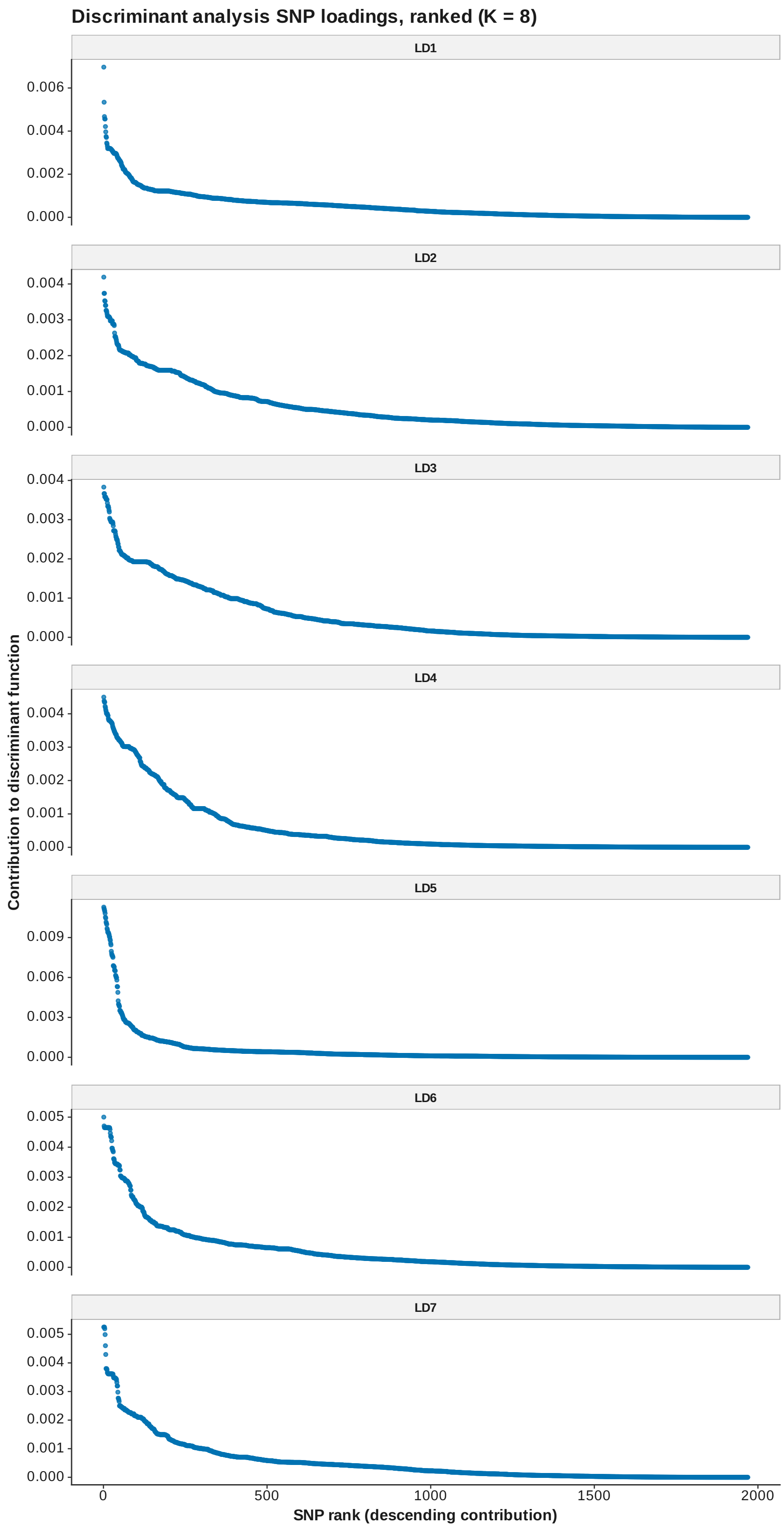

Figure 109: DAPC loadings ranked K 8

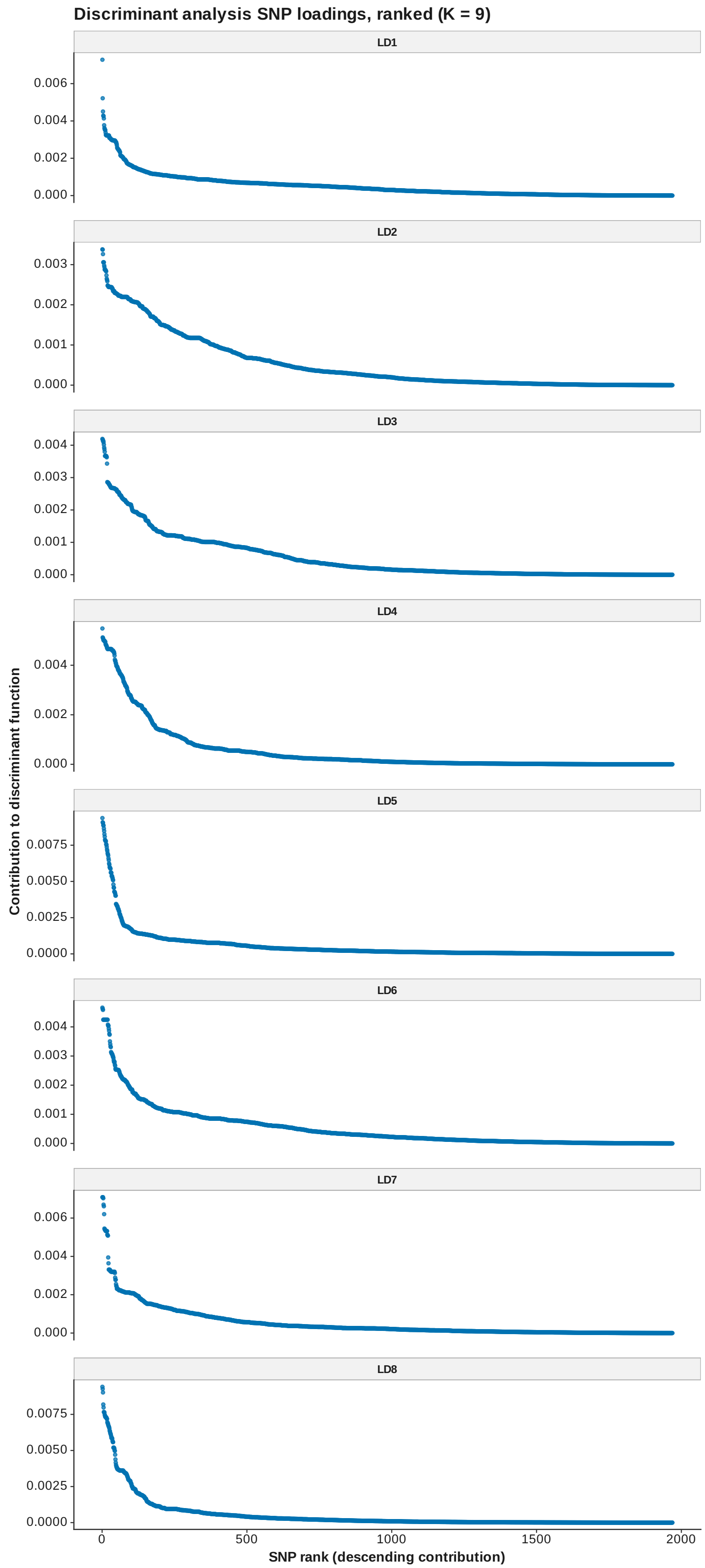

Figure 110: DAPC loadings ranked K 9

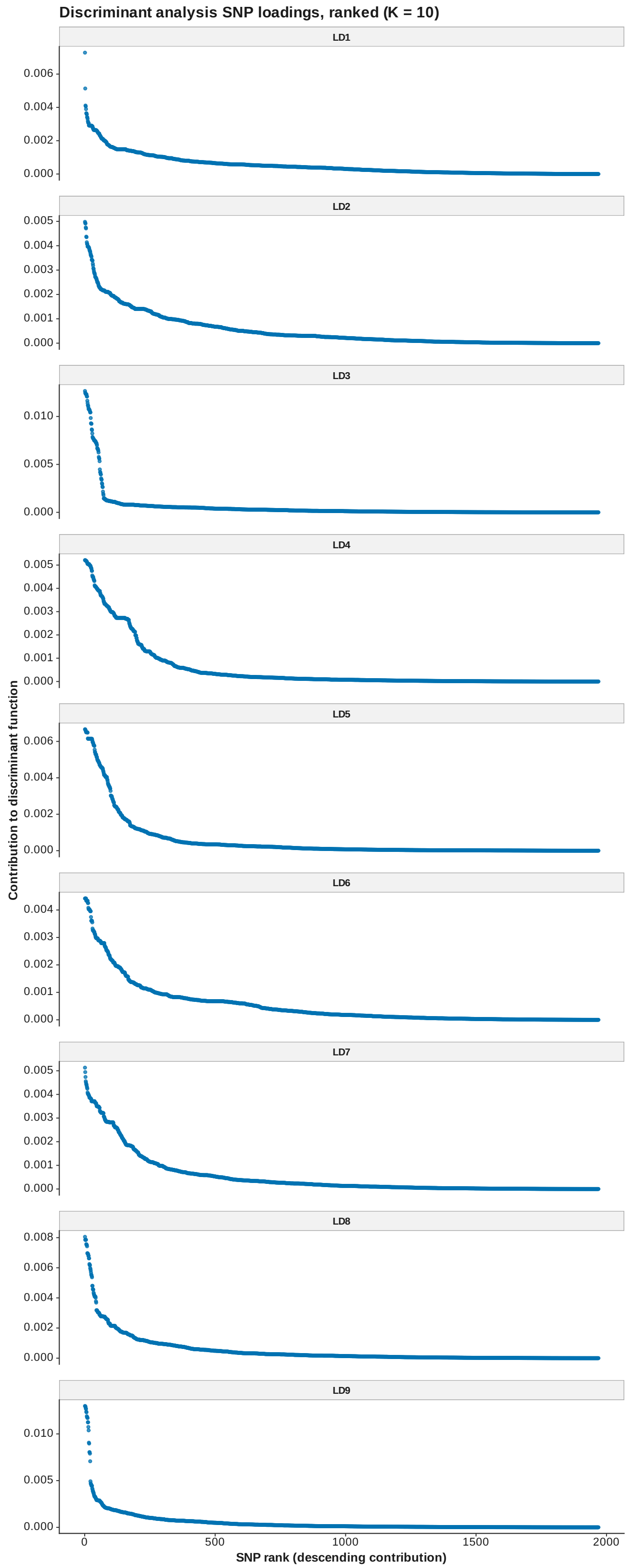

Figure 111: DAPC loadings ranked K 10

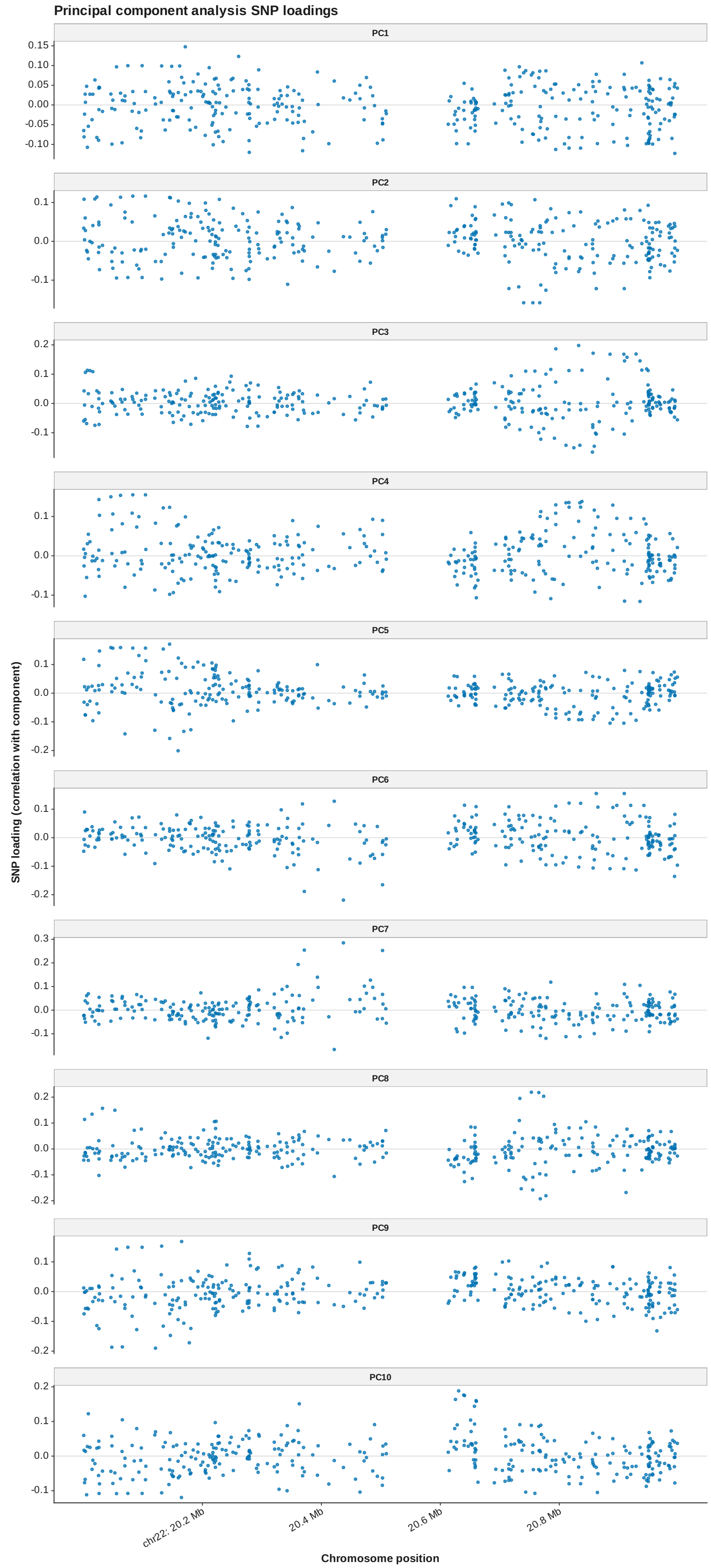

Figure 112: PCA loadings manhattan

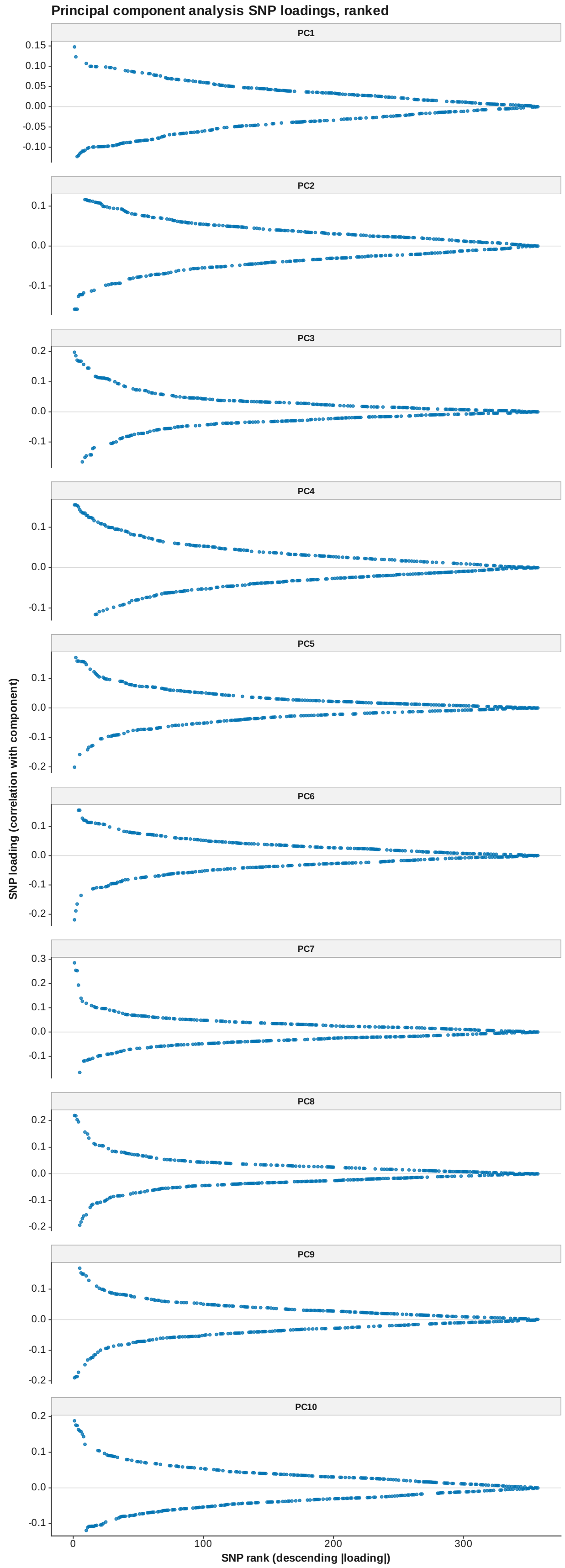

Figure 113: PCA loadings ranked

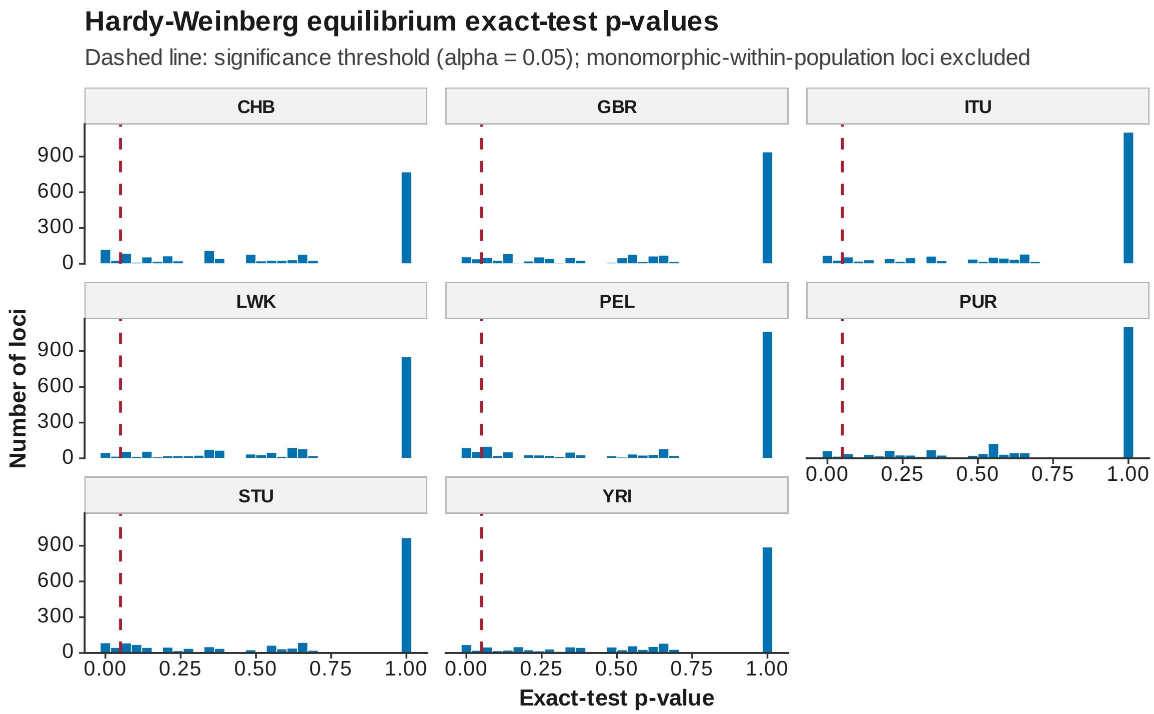

Figure 114: HWE pvalues

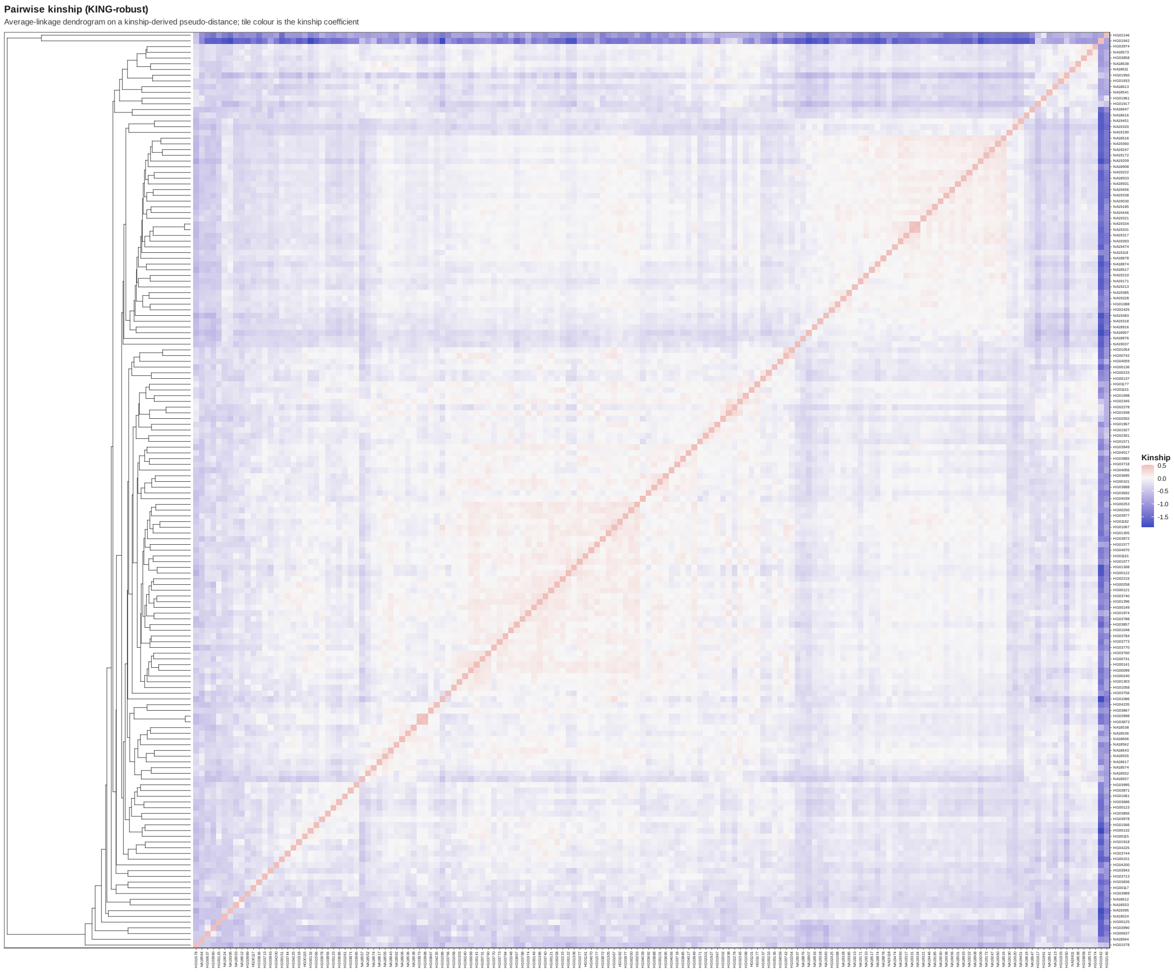

Figure 115: Kinship heatmap

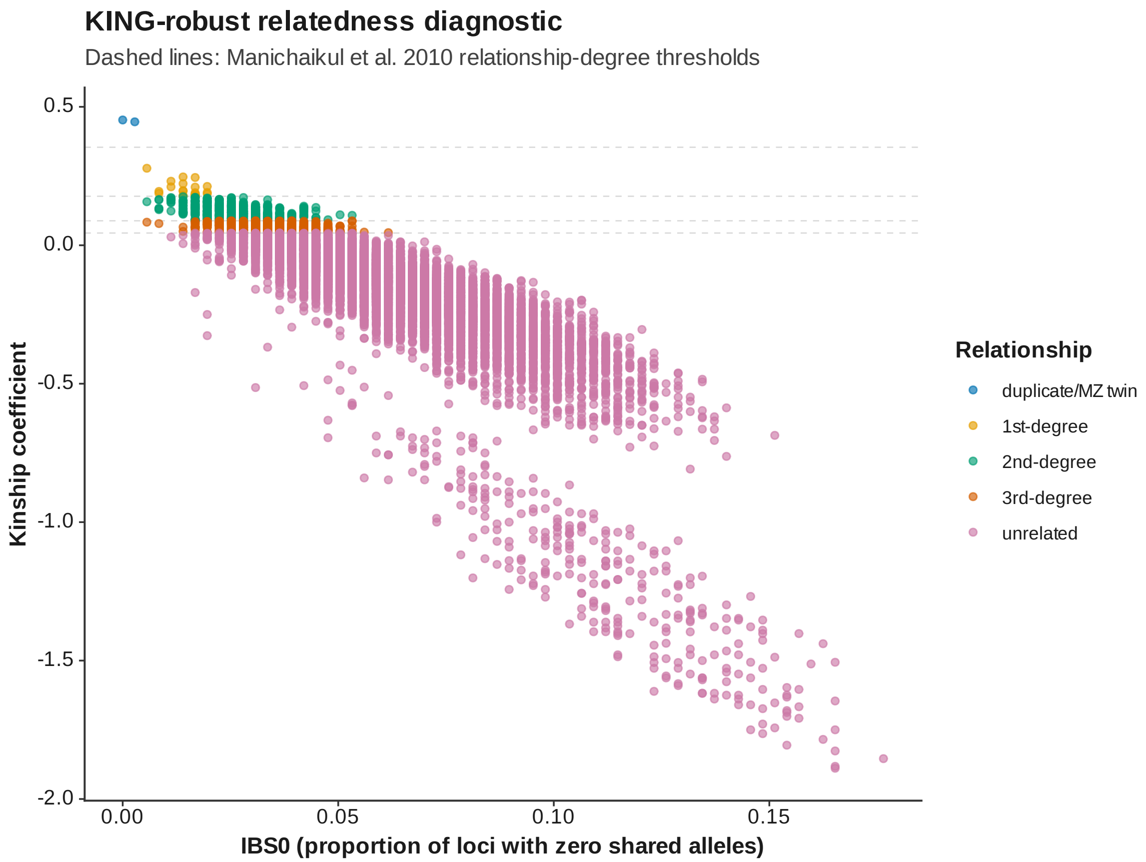

Figure 116: Kinship IBS 0 vs kinship

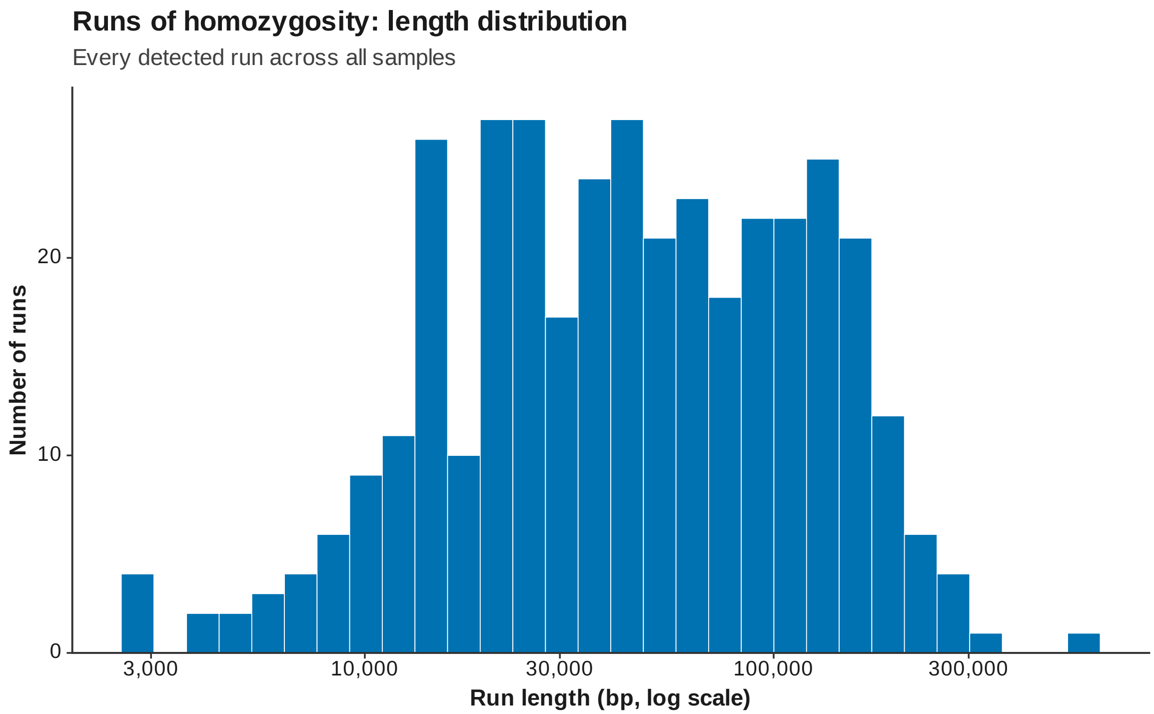

Figure 117: ROH length distribution

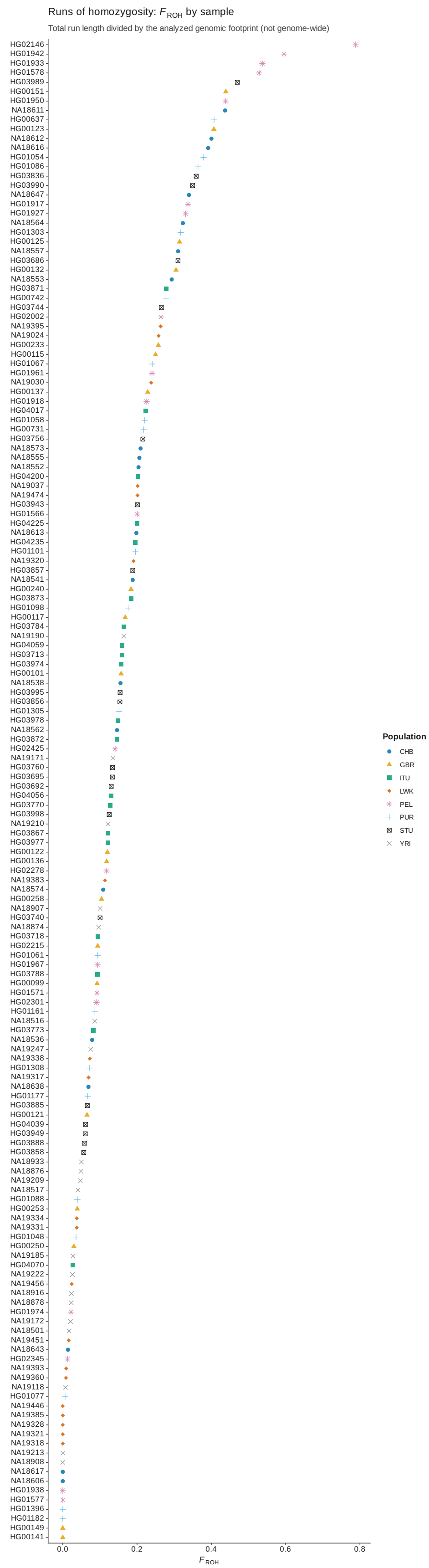

Figure 118: ROH FROH by sample

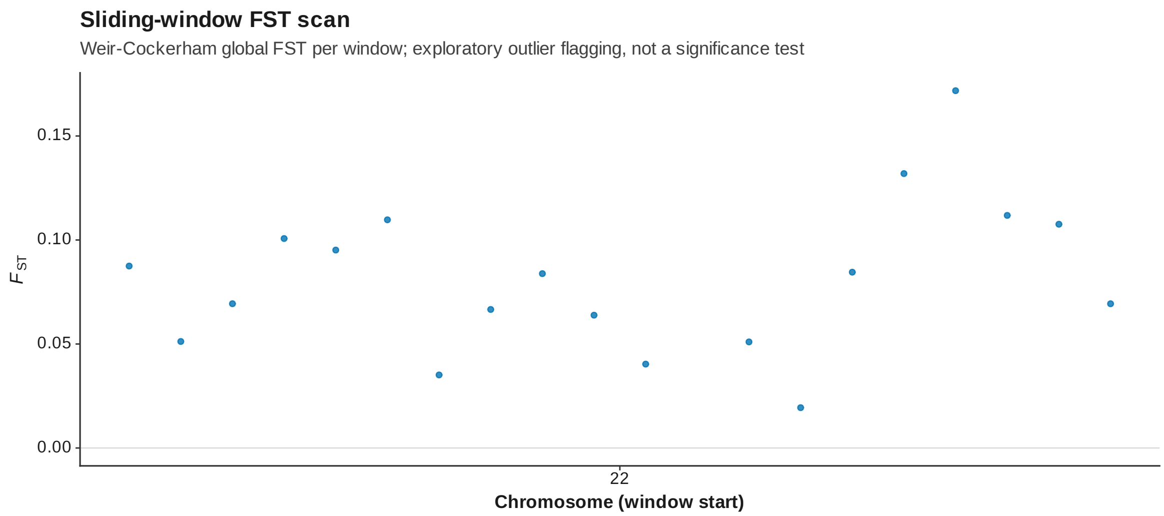

Figure 119: Genome scan FST manhattan

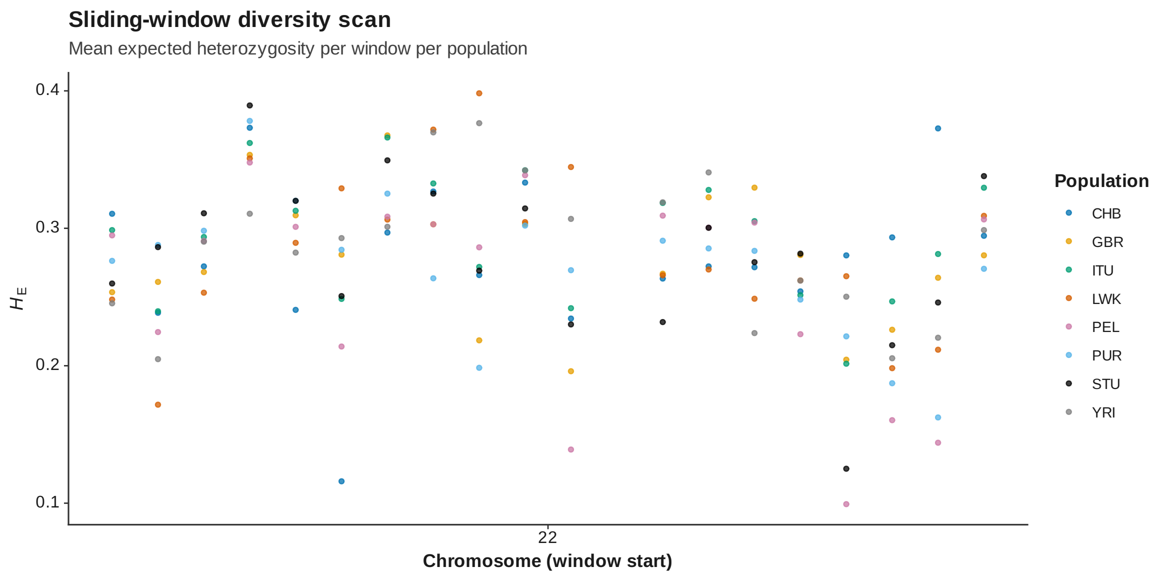

Figure 120: Genome scan diversity manhattan

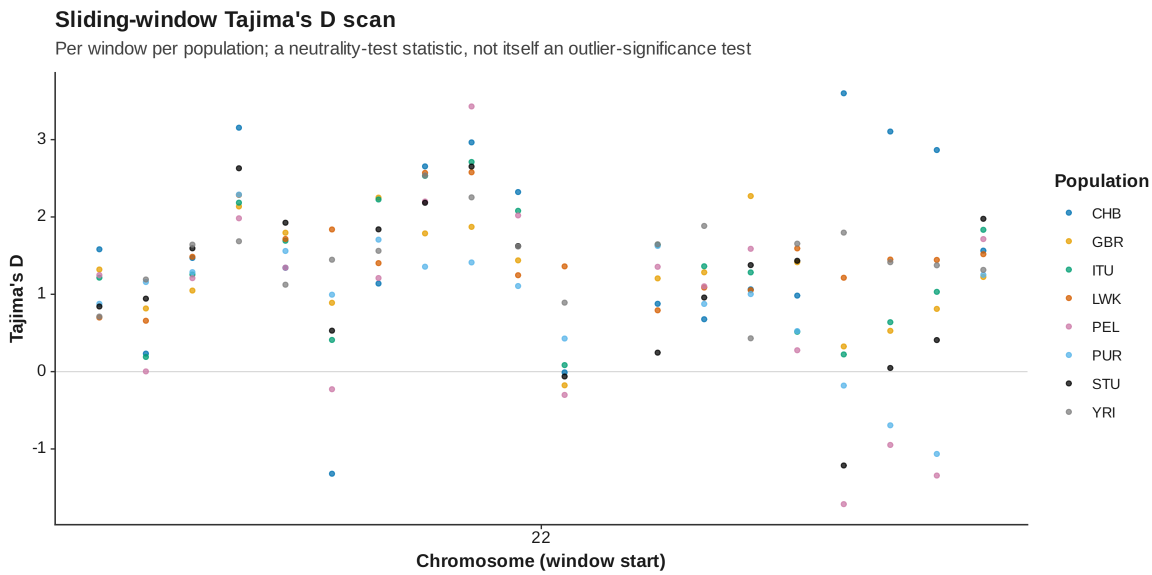

Figure 121: Genome scan tajima d manhattan

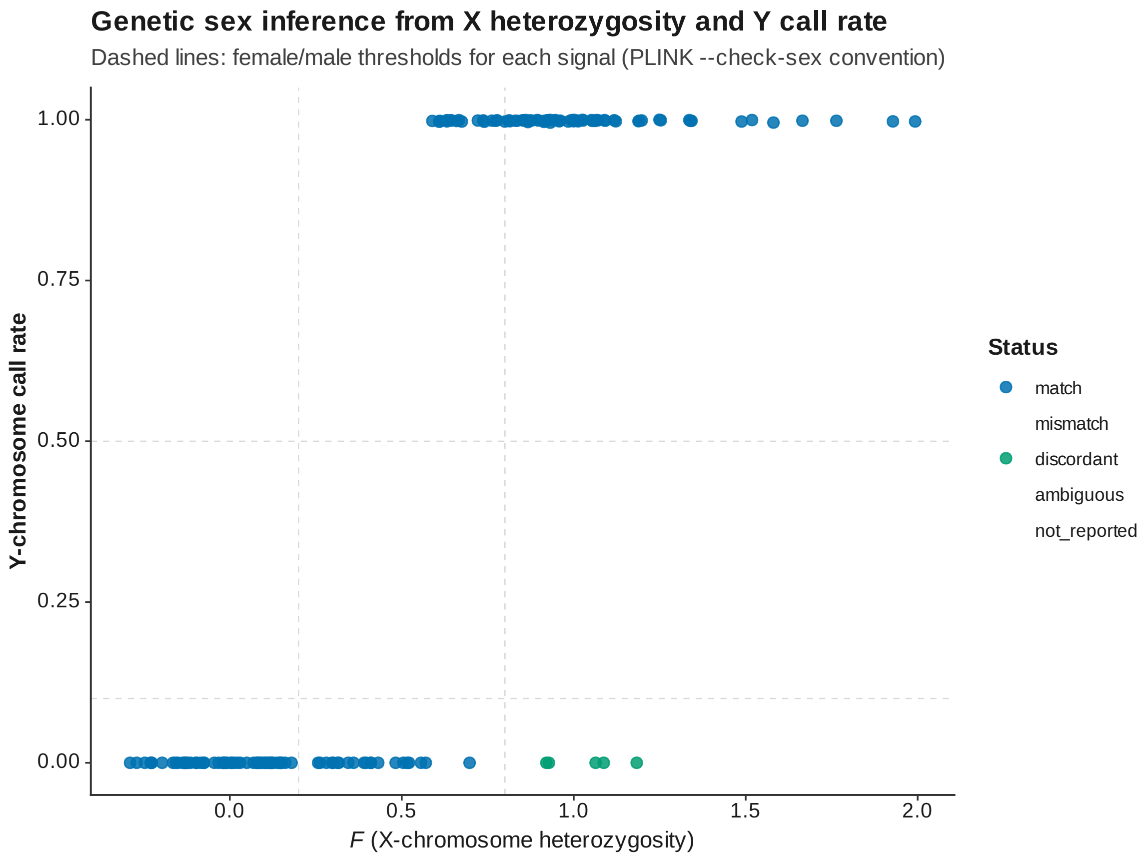

Figure 122: Sex check F by sample

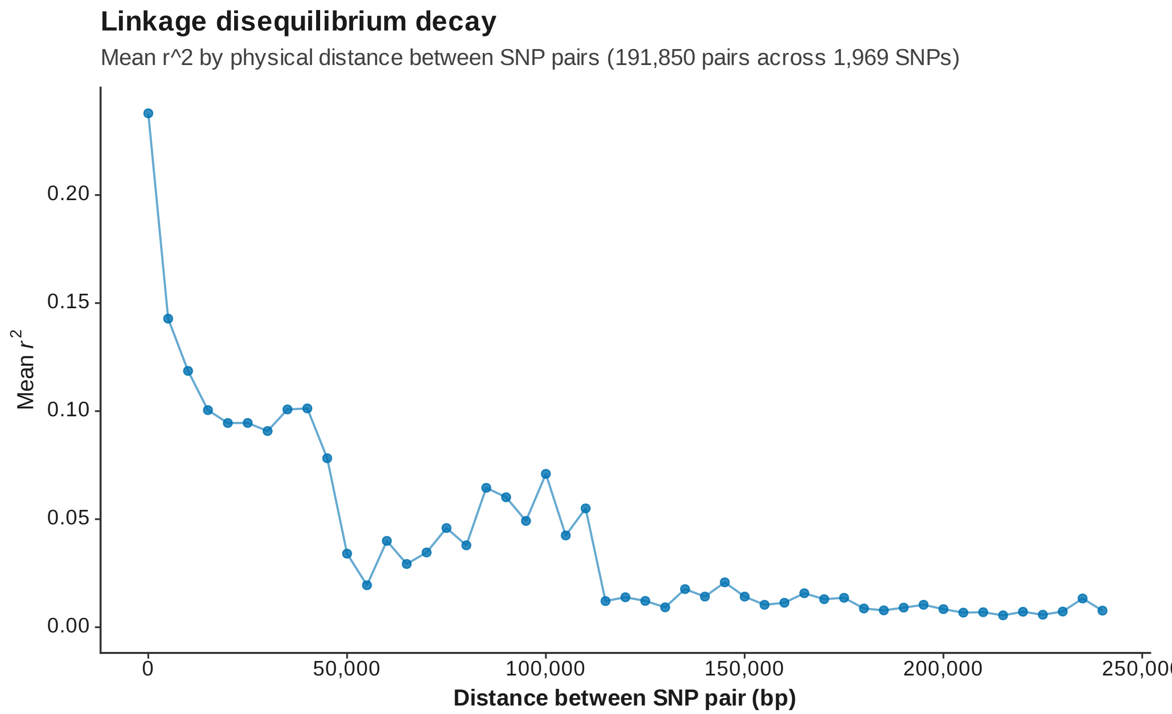

Figure 123: LD decay

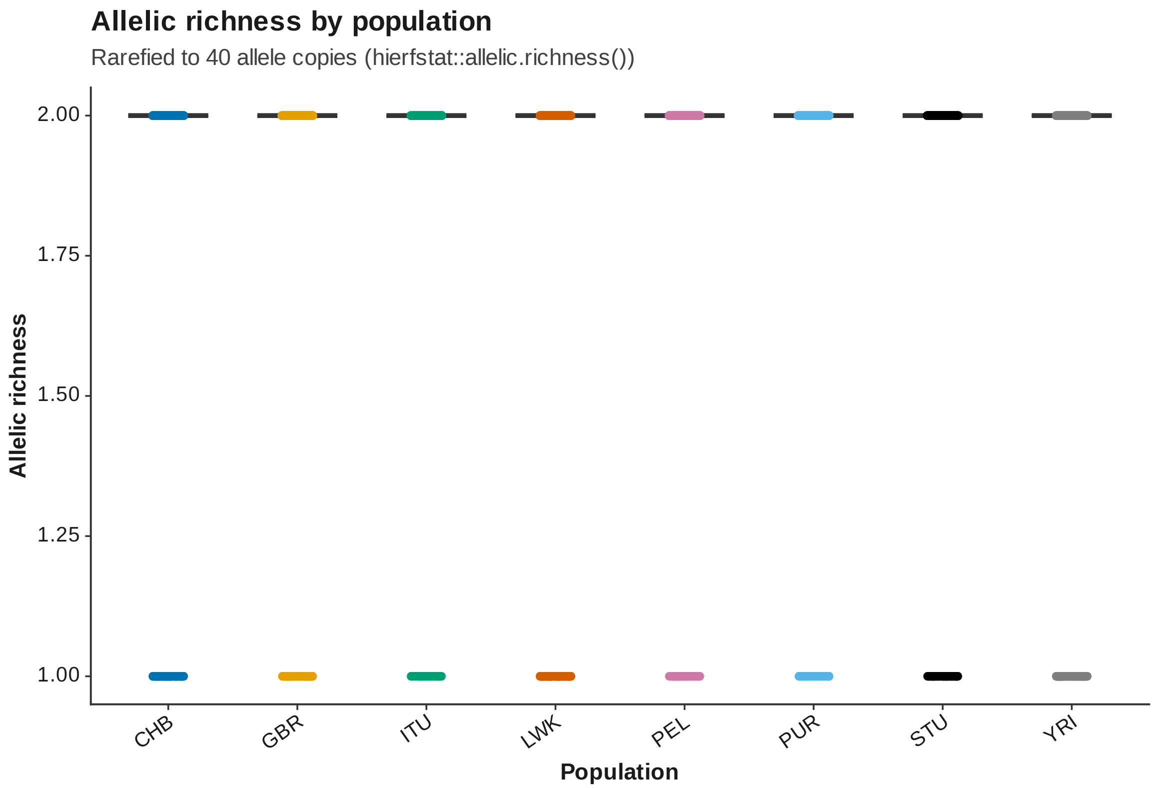

Figure 124: Allelic richness

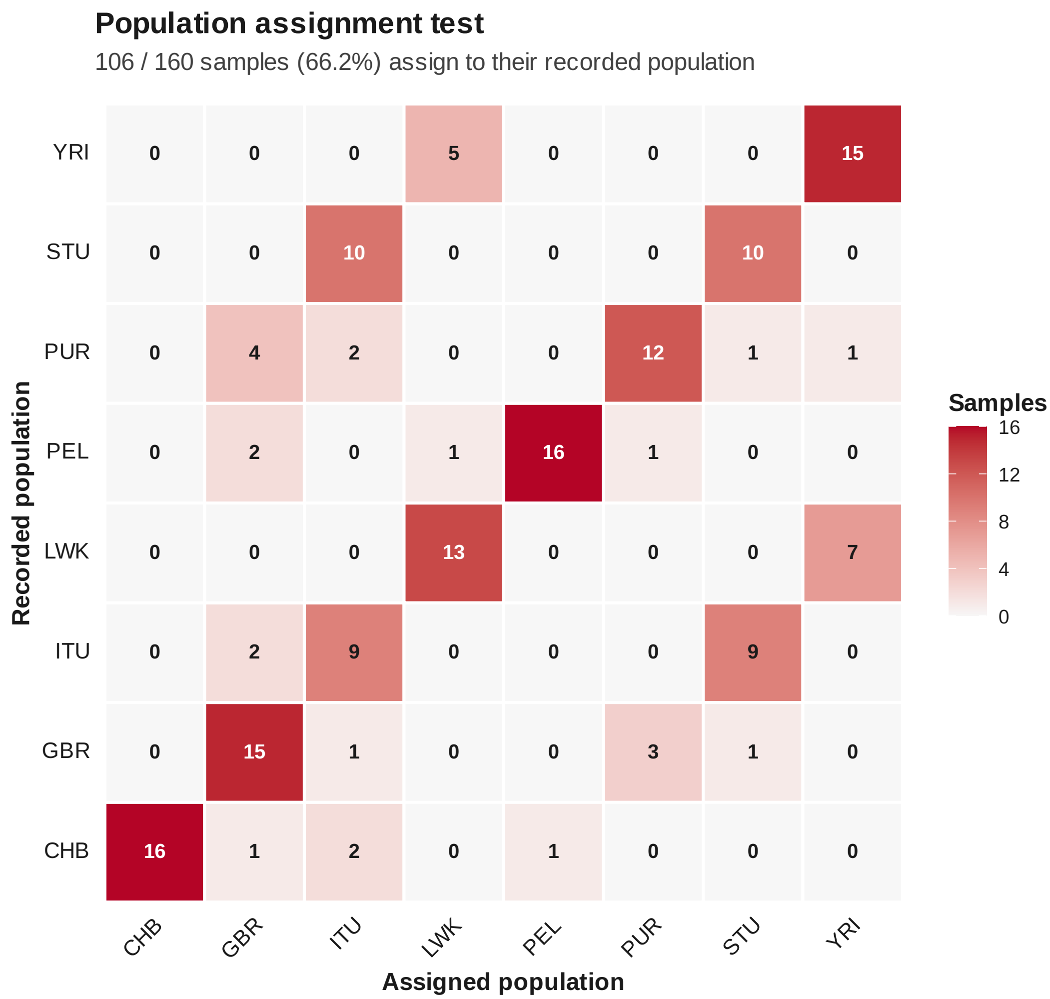

Figure 125: Population assignment

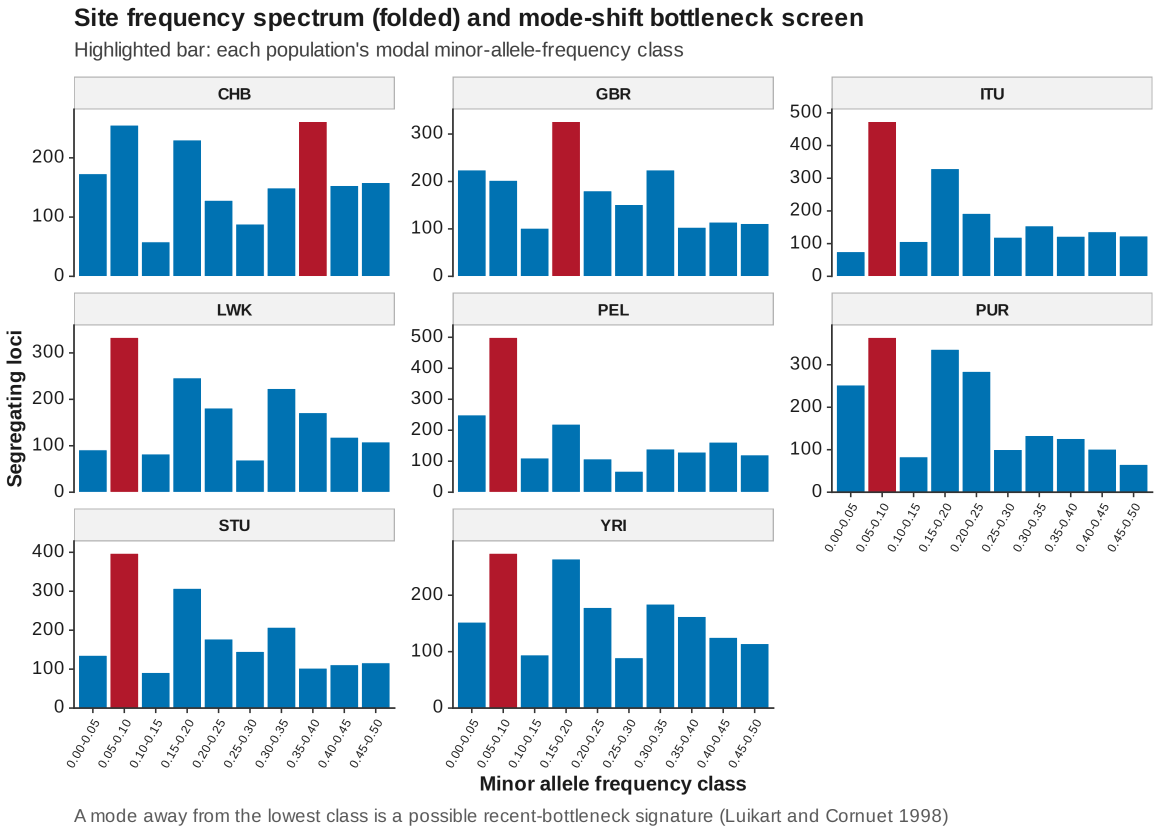

Figure 126: Site frequency spectrum

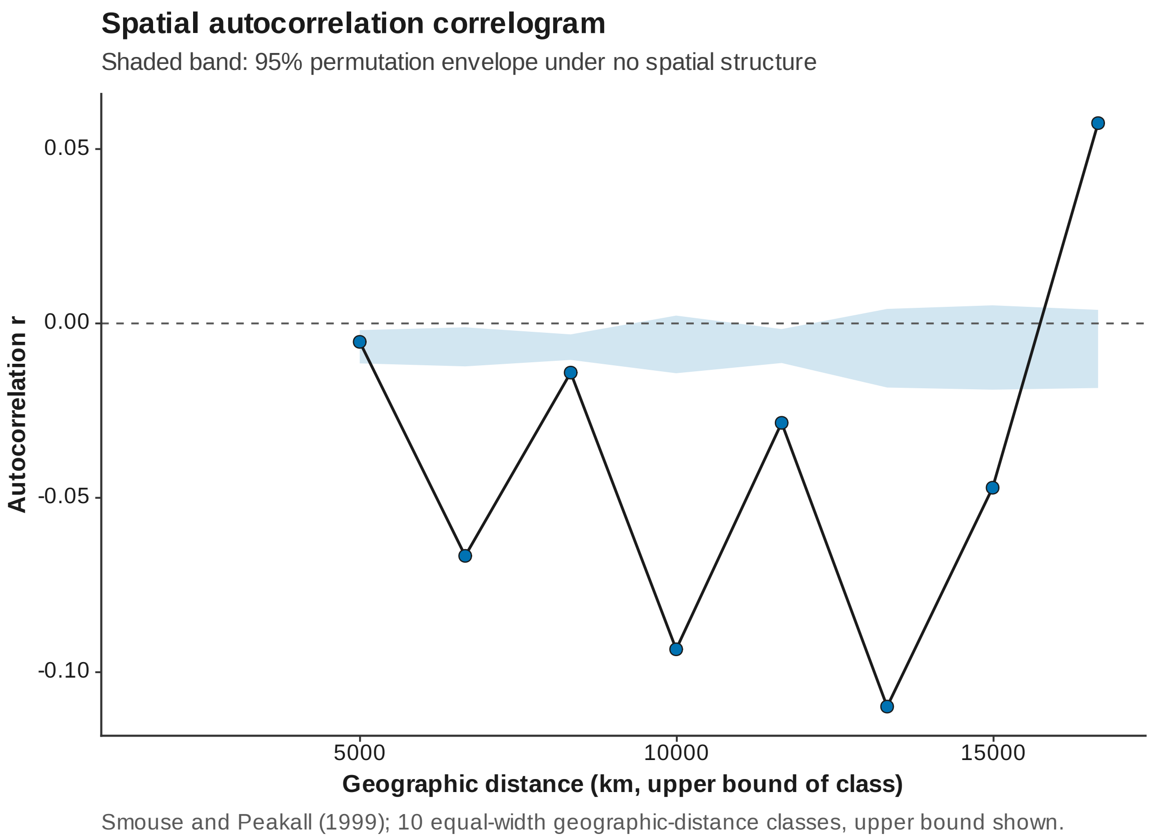

Figure 127: Spatial autocorrelation

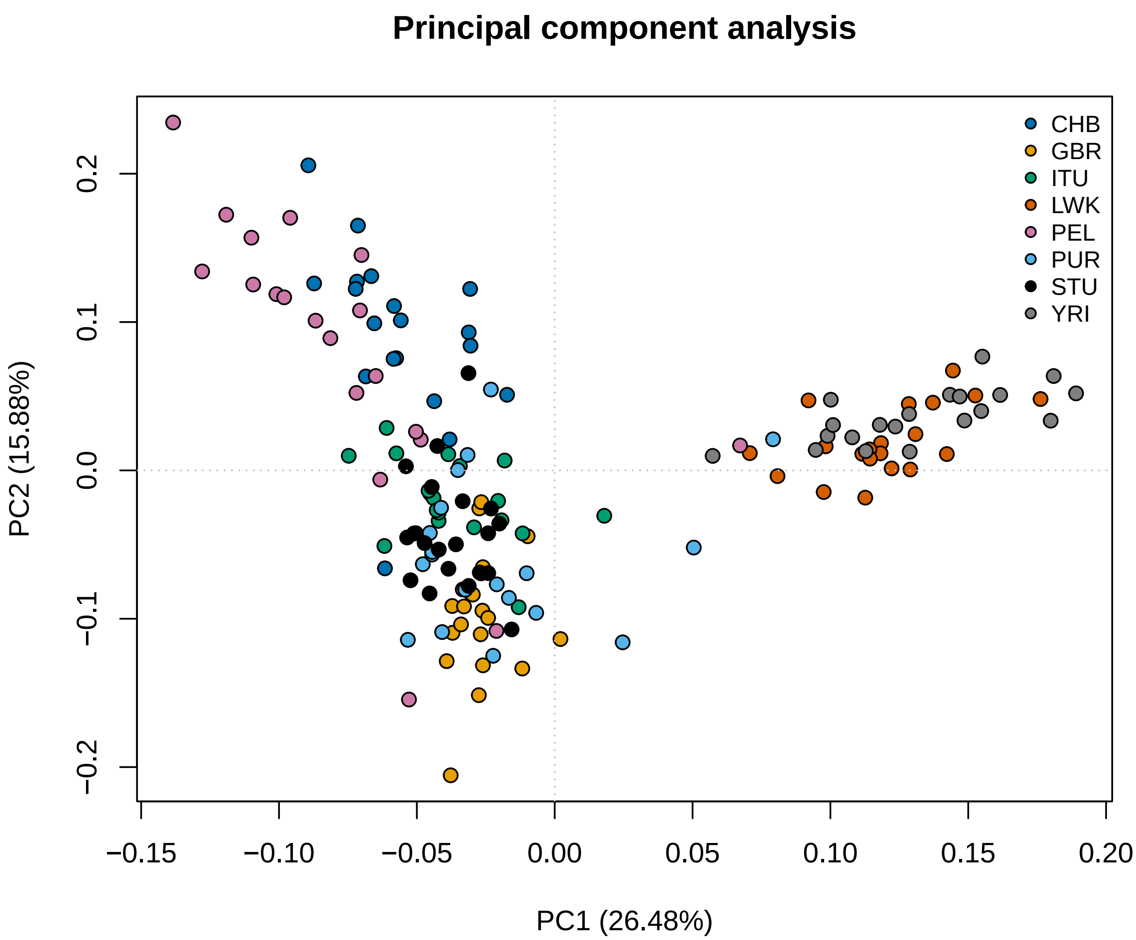

Figure 128: PCA PC 1 PC 2

21 Reproducibility

This report was generated from the serialized pipeline result object. Exact input hashes, settings, package versions, and stage runtimes are stored alongside the report.

21.1 Population-structure reproducibility

K	assignment_accuracy	replicate_max_rmse
2	0.2500	0.0971
3	0.2938	0.0000
4	0.3562	0.3841
5	0.4062	0.3512
6	0.3938	0.2944
7	0.4812	0.0727
8	0.4625	0.1614
9	0.4688	0.1830
10	0.3812	0.2373

21.2 fastStructure

K	exit_status	executable	log_file	q_file	marginal_likelihood
2	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K2.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.2.meanQ	-0.8537830
3	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K3.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.3.meanQ	-0.8492008
4	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K4.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.4.meanQ	-0.8545312
5	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K5.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.5.meanQ	-0.8604124
6	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K6.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.6.meanQ	-0.8672863
7	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K7.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.7.meanQ	-0.8723592
8	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K8.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.8.meanQ	-0.8724382
9	0	/home/bioinfo/miniconda3/envs/popgenvcf/bin/structure.py	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/fastStructure_K9.log	/tmp/claude-1000/-home-bioinfo-prog-popgenVCF/1e0fdcc8-bd1c-40a3-8318-9c5bb1b0a35c/scratchpad/quickstart-reference-run-final/structure/faststructure.9.meanQ	-0.8710042

21.3 Sparse non-negative matrix factorization

K	run	cross_entropy
2	1	0.658188
2	2	0.641827
2	3	0.632785
2	4	0.665263
2	5	0.652600
3	1	0.652267
3	2	0.621568
3	3	0.625406
3	4	0.650066
3	5	0.636630
4	1	0.645990
4	2	0.618994
4	3	0.620640
4	4	0.646500
4	5	0.642422
5	1	0.642907
5	2	0.617357
5	3	0.619569
5	4	0.643777
5	5	0.631210
6	1	0.636456
6	2	0.616501
6	3	0.614301
6	4	0.639528
6	5	0.628516
7	1	0.639879
7	2	0.617185
7	3	0.609143
7	4	0.640496
7	5	0.633083
8	1	0.637875
8	2	0.620175
8	3	0.606337
8	4	0.643657
8	5	0.643623
9	1	0.639730
9	2	0.623835
9	3	0.597450
9	4	0.647969
9	5	0.632479